



Adama Science and Technology University
School of Electrical Engineering and Computing
Department of Computer Science and
Engineering

PROJECT TITLE:

Driving and Vehicles License Department Management
System (DVLD)

DECLARATION

We declare that this project is the result of our own original work carried out as students of Adama Science and Technology University, College of Electrical Engineering and Computing, in the Department of Computer Science and Engineering. All information, analyses, designs, and implementations presented in this project are our independent efforts and have not been submitted to any other institution for academic credit. Any materials, data, ideas, or references obtained from other sources have been properly acknowledged and cited in accordance with academic and ethical standards.

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ACRONYM

ASTU : Adama Science and Technology University
 DVLD : Driving and Vehicle License Department
 UI/UX : User Interface / User Experience
 API : Application Programming Interface
 DBMS : Database Management System
 SQL : Structured Query Language
 IDE : Integrated Development Environment
 UML : Unified Modeling Language
 SDLC : Software Development Life Cycle
 SRS : Software Requirements Specification
 TCP/IP : Transmission Control Protocol / Internet Protocol
 CRUD : Create, Read, Update, Delete
 DBA : Database Administrator
 API : Application Programming Interface
 ASP.NET : Active Server Pages .NET
 ADO.NET : ActiveX Data Objects .NET
 .NET : Dot NET Framework
 VS Code : Visual Studio Code
 SQL Server : Structured Query Language Server
 RDBMS : Relational Database Management System
 RBAC : Role-Based Access Control

ABSTRACT

The Driving and Vehicle License Department (DVLD) Management System is developed to modernize and automate the traditionally manual driving license administration process. In many regions, including Ethiopia, licensing services rely heavily on paper-based procedures, physical visits, and fragmented record management, which result in long processing times, data inconsistency, limited transparency, and high operational workload. These challenges negatively affect applicants, driving schools, and DVLD administrative staff, creating a strong need for a centralized and efficient digital solution.

This project proposes a secure, scalable, and integrated DVLD Management System that connects applicants, driving schools, and DVLD authorities through a unified digital platform. The system provides a mobile application for applicants and a desktop-based application for driving schools and DVLD staff, enabling online registration, application tracking, document management, automated scheduling, and notification services. Developed using a three-tier architecture and modern technologies such as C#, .NET, Flutter, and SQL Server, the system ensures data security through Role-Based Access Control (RBAC) and centralized database management. The proposed solution enhances efficiency, reduces manual errors, improves service transparency, and supports future expansion toward comprehensive e-government services.

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Chapter 1: Introduction

1.1. Introduction

The Driving and Vehicle License Department (DVLD) Management System is intended to streamline and computerize the formerly manual processes of driving license issuances, driving schools and communication with the applicants. The project unveils a more centralized and organized digital platform bringing together an administrative system based on desktops and mobile access by the applicants.

The system allows applicants to apply, monitor application status, and get notifications as well as handle license-related services through a mobile-based application. A desktop based system is used by driving schools and DVLD staff to run through registrations, training schedules, examination and issuing of licenses. The system provides adequate coordination between the applicants, driving institutes, as well as the DVLD, by ensuring the existence of a centralized database and a standardized workflow.

The DVLD Management System enhances efficiency, transparency, and service delivery by automating scheduling and notifications and reducing manual paperwork and imposing structured business rules. The project helps to continue enhancing the digital transformation processes in the administration and maintain the security of the work as it offers a scalable, secure and user-friendly solution to address driving license operations.

1.2. Background of the Project

In most areas, driving license administration remains in the manual systems by use of physical documents, physical application and disjointed record storage systems across multiple offices and driving schools. This has a tendency of making applicants wait a lot, visit several times and have less visibility of the progress of the applications. Driving schools also have independent records that are not always aligned with the DVLD which increases the risk of errors and data inconsistency.

As e-government projects expanded and more people started to use mobile technologies, it became necessary to have a single chain that may link applicants, driving schools and the DVLD into a single digital space. The suggested DVLD Management System provides solutions to these issues by proposing the idea of mobile-based communication with the applicants, desktop-based administrative control, centralized information management, and automated communication systems.

The system substitutes the separated, paper-based processes with the organized computer processes including electronic application processing, planned tests, training status monitoring, and license management. This will enhance better coordination between stakeholders and also guarantee long term scalability and accuracy of data.

1.3. Statement of the Problem

Existing systems suffer from:

- Manual verification and data entry leading to long processing times
- No unified platform connecting applicants, driving schools, and the DVLD
- Lack of automated notifications and scheduling systems
- Difficulty tracking application status across different offices
- Lack of centralized digital records shared across institutions
- Fragmented and inconsistent record management
- Limited transparency for applicants during the licensing process
- Limited scalability for future nationwide expansion

There is a strong need for an automated system that:

- Integrates applicants, driving schools, and the DVLD into a single platform
- Supports mobile access for applicants and desktop dashboards for administrators
- Automates application tracking, scheduling, and notifications
- Ensures reliable and centralized data storage
- Improves operational efficiency and service transparency
- Allows future extensibility for additional digital services

1.4. Objective of the Project

1.4.1. General Objectives

To design and develop a secure, scalable, and fully automated Driving and Vehicle License Department (DVLD) Management System that streamlines and modernizes the entire licensing process. The system will enhance coordination and communication among applicants, driving schools, and the DVLD by providing an integrated platform accessible through both mobile and desktop applications. It will ensure data security, support high user volumes, and automate key workflows such as application submission, scheduling of driving tests, issuance of licenses, and renewal processes. By leveraging modern technologies, the platform will reduce manual errors, minimize processing time, and improve transparency. Additionally, it will offer real-time tracking of application status, digital notifications, and centralized record management, thereby fostering efficiency, accountability, and user satisfaction across all stakeholders involved in the licensing ecosystem.

1.4.2. Specific Objectives

- Develop a centralized database to securely store applicant, school, exam, and license records for easy access and consistency.
- Enable mobile registration so applicants can sign up with driving schools directly through a user-friendly app.
- Provide a desktop dashboard for schools to manage trainees, schedules, and progress with simple reporting tools.
- Support the management and recording of theory and practical test results within the system.
- Automate test scheduling and application status updates to reduce manual coordination.
- Provide timely notifications to applicants regarding schedules, approvals, and service updates through in app alerts and messaging.
- Give DVLD administrators dashboards for oversight, license management, and monitoring of schools and applicants.
- Ensure secure data synchronization between driving schools and the DVLD to maintain accuracy and trust.

- Implement role-based access control to ensure secure and authorized system usage.

1.5. Scope and Limitation

1.5.1. Scope of the Project

The system includes the following modules:

- Applicant mobile application for registration, school selection, application tracking, and notifications
- Driving school desktop dashboard for trainee management, scheduling, and reporting
- DVLD administrative desktop and web dashboard for application processing and license management
- Centralized SQL Server database for storing and managing system data
- Vision, theory and practical test scheduling and result recording
- Automated reminders and system notifications (in-app and system-generated messages)
- Role-based access control for applicants, driving schools, and DVLD administrators
- Synchronization of data across all system components

1.5.2. Limitations of the Project

- Connectivity with the applicant mobile application may be limited due to the offline nature of the DVLD desktop system.
- The system focuses on process management and record handling rather than real-time online examination delivery.
- Identity verification relies on standard document submission and administrative review rather than biometric or computer vision techniques.
- SMS-based notifications may depend on third-party service providers.
- Nationwide deployment would require additional infrastructure, policy alignment, and integration with external government systems.

1.6. Deliverables

1. Fully functional DVLD desktop for administration and Driving Institute dashboards
2. Mobile application for applicants
3. Centralized SQL Server database with defined schemas and relationships
4. Test scheduling and result management modules
5. Notification and reminder system
6. Role-based access control implementation
7. Complete project documentation and technical report
8. Testing and evaluation results

1.7. Feasibility Study

This section assesses whether the project is practical and appropriate, considering its goals, resources, and potential impact. It outlines the main benefits, such as efficiency gains, cost savings, or improved outcomes, and weighs them against the expected costs, including financial investment, time, and risks. The analysis also considers how well the project fits with organizational priorities and whether it can be sustained over time.

1.7.1. Technical Feasibility

The system is technically feasible, relying on reliable and widely supported technologies such as C#, .NET, ASP.NET, Flutter/Xamarin, and SQL Server. These platforms enable secure data handling, multi-platform development, and smooth integration with existing infrastructure. Their widespread use ensures long-term stability, strong community support, and access to extensive documentation, which reduces risks during implementation. Adopting a three-tier architecture further strengthens the design by ensuring modularity, scalability, and easier maintenance, while also allowing for future enhancements such as integration with national ID systems or broader government services.

In addition, the availability of essential development tools including Visual Studio, SQL Server Management Studio, GitHub, and open source libraries advances functionality. These resources are freely accessible and widely used, lowering costs and enabling features like image recognition

or automated verification. Together, the chosen technologies and tools provide a solid technical foundation that minimizes risks, supports long-term adaptability, and maximizes the likelihood of successful project delivery.

1.7.2. Operational Feasibility

The system aligns with existing DVLD workflows while introducing automation and structured communication between applicants, schools, and administrators. Driving schools can manage students online and send training records to the DVLD. Applicants can apply, study, take theory tests, and receive updates directly from their mobile devices. The DVLD can manage operations through both online and offline dashboards, ensuring reliability even in areas with unstable internet access.

1.7.3. Economic Feasibility

The system is economically feasible because it relies on widely available development tools and technologies that do not require significant upfront investment, reducing barriers to adoption for organizations of varying sizes. Its modular architecture makes it scalable at a national level without causing major financial strain, allowing institutions to expand server capacity, integrate Computer Vision models, and strengthen security measures gradually as the project grows. This phased approach ensures that resources are allocated efficiently, minimizing risks while maximizing long-term value. As a result, the solution remains sustainable, practical, and suitable for long-term implementation across the Driving and Vehicles Department, supporting both immediate operational needs and future innovation.

1.8. Significance of the Project

The system delivers major benefits to applicants, driving schools, and the government:

- Faster processing through automation of registration, exam management, and license issuance
- Better transparency and fraud prevention using centralized records and managed process
- Improved applicant experience with mobile access, scheduling, and reminders

- Stronger coordination between schools and DVLD via real-time data exchange
- Foundation for future e-government services such as integration with traffic police or national ID databases

1.9. Beneficiaries of the project

- Applicants: Online applications, exam schedules, reminders, and results
- Driving Schools: Dashboard for managing students and communicating with DVLD
- DVLD Offices: Reduced workload, fewer errors, centralized oversight
- Government: Improved transparency, reduced fraud, and better reporting
- Society: Competent and well-tested drivers, leading to safer roads

1.10. Methodology

Agile methodology will be used to ensure continuous refinement and iterative development across all system modules.

Phases Include:

1. Requirement Analysis – Identify the needs of applicants, schools, and DVLD administrators.
2. System Design – Develop the three-tier architecture, design UI/UX for the mobile and desktop dashboards, and plan data synchronization structures.
3. Implementation – Build mobile, and desktop applications; implement scheduling for Exams, and a centralized database.
4. Integration & Testing – Perform functional, performance, and security testing for all components.
5. Deployment & Documentation – Prepare manuals, deployment plans, and conduct user training.

1.11. Development tools

- Languages: C# (.NET), Dart (Flutter)
- Database: SQL Server
- Frameworks: ASP.NET, WinForms
- Data Access: ADO.NET
- Version Control: GitHub
- IDEs: Visual Studio, VS Code

1.12. Required resources with cost

The project primarily relies on standard development tools, personal computing devices, and freely available software frameworks, which keeps the overall development effort affordable and accessible to a wide range of teams. While the initial build can be completed with minimal expense, a fully scalable deployment would naturally require additional resources such as stable hosting environments, stronger infrastructure, and long-term maintenance strategies to ensure reliability. Importantly, these requirements grow gradually in line with system adoption, allowing costs to be distributed over time rather than concentrated upfront. This incremental approach makes the overall financial commitments both manageable and realistic for a government institution seeking to modernize its services, ensuring that the investment supports sustainable growth, operational efficiency, and future innovation.

1.13. Task and Schedule

The development of the system is organized across five months, ensuring steady progress and clear milestones.

1. **Month 1:** Requirement Analysis and Planning Focus on understanding what the system should achieve, outlining the main features, and shaping the overall direction of the project.
2. **Month 2:** System Design Create the system's architecture, plan the database, and prepare the layouts and workflows for the mobile and web platforms.

3. **Month 3:** Implementation – Backend and Core Functionalities Develop the essential system logic, main modules, and the core operations that support communication between the user, schools, and the department.
4. **Month 4:** Frontend Development and Integration Build and connect the interfaces for all platforms, making sure they work smoothly with the backend and database.
5. **Month 5:** Testing, Debugging, and Deployment Preparation Test the system, make necessary adjustments, and finalize everything needed for deployment, documentation, and project submission.

1.14. Team Composition

No.	Student Name	Role
1.	Bilal Mohammedhasen	Team Leader, Project Manager, Full-stack Developer
2.	Newal Hussen	Back-End Developer, Database Architecture and design
3.	Multezem Mohammed	Front-End Developer, UI/UX Designer
4.	Eman Mohammed	Front-End Developer, UI/UX Designer
5.	Eyob Tesfaye	Back-End Developer, Database Architecture and design

Chapter 2: Description of Existing System and Literature Review

2.1. Major Function of Existing System

The current driving license management process in Ethiopia is largely manual, paper based, and dependent on physical interactions between applicants and DVLD offices. Although some offices have introduced partial computerization , the majority of core operations still rely on human driven procedures. The existing system performs several major functions ,but each function is executed through traditional work flows that lack automation, centralized data access , or real time communication.

Applicant Registration and Document Collection

Applicants must physically visit the DVLD office to submit registration forms and provide supporting documents such as ID cards, previous license levels, medical certificates, and payment receipts. The staff manually check these documents and record applicant details in logbooks or basic computer files.

Input: Paper forms, applicant documents

Process: Manual verification and data entry

Output: Paper-based applicant file

I. Written, Vision, and Practical Test Scheduling

The scheduling of vision tests, written exams, and street driving tests is handled manually by staff. Applicants are assigned examination dates based on availability, often recorded in notebooks or Excel sheets. Applicants must return or call to confirm their exam dates.

Input: Applicant information and preferred exam dates

Process: Staff manually assign test slots

Output: Handwritten or verbally communicated test schedules

II. Conducting Tests and Recording Results

Test results are recorded by evaluators on printed score sheets during the assessment process. These sheets are then collected and submitted to administrative personnel, who are responsible for manually entering the data into local files or official registers. Because this process relies heavily on manual transcription, errors can occasionally occur when transferring information. When such mistakes are identified, corrections must be handled through a separate procedure, which not only increases the workload but also introduces delays in finalizing and validating the records

Input: Test score sheets

Process: Manual scoring and manual entry

Output: Paper records and occasional scattered digital files

III. License Issuance and Renewal

Once an applicant passes the required tests, their information is manually processed to issue a new license or renew an existing one. Staff prepare physical license cards, often relying on basic printing tools without centralized validation across regions.

Input: Approved applicant records

Process: Manual license card preparation

Output: Physical driving license

IV. Handling Complaints and Communication

Communication between applicants and staff occurs mainly through in-person visits, phone calls, or notice boards. There is no automated notification system for exam reminders, document updates, or result announcements.

Input: Applicant complaints and inquiries

Process: Staff response via face-to-face or phone communication

Output: Informal issue handling and resolutions

V. Record Keeping and Reporting

DVLD offices maintain large volumes of physical files and scattered digital documents. Generating reports for performance analysis, fraud investigation, or policy evaluation requires extensive manual effort, and often the data is incomplete or inconsistent.

Input: Paper files, Excel sheets, local computer records

Process: Manual sorting and compilation

Output: Delayed and sometimes inaccurate reports

2.2. Users of the Current System

The existing driving license management process serves multiple stakeholders who directly interact with the system for various purposes. These users depend on the current manual procedures to perform tasks such as registration, verification, testing, license issuance, and administrative management. The needs and expectations of these users vary based on their roles, but most require faster processing, transparency, and reliable communication.

Below are the primary users of the current system:

I. Applicants (New and Renewal Candidates)

Applicants include individuals seeking a new driving license, upgrading their license level, or renewing an existing license. They form the largest user group of the system.

Needs and expectations:

- Simple and clear registration process
- Accurate information about test dates
- Quick results and feedback
- Reduced waiting times and queues
- Reliable communication about requirements, exam schedules, and status updates
- Transparent procedures with minimal manual errors

II. DVLD Administrative Staff

These employees are responsible for handling key administrative tasks such as applicant registration, document verification, accurate data entry, card printing, and maintaining

applicant records. They perform most of the manual work required in the current system, ensuring that essential processes are completed efficiently.

Needs and expectations:

- Streamlined workflows to reduce workload
- Centralized applicant records for easy access
- Tools to minimize repetitive manual tasks
- Faster verification methods
- Reduced risk of errors during data entry
- improved accuracy and reduced paperwork

III. Management and Supervisors (DVLD Officials)

Higher-level officials are tasked with overseeing the overall operations of the system, ensuring that processes run smoothly and efficiently. Their responsibilities include preparing detailed reports, making key administrative and strategic decisions, and monitoring adherence to national driving regulations. By fulfilling these duties, they play an essential role in maintaining accountability, providing direction, and guiding the system toward consistent and effective performance

Needs and expectations:

- Accurate and up-to-date reports
- Reliable record-keeping for auditing and monitoring
- Tools to detect fraud or duplication
- Insights for planning and resource allocation

2.3. Drawbacks of the Current System

The existing driving license management system in Ethiopia relies heavily on manual procedures, paper-based documentation, and fragmented digital tools. While it fulfills basic operational needs, it suffers from numerous drawbacks that limit efficiency, accuracy, transparency, and scalability.

These limitations directly affect applicants, DVLD staff, and other stakeholders who depend on reliable and timely services.

The major drawbacks of the current system include:

I. Inefficiency and Slow Processing

Most procedures including registration, test scheduling, verification, and license issuance are performed manually.

This leads to:

- Long queues and extended waiting times
- Delays in examination and result processing
- Difficulty handling large numbers of applicants
- Repetitive tasks that increase staff workload

II. High Risk of Errors and Inaccuracy

Manual data entry and paper-based record handling increase the chances of:

- Typographical mistakes
- Misplaced or incomplete files
- Incorrect exam results due to transcription errors
- Duplicate or conflicting records

III. Lack of Centralized Data Management

Different DVLD branches maintain their own local files, spreadsheets, or handwritten logbooks to record and track applicant information. Because each branch relies on separate systems, this fragmented approach often leads to duplication, inconsistencies, and difficulties in consolidating records across the organization. This creates issues such as:

- Inconsistent records across regions
- Difficulty tracking applicants who register in multiple branches
- Limited ability to detect duplicate or fraudulent licenses
- Poor sharing of information between offices

IV. Limited Communication and No Notification System

Applicants receive updates manually through:

- Physical visits
- Phone calls
- Notice boards

This communication method causes:

- Missed exam dates
- Lack of reminders for document submission
- Poor applicant-staff interaction
- Increased confusion and repeated office visits

V. Inadequate Fraud Prevention

Because records are stored manually, the system is vulnerable to:

- Fake documents
- Duplicate license issuance
- Altered or lost paper records
- Unauthorized access or manipulation

VI. Poor Reporting and Decision-Making

Generating reports for management requires:

- Sorting through large volumes of paper files
- Manually compiling data
- Cross-checking inconsistent records

This results in:

- Delayed decisions
- Incomplete reports
- Inability to conduct real-time monitoring

- Lack of reliable data for planning and auditing

VII. Lack of Scalability and Modernization

The manual system does not support:

- Online services
- Remote access
- Mobile-based applications
- Integration with other systems (driving schools)

VIII. Dependence on Human Labor

Nearly every stage of the licensing process requires staff involvement. This makes the system vulnerable to:

- Human error
- Staff shortages
- Fatigue and burnout
- Inconsistent handling of cases

IX. Poor Applicant Experience

For many applicants, the current system is frustrating due to:

- Long processing times
- Repeated trips to DVLD offices
- Unclear procedures
- No digital tracking of progress

2.4. Literature Review

A literature review examines existing studies, theories, and systems relevant to the development of a Driving License Management System. It highlights previous efforts in digitalizing licensing processes, explores technological frameworks applied in similar projects, and identifies gaps that this project aims to address. This review is organized into thematic areas that reflect major

components of the system: e-government digitalization, license management systems, database-driven public services, AI-based automation, and software engineering methodologies.

I. E-Government and Digital Public Service Delivery

Globally, governments have been transitioning from manual administrative processes to automated digital platforms. According to studies on e-government systems, digitalization improves transparency, reduces corruption, and enhances service delivery efficiency (Sommerville, 2020). Research indicates that public institutions adopting electronic systems benefit from reduced bureaucracy, faster communication, and better accountability due to centralized data handling (Pressman, 2020).

In developing countries, including Ethiopia, the shift toward electronic public services remains a priority. Literature emphasizes that digitized platforms significantly minimize the challenges associated with paper-based systems, such as data loss, long queues, and inconsistent service quality. This creates an enabling context for developing an intelligent DVLD system that aligns with national digital transformation goals.

II. Driving License Management Systems in Other Countries

Several countries have implemented automated driving license systems that replace manual registration and verification. For example, India and Kenya have adopted digital license issuance systems integrated with centralized government databases, reducing fraud and enabling online verification. Similarly, Rwanda and Ghana use e-license platforms that allow applicants to schedule tests, submit documents, and receive updates through mobile interfaces. Studies of these systems show that automation leads to measurable improvements in accuracy, workflow efficiency, and record management. However, many African nations still face limitations in scalability, offline support, cross-platform access, and AI integration. This suggests a gap in standard license systems that your project aims to fill by combining desktop, mobile, and intelligent functionalities.

III. Role of Databases in Public Administration

Literature on database systems emphasizes the importance of centralized data for ensuring data consistency, accuracy, and secure access (Microsoft SQL Server Documentation, 2025). In traditional government processes, decentralized data storage leads to duplication, inconsistency, and difficulty in auditing. Research highlights that structured database management systems (DBMS) enable:

- Secure storage of citizen data
- Reliable retrieval and updating
- Prevention of unauthorized access
- Support for generating analytical reports

SQL-based systems are commonly used in government applications due to their robustness, scalability, and support for complex queries. These insights justify the adoption of SQL Server in this project to enhance the reliability and transparency of driving license records.

IV. Digital Identity, Validation, and Fraud Prevention

Research in digital identity systems highlights common challenges such as identity theft, duplicate registrations, and document forgery. Systems that rely heavily on manual verification often struggle to detect inconsistencies in applicant data. Studies recommend the use of automated biometric verification, document scanning, and machine learning to enhance integrity. While Ethiopia has implemented digital ID programs (Fayda ID), integration with DVLD systems remains limited. Literature suggests that linking identity records with intelligent license management systems improves:

- Verification accuracy
- Data integrity
- National record consistency

V. Identified Gaps in Existing Research and Systems

The literature highlights several notable gaps in existing driving license management solutions:

- **Lack of combined desktop and mobile support** – Current systems often operate on a single platform, limiting accessibility and flexibility for both administrators and applicants.

- **Insufficient offline capabilities** – In regions with poor or unstable internet connectivity, users face significant challenges, as most solutions rely heavily on continuous online access.
- **Limited integration between public service platforms** – Fragmented systems prevent seamless data sharing and coordination across government agencies, reducing efficiency and transparency.
- **Weak fraud detection mechanisms** – Reliance on manual checks makes it difficult to identify and prevent fraudulent activities, leaving the system vulnerable to misuse.
- **Limited user experience design** – Interfaces are often not tailored to applicants with low digital literacy, creating barriers to access and increasing the likelihood of errors during application or renewal processes.

Chapter 3: Proposed System Documentation

3.1 Overview

The proposed Driving and Vehicle License Department (DVLD) Management System aims at integrating all key operations of licensing under the same digital platform. It links the applicants, the driving schools and the DVLD in a unison of mobile and desktop applications where all three parties are capable of accessing the services they require at ease.

The aim of the system is to enhance the whole system of licensing that includes registration and document filing to the school approval, DVLD check, exam arrangement and issuing of the final license. It also aids in online theory tests, automatic reminders to the exam sessions, and safe transfer of data between the driving schools and the main DVLD office.

The system is technically constructed on three tiers (presentation layer, business logic layer, and data layer) and role-based access control. All the clients interact safely with a centralized SQL Server database which promotes consistency and easy synchronization of data with mobile and desktop interfaces.

3.2 Functional Requirements

The functional requirements describe what the system must be able to do for each major user group, as well as the shared services that support all operations.

3.2.1 Applicant (Mobile App)

- Register an account using personal information and upload required documents.
- Select a license category and submit a new driving license application.
- Provision supporting documentation through the Application Management Service for all application types (new, renewal, lost, damaged, international).
- Schedule and take required exams (vision, written, street).
- View exam schedules, results, and retake options.
- Make payments for application fees, test fees, or renewal fees(not integrated with Banking System but applied in the DVLD System)

- Receive notifications for exam dates, approvals, rejections, and license status updates.
- Track the full progress of license applications.
- Apply for license renewal and upload updated documents.
- Request replacement for lost or damaged licenses.
- Apply for an international driving license.
- View license details, expiry dates, and status.
- Access driving history and past test attempts.
- View and download payment receipts.
- Update profile information and re-upload documents if required.

3.2.2 Driving School and DVLD Administrator (desktop app)

- Consider applications provided by applicants and accept, reject or demand more documentation or explanation.
- Check applications sent by driving schools, prior to formal testing and licensing.
- Accept or designate dates on official theories and practical examinations depending on application status and qualification requirements.
- Control and schedule test events and document test results in the system.
- Issue driving licenses after all the requirements are completed and record the issue details.
- Operate driving school accounts such as the creation, updating and allocation of relevant roles and permissions.
- Manage user activities including creating accounts, resetting passwords and assigning roles.
- Create and access application status, test results, issued licenses, and institutional activity reports.
- Keep databases in sync with the central database of DVLD SQL Server to have unified data in the system.
- Monitor complaints or notifications that have been submitted by applicants or staff and make the required administrative decisions.

3.2.3 System Services (Shared)

- Maintain secure, centralized storage for all applicant, exam, license, and school data.

- Handle scheduling for all exams and prevent timing conflicts.
- Deliver notifications through in-app alerts, SMS, or email when necessary.
- Process documents and identity checks using Computer Vision for face matching and tamper detection.
- Provide secure APIs for all client applications, including proper authentication and access control.

3.3 Non-Functional Requirements

The non-functional requirements describe the overall qualities the system must maintain to ensure a smooth, reliable, and secure experience for all users.

❖ Performance

The system should respond quickly to user actions, handle multiple applicants and schools working at the same time, and keep data processing such as scheduling or verification efficient even during peak periods.

❖ Reliability

All core services should work consistently without frequent interruptions or unexpected downtime. Data must remain intact and protected, avoiding loss or corruption. Critical operations like exam scheduling and document verification should run smoothly without failures.

❖ Availability

The system should remain accessible at all times, ensuring users can depend on its availability whenever needed. Driving schools and DVLD staff must be able to carry out their work without interruptions or downtime. Applicants should also have reliable access to mobile services, allowing them to complete tasks smoothly at any time.

❖ Security

Sensitive data such as IDs, personal details, and exam records must be safeguarded at all times. This protection should rely on secure authentication, strong encryption, and carefully controlled

access mechanisms. Only authorized users should be permitted to view, update, or modify records, ensuring confidentiality and integrity

❖ **Usability**

All interfaces (mobile and desktop) should be simple to navigate, allowing users to complete tasks without confusion. Applicants should find the mobile app straightforward, while staff dashboards should support clear workflows.

❖ **Scalability**

The system should be designed to grow and expand over time, supporting an increasing number of users and driving schools as demand rises. It must also be capable of scaling toward wider national integration, ensuring smooth adoption across regions. This growth should be achieved without requiring major redesigns, preserving stability while accommodating future needs.

❖ **Maintainability**

The system should be designed in a way that makes updates, bug fixes, and feature additions simple to manage. By using a three-tier architecture, components remain well-structured, organized, and easier to maintain over time. This approach ensures long-term flexibility, reduces complexity, and supports efficient improvements as the system evolves.

3.4. System model

3.4.1 Scenarios

Scenario 1: Applicant Registration

- **Initial Assumption:**

The applicant can register either remotely using the DVLD mobile application or in person at a DVLD office or affiliated driving school through the desktop system.

- **Normal Flow (Mobile Registration):**

The applicant opens the DVLD mobile application, enters personal information (full name, date of birth, phone number, address), uploads a national ID and photograph, and

submits the registration form. If all data is valid, the system redirects the applicant to the verification page.

- **Alternative Flow (In-Person Registration via Desktop Application):**

A DVLD staff member or driving school officer registers the applicant using the desktop application by entering personal details and uploading required documents on behalf of the applicant.

- **What Can Go Wrong:**

Invalid phone number format, missing mandatory fields, duplicate records, or unsupported document formats. In such cases, the system displays an error message and requires correction.

- **System State on Completion:**

Applicant data is saved with an unverified or pending verification status, and a verification code or confirmation process is initiated.

Scenario 2: Applicant Login

- **Initial Assumption:**

The applicant is registered and verified in the system.

- **Normal Flow:**

The applicant enters their phone number and password. If credentials are valid, the system authenticates the user and redirects them to the home dashboard.

- **What Can Go Wrong:**

Invalid credentials or locked account. The system displays an appropriate error message.

- **System State on Completion:**

A secure authenticated session is created for the applicant.

Scenario 3: Selecting a Driving School and Submitting Application

- **Initial Assumption:**

The applicant is logged into the system.

- **Normal Flow:**

The applicant selects a preferred driving school from the available list, reviews school

details, and submits a driving license application. Required documents are verified, and the application is forwarded to the selected school.

- **What Can Go Wrong:**

Missing documents, invalid selections, or unavailable training slots.

- **System State on Completion:**

The application is stored in the system and marked as **Pending School Review**.

Scenario 4: Driving School Reviewing Applications

- **Initial Assumption:**

A driving school administrator is authenticated using the desktop application.

- **Normal Flow:**

The administrator reviews submitted applications, checks applicant details, and either approves, rejects, or requests additional documents.

- **What Can Go Wrong:**

Incomplete school authorization, missing applicant data, or system interruptions.

- **System State on Completion:**

Application status is updated, and the applicant is notified of the decision.

Scenario 5: Training Schedule and Notifications

- **Initial Assumption:**

The applicant has been accepted by a driving school.

- **Normal Flow:**

The driving school administrator creates a training schedule and assigns instructors. The applicant receives notifications and can view training dates through the mobile application or via office communication.

- **What Can Go Wrong:**

Schedule conflicts, unavailable instructors, or data synchronization issues.

- **System State on Completion:**

Training schedule is stored and synchronized with the DVLD system.

Scenario 6: Vision Test Scheduling and Evaluation (Local License)

- **Initial Assumption:**
The applicant has completed training and is eligible for testing.
- **Normal Flow:**
DVLD staff schedules a vision test through the desktop application. The applicant is notified of the schedule via the mobile app or in person. A vision test officer records the test result in the system.
- **What Can Go Wrong:**
Applicant fails the vision test, misses the appointment, or medical requirements are not met.
- **System State on Completion:**
Vision test results are stored. **Passing the vision test is a prerequisite for the theory test.**

Scenario 7: Theory Test Scheduling and Attendance

- **Initial Assumption:**
The applicant has passed the vision test.
- **Normal Flow:**
DVLD staff schedules a theory exam using the desktop system. The applicant receives the exam details through notification. The applicant attends the exam as scheduled.
- **What Can Go Wrong:**
Unavailable slots, applicant absence, or scheduling conflicts. If the vision test is not passed, theory scheduling is blocked.
- **System State on Completion:**
Theory test attendance is recorded and awaits result entry.

Scenario 8: Theory Test Result and Retake (If Required)

- **Initial Assumption:**
The applicant has attended the theory test.
- **Normal Flow:**
A DVLD test officer records the theory test result. If the applicant passes, they become eligible for the practical test. If failed, the applicant may request a retake.

- **What Can Go Wrong:**
Applicant fails the test, exceeds retake limits, or has unpaid retake fees.
- **System State on Completion:**
Applicant status updated to Theory Passed or Retake Required.
Passing the theory test is a prerequisite for the practical test.

Scenario 9: Practical Test Scheduling

- **Initial Assumption:**
The applicant has **passed both the vision and theory tests**.
- **Normal Flow:**
DVLD staff assign a practical driving exam date using the desktop application. The applicant receives notification via the mobile app.
- **What Can Go Wrong:**
No available examiners, incomplete records, or scheduling conflicts.
- **System State on Completion:**
Practical test appointment stored and confirmed.

Scenario 10: Practical Test Result and Retake (If Required)

- **Initial Assumption:**
The applicant has attended the practical driving exam.
- **Normal Flow:**
The examiner records the result in the system. If the applicant passes, they become eligible for license issuance. If failed, a retake can be requested.
- **What Can Go Wrong:**
Applicant fails the test, examiner report missing, or misconduct during the exam.
- **System State on Completion:**
Applicant status updated to **Practical Passed or Retake Required**.

Scenario 11: Local Driving License Issuance

- **Initial Assumption:**
The applicant has **passed all required tests**.

- **Normal Flow:**
DVLD staff verifies the applicant's records and issues a local driving license using the desktop system. The applicant is notified and can view license details through the mobile app.
- **What Can Go Wrong:**
Outstanding fees, missing documents, or system processing errors.
- **System State on Completion:**
License status updated to **Issued and Active**.

Scenario 12: Driving License Renewal

- **Initial Assumption:**
The applicant owns an active or expiring driving license.
- **Normal Flow:**
The applicant submits a renewal request via the mobile app or in person. Required documents and medical reports are uploaded or verified. DVLD staff approves the request and updates the license validity.
- **What Can Go Wrong:**
Expired license beyond renewal limits, invalid medical report, or unpaid fees.
- **System State on Completion:**
License validity updated or renewal request rejected.

Scenario 13: Lost or Damaged License Replacement

- **Initial Assumption:**
The applicant owns a valid license that is lost or damaged.
- **Normal Flow:**
The applicant submits a replacement request and pays the required fee. DVLD staff verifies the record and issues a replacement license.
- **What Can Go Wrong:**
Incorrect license details, identity mismatch, or unpaid replacement fees.
- **System State on Completion:**
Replacement license issued and old license marked invalid or inactive.

Scenario 14: International Driving License Application

- **Initial Assumption:**

The applicant holds a **valid local driving license**.

- **Normal Flow:**

The applicant submits an international license request along with required documents and payment. DVLD staff reviews and approves the application.

- **What Can Go Wrong:**

Expired local license, missing passport, or payment issues.

- **System State on Completion:**

International license application saved with **Approved** or **Pending** status.

Scenario 15: License Detainment and Release

- **Initial Assumption:**

A traffic violation requiring license detainment has occurred.

- **Normal Flow:**

Authorized personnel record the detainment in the system. After fine payment, the applicant requests release. DVLD staff verifies payment and restores the license.

- **What Can Go Wrong:**

Incorrect violation data, unpaid fines, or record mismatch.

- **System State on Completion:**

License status restored to **Active**.

3.4.2. Use case model

The use case model serves as a foundational blueprint for understanding how different users interact with the system and what functionalities the system must support. It outlines the full range of operations by identifying each actor and the specific actions they can perform. In our project, the primary actors are the Applicant, Owner, System Administrator, and the System itself. Each actor has distinct responsibilities within the workflow: applicants may browse available properties, submit applications, or update their profiles; owners can manage property listings, review applications, and track tenant activities; system administrators oversee platform settings, maintain

user roles, and ensure system security; while the system actor represents automated processes such as sending notifications, validating data, and updating statuses.

By defining these interactions, the use case model provides a clear understanding of the functional requirements of the entire platform. It ensures that every process—from property browsing to application approval—fits into a structured workflow where each actor’s role is clearly outlined. This helps developers, designers, and stakeholders visualize how different parts of the system work together, enabling better planning, implementation, and control throughout the project lifecycle.



Figure 1: Use Case Diagram

Use Case Description

ACTOR 1: APPLICANT

Use Case 1: Register Account

Description:

Allows a new applicant to create an account in the DVLD system by providing personal and login information.

Pre-Conditions:

- The applicant does not already have an account.

Post-Conditions:

- A new applicant account is created and stored in the system.

Main Flow:

1. Applicant selects **“Register Account.”**
2. The system displays the registration form.
3. Applicant enters required personal and login details.
4. Applicant submits the form.
5. The system validates the information.
6. The system creates the account and confirms registration.

Exception Flows:

- Missing or invalid information → System requests correction.
- Username already exists → System prompts applicant to choose another.

Use Case 2: Login

Description:

Allows an applicant to authenticate and access the DVLD system.

Pre-Conditions:

- Applicant has a registered account.

Post-Conditions:

- The applicant is logged into the system.

Main Flow:

1. Applicant enters username and password.
2. System validates credentials.
3. The system grants access to the applicant dashboard.

Exception Flows:

- Invalid credentials → System displays an error message.

Use Case 3: Apply for New Driving License

Description:

Allows an applicant to submit a request for a new local driving license by selecting a license category and initiating the testing and approval workflow.

Pre-Conditions:

- The applicant is authenticated.
- Applicant profile information is complete.

Post-Conditions:

- A new driving license application is created and linked to the applicant.
- Application status is set to *Pending*.

Main Flow:

1. Applicant selects **“Apply for New Driving License.”**
2. The system displays available license categories.
3. Applicant selects a category and confirms.
4. The system validates age and eligibility rules.
5. The system creates the application record.
6. The system prompts the applicant to upload required documents.

Exception Flows:

- Applicant does not meet eligibility requirements → System rejects the application.
- System error occurs → Application is not created and applicant is notified.

Use Case 4: Upload Required Documents**Description:**

Allows the applicant to upload required documents for a driving license application.

Pre-Conditions:

- A valid application exists.
- The applicant is authenticated.

Post-Conditions:

- Documents are uploaded and linked to the application.

Main Flow:

1. Applicant selects **“Upload Required Documents.”**
2. System displays required document list.
3. Applicant uploads documents.
4. The system validates and stores the documents.

Exception Flows:

- Invalid or missing documents → System requests re-upload.

Use Case 5: Pay Application Fees

Description:

Allows the applicant to pay required application fees.

Pre-Conditions:

- A valid application exists.

Post-Conditions:

- Payment is recorded in the system.

Main Flow:

1. Applicant selects **“Pay Application Fees.”**
2. The system displays payment details.
3. Applicant completes payment.
4. The system confirms successful payment.

Exception Flows:

- Payment failure → System prompts applicant to retry.

Use Case 6: Schedule Vision Test

Description:

Allows the applicant to schedule a vision test appointment.

Pre-Conditions:

- Application is approved for testing.
- Required fees are paid.

Post-Conditions:

- Vision test appointment is scheduled.

Main Flow:

1. Applicant selects “**Schedule Vision Test.**”
2. The system displays available dates and locations.
3. Applicant selects a schedule.
4. The system confirms the appointment.

Use Case 7: Schedule Written Test

Description:

Allows the applicant to schedule a written test.

Pre-Conditions:

- Vision test is passed.

Post-Conditions:

- Written test appointment is scheduled.

Main Flow:

1. Applicant selects “**Schedule Written Test.**”
2. The system displays available schedules.
3. Applicant confirms a selected slot.
4. The system records the appointment.

Use Case 8: Schedule Street Test

Description:

Allows the applicant to schedule a street (practical) driving test.

Pre-Conditions:

- Written test is passed.

Post-Conditions:

- Street test appointment is scheduled.

Main Flow:

1. Applicant selects “**Schedule Street Test.**”
2. The system displays available test slots.
3. Applicant selects and confirms a slot.
4. The system schedules the test.

Use Case 9: Retake Test**Description:**

Allows the applicant to retake a failed test.

Pre-Conditions:

- Applicant has failed a previous test.

Post-Conditions:

- A new test appointment is scheduled.

Main Flow:

1. Applicant selects “**Retake Test.**”
2. The system displays available test dates.
3. Applicant selects a date.
4. The system confirms the retake schedule.

Use Case 10: View Test Results**Description:**

Allows the applicant to view the results of completed tests.

Pre-Conditions:

- Applicants have completed at least one test.

Post-Conditions:

- Test results are displayed.

Use Case 11: Check Application Status

Description:

Allows the applicant to view the current status of submitted applications.

Use Case 12: Receive Approval / Rejection Notification

Description:

Allows the applicant to receive notifications regarding application approval or rejection.

ACTOR 2: DRIVER USE CASE DESCRIPTIONS

Use Case 1: View License Details

Description:

Allows the driver to view detailed information about their driving license, including license class, issue date, expiration date, and current status.

Pre-Conditions:

- The driver is authenticated in the system.
- The driver has at least one license registered.

Post-Conditions:

- License details are displayed to the driver.

Main Flow:

1. The system displays the driver dashboard.
2. The driver selects “View License Details.”

3. The system retrieves license information.
4. The system displays the license details.

Exception Flows:

- No license found → The system displays an informational message.

Use Case 2: Check License Validity

Description:

Allows the driver to verify whether their driving license is valid, expired, or suspended.

Pre-Conditions:

- The driver is authenticated.

Post-Conditions:

- License validity status is displayed.

Main Flow:

1. The system displays the driver services menu.
2. The driver selects “Check License Validity.”
3. The system checks the license status.
4. The system displays the validity result.

Exception Flows:

- License record not found → The system displays an error message.

Use Case 3: Renew License

Description:

Allows the driver to request renewal of an existing driving license.

Pre-Conditions:

- The driver is authenticated.
- The driver has a license eligible for renewal.

Post-Conditions:

- A license renewal application is created.
- Application status is set to Pending.

Main Flow:

1. The system displays the driver services menu.
2. The driver selects “Renew License.”
3. The system validates renewal eligibility.
4. The system creates a renewal application.
5. The system prompts the driver to pay renewal fees.

Exception Flows:

- License not eligible for renewal → The system rejects the request.
- System error → Application is not created.

Use Case 4: Replace Lost License

Description:

Allows the driver to request a replacement license if the original license is lost.

Pre-Conditions:

- The driver is authenticated.
- The driver has a valid license.

Post-Conditions:

- A replacement application is created.

Main Flow:

1. The system displays the driver services menu.
2. The driver selects “Replace Lost License.”
3. The system validates license ownership.
4. The system creates a replacement request.
5. The system prompts the driver to pay replacement fees.

Exception Flows:

- License not found → The system rejects the request.

Use Case 5: Replace Damaged License

Description:

Allows the driver to request a replacement for a physically damaged or lost license.

Pre-Conditions:

- The driver is authenticated.
- The driver has a valid license.

Post-Conditions:

- A damaged or lost license replacement request is recorded.

Main Flow:

1. The system displays the driver services menu.
2. The driver selects “Replace Damaged or Lost License.”
3. The system verifies license details.
4. The system creates a replacement application.
5. The system prompts the driver to proceed with payment.

Exception Flows:

- Invalid license details → The system displays an error.

Use Case 6: Request License Release

Description:

Allows the driver to request the release of a detained license after fulfilling required conditions.

Pre-Conditions:

- The driver is authenticated.
- The driver has a detained license.

Post-Conditions:

- A license release request is submitted.

Main Flow:

1. The system displays the driver services menu.
2. The driver selects “Request License Release.”
3. The system checks detention records.
4. The system prompts the driver to pay detainment fines.
5. The system submits the release request.

Exception Flows:

- License not detained → The system rejects the request.

Use Case 7: Pay Detainment Fines

Description:

Allows the driver to pay fines associated with a detained license.

Pre-Conditions:

- The driver is authenticated.
- Outstanding detainment fines exist.

Post-Conditions:

- Fine payment is recorded.
- License release process can proceed.

Main Flow:

1. The system displays fine details.
2. The driver selects “Pay Detainment Fines.”
3. The driver completes the payment.
4. The system confirms successful payment.

Exception Flows:

- Payment failure → The system prompts the driver to retry.

ACTOR 3: TEST OFFICER

Use Case 1: Conduct Vision Test

Description:

Allows the Test Officer to conduct a vision test for an applicant and evaluate their eyesight according to DVLD rules.

Pre-Conditions:

- The applicant has scheduled a vision test.
- The Test Officer is authenticated in the system.

Post-Conditions:

- Vision test results are recorded in the system.

Main Flow:

1. Test Officer logs into the DVLD system.
2. The Test Officer views the list of scheduled vision tests.
3. The Test Officer selects an applicant.

4. Vision test is conducted.
5. The Test Officer records the test result.
6. The system saves the result.

Exception Flows:

- Applicant absent → Test marked as missed.
- System error → Test result not saved and error message displayed.

Use Case 2: Conduct Written Test

Description:

Allows the Test Officer to supervise and conduct the written driving knowledge test.

Pre-Conditions:

- The applicant has scheduled a written test.
- The Test Officer is logged in.

Post-Conditions:

- Written test results are stored in the system.

Main Flow:

1. Test Officer accesses scheduled written tests.
2. Applicants complete the written test.
3. The Test Officer evaluates the answers.
4. The Test Officer submits the test result.
5. The system saves the result.

Exception Flows:

- Cheating detected → Test marked as failed.
- Incomplete test → Test flagged for retake.

Use Case 3: Conduct Street Test

Description:

Allows the Test Officer to conduct and evaluate the applicant's on-road driving skills.

Pre-Conditions:

- Applicant has passed previous required tests.
- A street test is scheduled.

Post-Conditions:

- Street test results are recorded.

Main Flow:

1. The Test Officer selects the scheduled street test.
2. Applicant performs the driving test.
3. Test Officer evaluates driving performance.
4. Test Officer submits test result.
5. System updates applicant status.

Exception Flows:

- Unsafe driving → Immediate failure.
- Test interrupted → Reschedule required.

Use Case 4: Record Test Results

Description:

Allows the Test Officer to record and submit test results for any conducted test.

Pre-Conditions:

- A test has been completed.

Post-Conditions:

- Test results are saved and linked to the applicant's record.

Main Flow:

1. The Test Officer selects the completed test.
2. The Test Officer enters test results and notes.
3. The Test Officer submits the result.
4. The system stores the data.

Exception Flows:

- Missing result data → System requests completion.

ACTOR 4: REGISTRATION OFFICER**Use Case 1: Register Applicant****Description:**

Allows the Registration Officer to register a new applicant in the DVLD system.

Pre-Conditions:

- Applicant provides required personal information.
- The Registration Officer is logged in.

Post-Conditions:

- Applicant profile is created in the system.

Main Flow:

1. The Registration Officer selects "Register Applicant."
2. The system displays the applicant registration form.
3. Officer enters applicant details.
4. The system validates the information.
5. Applicant record is created.

Exception Flows:

- Missing information → System requests correction.
- Duplicate national ID → Registration rejected.

Use Case 2: Verify Applicant Details**Description:**

Allows the Registration Officer to verify applicant identity and submitted documents.

Pre-Conditions:

- The applicant has submitted required documents.

Post-Conditions:

- Applicant details are verified or rejected.

Main Flow:

1. Officer views pending applications.
2. The officer reviews applicant information and documents.
3. Officer verifies authenticity.
4. System updates verification status.

Exception Flows:

- Invalid documents → Application rejected.
- Incomplete documents → Applicant notified.

Use Case 3: Review Application**Description:**

Allows the Registration Officer to review license applications before approval or rejection.

Pre-Conditions:

- Application is submitted and verified.

Post-Conditions:

- Application status is updated.

Main Flow:

1. Officer opens application list.
2. The officer reviews application details.
3. Officer approves or rejects application.
4. System records decision.

Exception Flows:

- Missing requirements → Application returned for correction.

Use Case 4: Upload Required Documents

Description:

Allows the Registration Officer to upload or attach required documents to an applicant's application.

Pre-Conditions:

- Applicants have an active application.

Post-Conditions:

- Documents are stored and linked to the application.

Main Flow:

1. Officer selects applicant application.
2. Officer uploads documents.
3. System validates file format.
4. The system saves documents.

Exception Flows:

- Unsupported file format → Upload rejected.

Use Case 5: Update Application Status**Description:**

Allows the Registration Officer to update the status of an application (Pending, Approved, Rejected).

Pre-Conditions:

- Application exists in the system.

Post-Conditions:

- Updated application status is saved.

Main Flow:

1. Officer selects application.
2. Officer changes application status.
3. System records the update.
4. Applicants are notified.

Exception Flows:

- Invalid status change → System blocks update.

ACTOR 5: LICENSE ISSUER**Use Case 1: Issue License****Description:**

Allows the License Issuer to issue a driving license to an applicant who has successfully completed all required tests.

Pre-Conditions:

- Applicants have passed all mandatory tests.
- Application status is approved.
- License Issuer is authenticated.

Post-Conditions:

- A new driving license is issued and stored in the system.

Main Flow:

1. License Issuer logs into the system.
2. License Issuer searches for approved applications.
3. License Issuer selects an applicant.
4. The system displays applicant and test details.
5. License Issuer issues the license.
6. The system generates license records.

Exception Flows:

- Incomplete test results → License issuance blocked.
- System error → License not generated.

Use Case 2: Renew License**Description:**

Allows the License Issuer to renew an existing driving license.

Pre-Conditions:

- Driver submits renewal request.
- License is eligible for renewal.

Post-Conditions:

- License expiration date is updated.

Main Flow:

1. License Issuer opens renewal requests.
2. The system displays license details.
3. License Issuer verifies eligibility.
4. License Issuer renews license.
5. System updates expiration date.

Exception Flows:

- License expired beyond allowed period → Renewal rejected.
- Outstanding fines → Renewal blocked.

Use Case 3: Replace Lost License**Description:**

Allows the License Issuer to issue a replacement for a lost license.

Pre-Conditions:

- Driver reports lost license.
- Driver identity is verified.

Post-Conditions:

- Replacement license is issued.

Main Flow:

1. License Issuer receives replacement request.
2. System verifies driver details.
3. License Issuer issues replacement license.
4. The system invalidates old licenses.

Exception Flows:

- Identity mismatch → Replacement denied.

Use Case 4: Replace Damaged License**Description:**

Allows the License Issuer to replace a damaged driving license.

Pre-Conditions:

- Driver submits damaged license.
- Replacement fee is paid.

Post-Conditions:

- A new license is issued.

Main Flow:

1. License Issuer reviews replacement requests.
2. License Issuer verifies license condition.
3. System issues new license.
4. Old license marked as replaced.

Exception Flows:

- License not verifiable → Replacement rejected.

Use Case 5: Detain License**Description:**

Allows the License Issuer to detain a driver's license due to violations.

Pre-Conditions:

- Valid violation records exist.

Post-Conditions:

- License is marked as detained.

Main Flow:

1. License Issuer selects driver record.
2. License Issuer enters detention details.
3. System records detention.
4. License status updated to detained.

Exception Flows:

- Missing violation details → Detainment canceled.

Use Case 6: Release Detained License**Description:**

Allows the License Issuer to release a detained license after fines are paid.

Pre-Conditions:

- License is detained.
- All fines are paid.

Post-Conditions:

- License status updated to active.

Main Flow:

1. License Issuer checks detained license records.
2. The system verifies fine payment.
3. License Issuer releases license.
4. System updates license status.

Exception Flows:

- Unpaid fines → License not released.

Use Case 7: Check License Validity

Description:

Allows the License Issuer to verify whether a license is valid.

Pre-Conditions:

- License exists in the system.

Post-Conditions:

- License validity status is displayed.

Main Flow:

1. License Issuer enters license number.
2. The system retrieves license records.
3. The system displays validity status.

Exception Flows:

- License not found → Error message displayed.

ACTOR 6: ADMINISTRATOR

Use Case 1: Add User

Description:

Allows the Administrator to add new system users.

Pre-Conditions:

- Administrator is authenticated.

Post-Conditions:

- A new user account is created.

Main Flow:

1. Administrator selects “Add User.”
2. The system displays user forms.
3. Administrator enters details.
4. The system saves user accounts.

Exception Flows:

- Duplicate username → Creation rejected.

Use Case 2: Add Test Officer**Description:**

Allows the Administrator to add a Test Officer to the system.

Pre-Conditions:

- Administrator is logged in.

Post-Conditions:

- A Test Officer account is created.

Main Flow:

1. Administrator selects “Add Test Officer.”
2. Enter officer details.
3. The system validates and saves data.

Exception Flows:

- Missing required data → Request correction.

Use Case 3: Manage License Classes

Description:

Allows the Administrator to manage driving license classes.

Pre-Conditions:

- Administrator authenticated.

Post-Conditions:

- License classes updated.

Main Flow:

1. Administrator opens license class management.
2. Adds or updates class details.
3. The system saves changes.

Exception Flows:

- Invalid class data → Update rejected.

Use Case 4: View Reports**Description:**

Allows the Administrator to view system reports and statistics.

Pre-Conditions:

- System data available.

Post-Conditions:

- Reports displayed.

Main Flow:

1. Administrator selects report type.
2. The system generates reports.

3. Administrator views data.

Exception Flows:

- No data available → Empty report shown.

Use Case 5: Add Driving School Institute**Description:**

Allows the Administrator to register and manage driving school institutes in the system.

Pre-Conditions:

- Administrator is authenticated.

Post-Conditions:

- Driving institute details are stored in the system.

Main Flow:

1. Administrator selects “Add Driving School Institute.”
2. The system displays the institute registration form.
3. Administrator enters institute details.
4. Administrator submits the form.
5. The system validates and saves the information.

Exception Flows:

- Missing or invalid data → System requests correction.

ACTOR 7: INSTITUTE MANAGER**Use Case 1: Manage Driving Institute****Description:**

Allows the Institute Manager to manage driving school details.

Pre-Conditions:

- Institute Manager authenticated.

Post-Conditions:

- Institute details updated.

Main Flow:

1. Manager opens institute profile.
2. Updates institute information.
3. The system saves changes.

Exception Flows:

- Invalid data → Update rejected.

Use Case 2: Create Training Schedule**Description:**

Allows the Institute Manager to create training schedules for students.

Pre-Conditions:

- Students enrolled.

Post-Conditions:

- Training schedule created.

Main Flow:

1. Manager selects “Create Schedule.”
2. Enters training dates and times.
3. The system saves schedules.

Exception Flows:

- Schedule conflict → System warns manager.

Use Case 3: Submit Training Completion Report

Description:

Allows the Institute Manager to submit training completion reports.

Pre-Conditions:

- Students completed training.

Post-Conditions:

- Training completion recorded.

Main Flow:

1. The manager selects students.
2. Uploads completion report.
3. System stores report.

Exception Flows:

- Missing report → Submission blocked.

Use Case 4: View Student Progress

Description:

Allows the Institute Manager to monitor student progress.

Pre-Conditions:

- Students registered under the institute.

Post-Conditions:

- Progress information displayed.

Main Flow:

1. The manager selects students.
2. The system displays training progress.

Exception Flows:

- Student not found → Error displayed

3.5. Object Model

3.5.1. Data Dictionary

1.Applicant

Attributes	Data Type	Description
applicant_id	INT(PK)	Unique identifier for each applicant
first_name	VARCHAR(50)	Applicant's first name
dob	DATE	Date of birth
phone	VARCHAR(15)	Contact phone number
email	VARCHAR(100)	Email address (optional)
address	VARCHAR(200)	Residential address
status	ENUM('submitted', 'school accepted', 'DVLDVerified', 'ExamScheduled', 'Completed', 'Rejected')	Current application status
created_at	DATETIME	Record creation timestamp
updated_at	DATETIME	Last update timestamp

Table 2: Applicant data dictionary

2.Documents

Attribute	Data type	Description
document_id	INT(PK)	Unique document identifier
applicant_id	INT(FK)	Reference to applicant
type	ENUM('ID', 'Medical', 'Photo')	Type of document
file_path	VARCHAR(255)	Storage path of uploaded file
verified	BOOLEAN	Verification status
uploaded_at	DATETIME	Upload timestamp

Table 3: Document data dictionary

3.Driving Institute

Attribute	Datatype	Description
school_id	INT(PK)	Unique school identifier
name	VARCHAR(100)	School name
address	VARCHAR(200)	School physical address
contact_phone	VARCHAR(15)	School contact number
contact_email	VARCHAR(100)	School email address
status	ENUM('Registered', 'Blocked')	School registration status

created_at	DATETIME	Registration date
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Table 4: Driving institute data dictionary

4.Application

Attribute	Datatype	Description
application_id	INT(PK)	Unique application identifier
applicant_id	INT(FK)	Reference to applicant
school_id	INT(FK)	Reference to selected school
applied_on	DATE	Application submission date
current_status	VARCHAR(50)	Current application status
remarks	TEXT	Additional notes/comments

Table 5: Application data dictionary

5.Application Types

Attribute	Datatype	Description
type_id	INT(PK)	Unique type identifier
title	VARCHAR(100)	Type title (New, Renewal, Replacement, International, Release)
fees	DECIMAL(10,2)	Fee amount

Table 6: application type data dictionary

6.Exam

Attribute	Datatype	Description
exam_id	INT (PK)	Unique exam identifier
application_id	INT (FK)	Reference to application
exam_type	ENUM('Theory', 'Practical')	Type of examination
scheduled_at	DATETIME	Scheduled date and time
location	VARCHAR(100)	Exam venue/location
status	ENUM('Scheduled', 'Completed', 'Canceled')	Exam status
score	INT (NULLABLE)	Exam score (if completed)
result	ENUM('Pass', 'Fail', 'Pending')	Exam result

Table 7: Exam data dictionary

7.License

Attribute	Datatype	Description
license_id	INT(PK)	Unique license identifier
applicant_id	INT(FK)	Reference to applicant
license_number	VARCHAR(20)	Official license number
issue_date	DATE	License issuance date
expiry_date	DATE	License expiration date

status	ENUM('Active', 'Expired', 'Suspended', 'Revoked')	License status
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Table 8: Licence data dictionary

8.Notification

Attribute	Datatype	Description
notification_id	INT(PK)	Unique notification identifier
recipient_id	INT	ID of recipient (applicant/school/DVLD)
channel	ENUM('In-App', 'SMS', 'Email')	Delivery channel
message	TEXT	Notification content
sent_at	DATETIME	Send timestamp
status	ENUM('Sent', 'Delivered', 'Failed')	Delivery status

Table 9: Notification data dictionary

9.Fraudcheck

Attribute	Attribute	Description
check_id	INT(PK)	Unique check identifier
application_id	INT(FK)	Reference to application
confidence_score	DECIMAL(5,2)	Match confidence percentage
performed_at	DATETIME	Check execution timestamp
notes	TEXT	Additional observations

Table 10: Fraud check data dictionary

10.Complaint

Attribute	Attribute	Description
complaint_id	INT(PK)	Unique complaint identifier
application_id	INT(FK)	Reference to application
institute_id	INT(FK)	Reference to institute
description	TEXT	complain text

Table 11: Complaint data dictionary

3.5.2. Class diagram

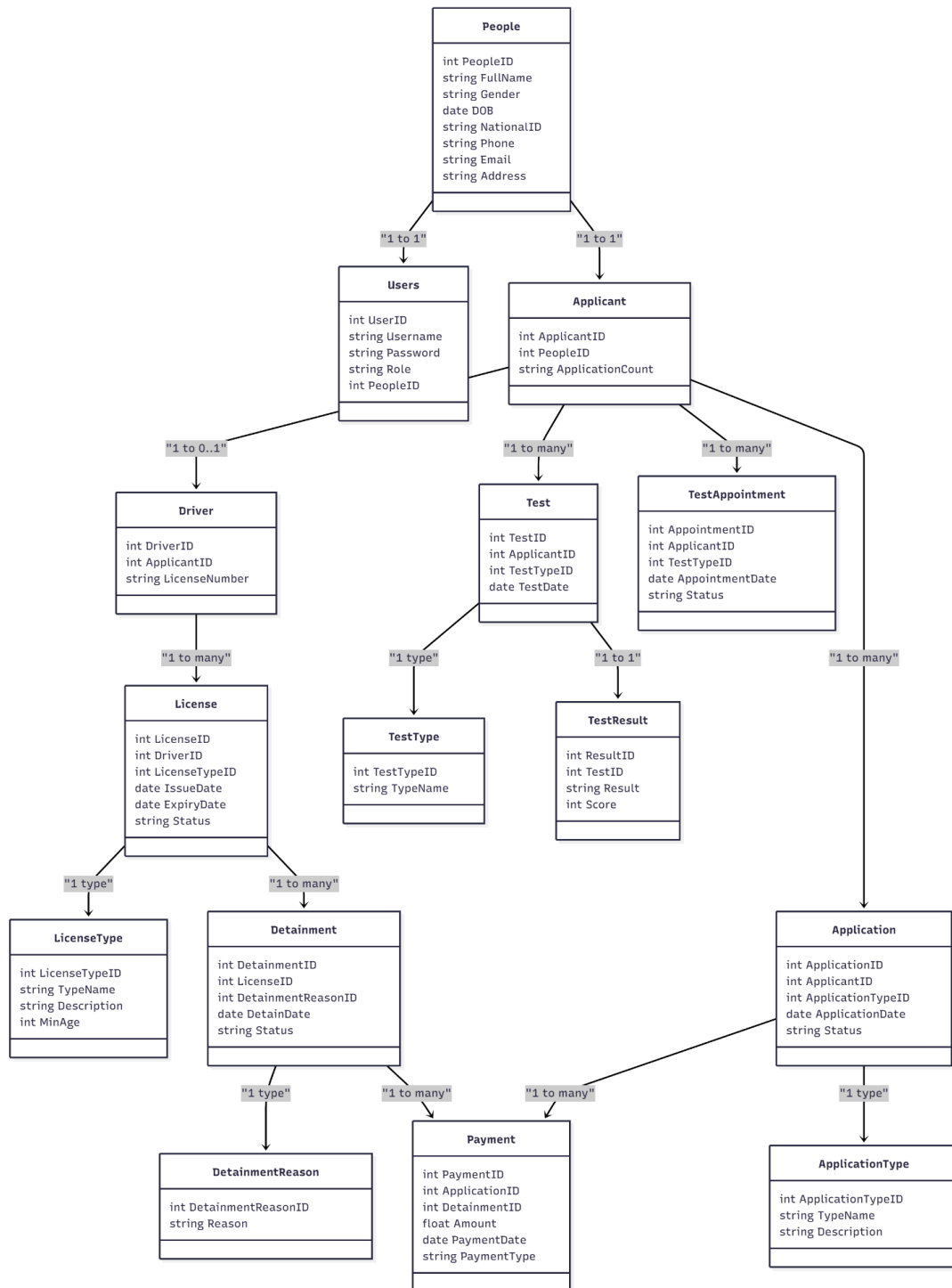


Figure 2: Class Diagram of DVLD Management System

3.6. Dynamic model

The Dynamic Model describes the behavior of the DVLD Management System over time by illustrating how different actors interact with the system during various operational workflows. It focuses on the sequence of actions, data exchanges, and state transitions that occur while performing core processes such as applicant registration, training management, test scheduling, examination handling, license issuance, renewal, and detention.

This model helps in understanding how system components respond to user actions, how decisions are made at different stages, and how information flows between the mobile application, desktop system, business logic, and database. By capturing these time-dependent interactions, the Dynamic Model provides a clear view of the system's operational logic and ensures that all functional requirements are executed in the correct order.

3.6.1. Sequence diagram

The Sequence Diagram illustrates the chronological order of interactions between system components and actors over time, showing how requests move from the applicant through mobile or desktop interfaces, into the business logic layer, and finally to the database where information is stored or retrieved. It highlights key processes such as registration, test scheduling, and license issuance, making clear how each step depends on the proper exchange of messages. By mapping these interactions, the diagram clarifies both the flow of communication and the system's response at each stage, helping stakeholders understand dependencies, identify potential bottlenecks, and ensure that workflows remain consistent and reliable.

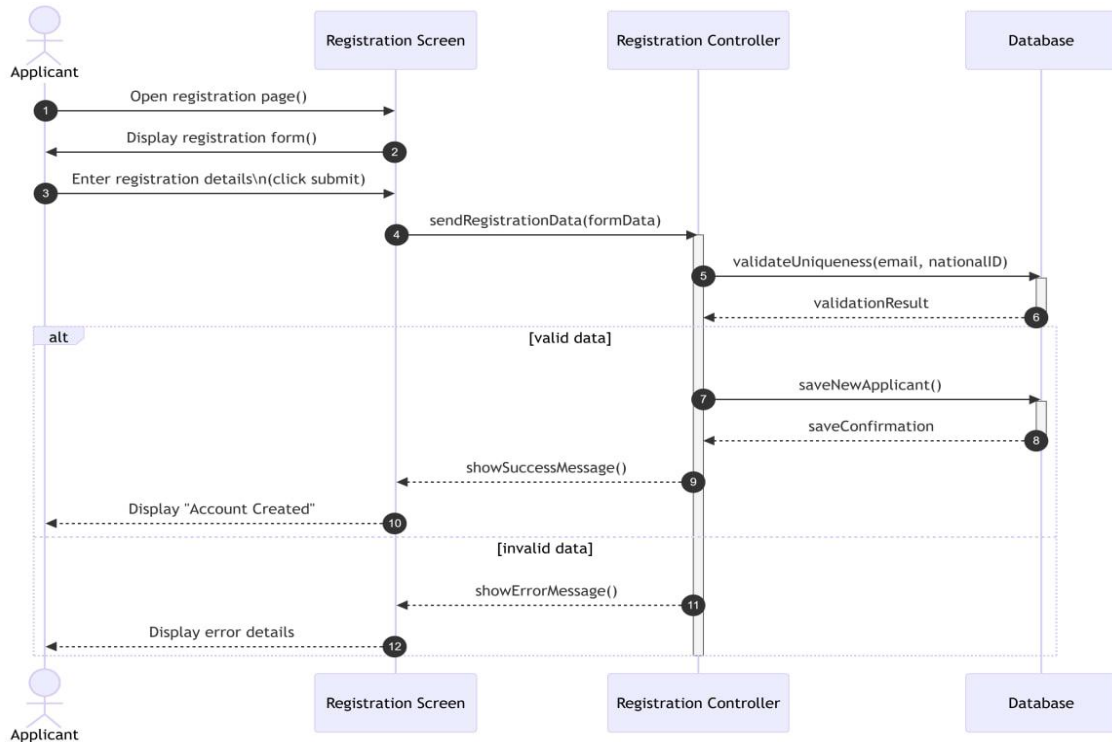


Figure 3: Sequence diagram of Applicant Registration

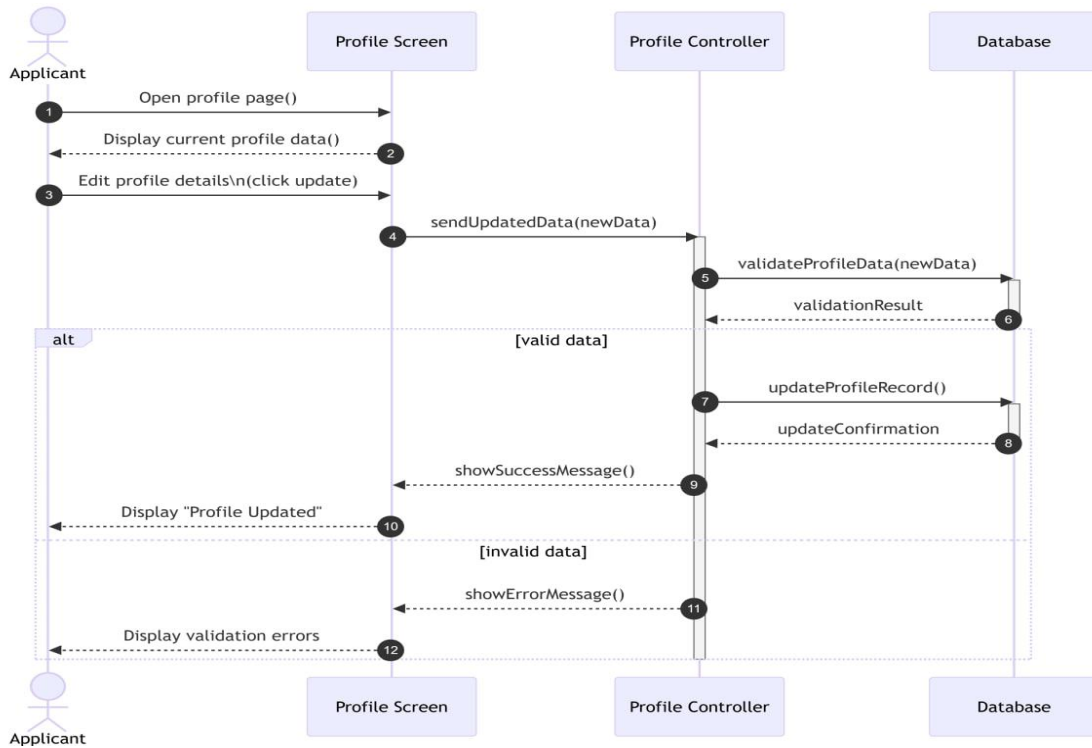


Figure 4: Sequence diagram of Applicant Update Profile

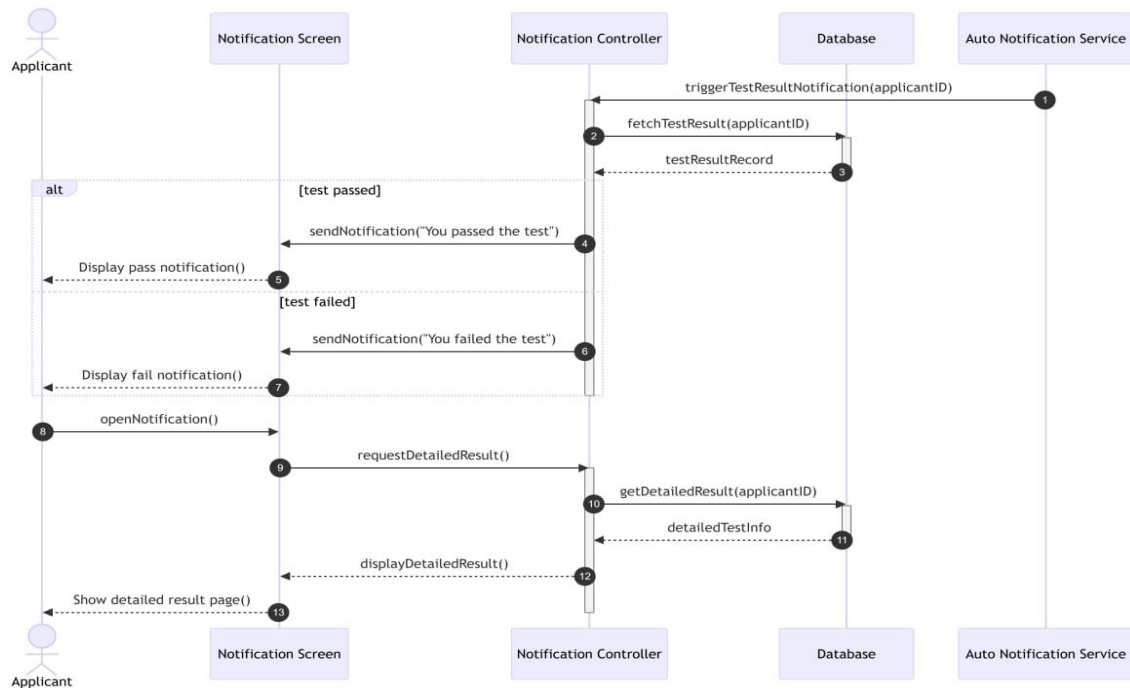


Figure 5: Sequence diagram of Receive Test Result

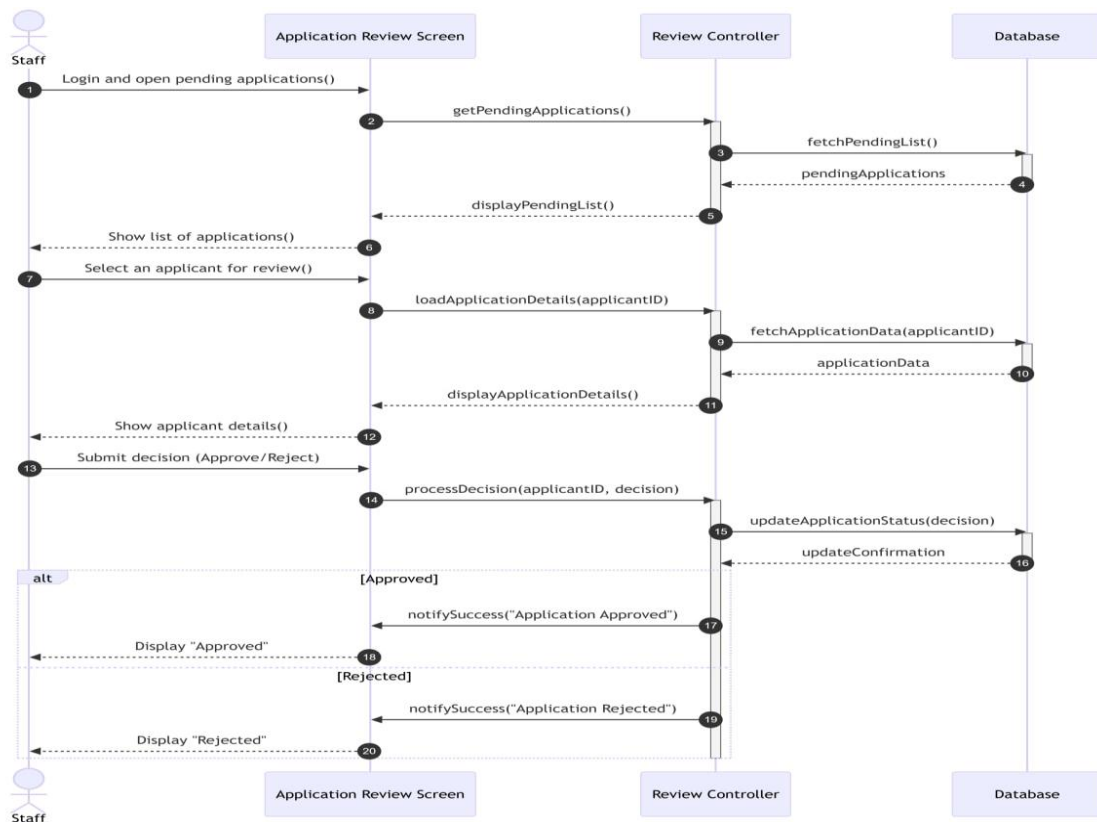


Figure 6: Sequence diagram Staff Review Application

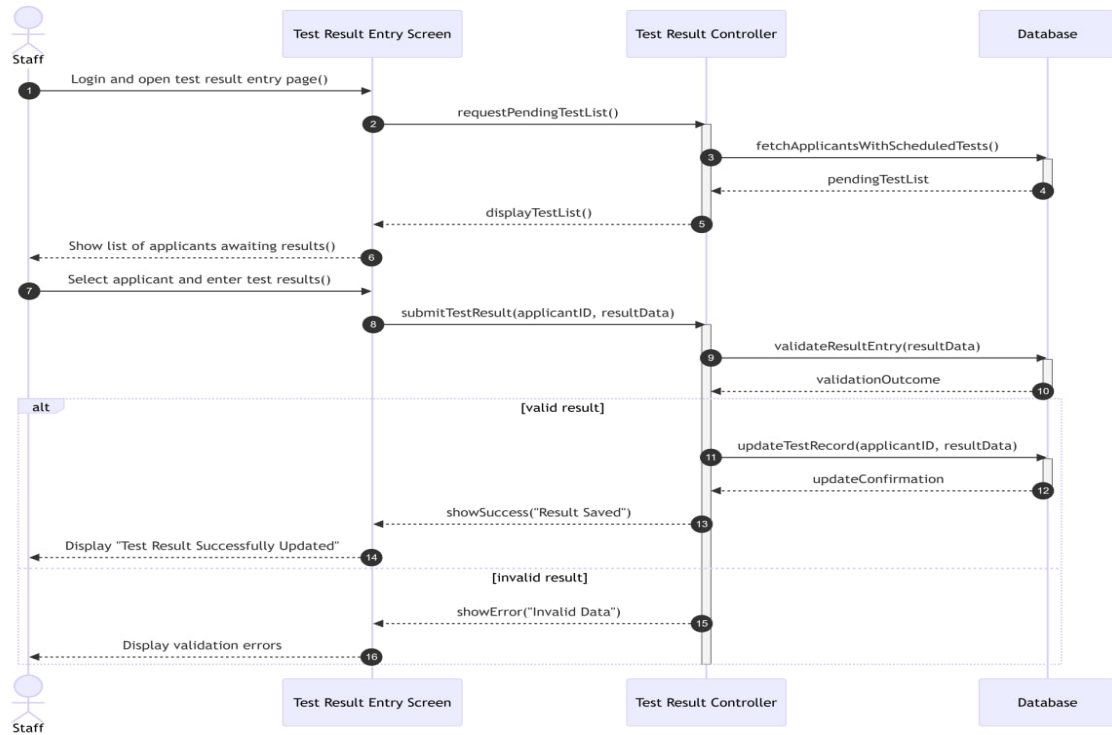


Figure 7: Sequence diagram of Update Test Result

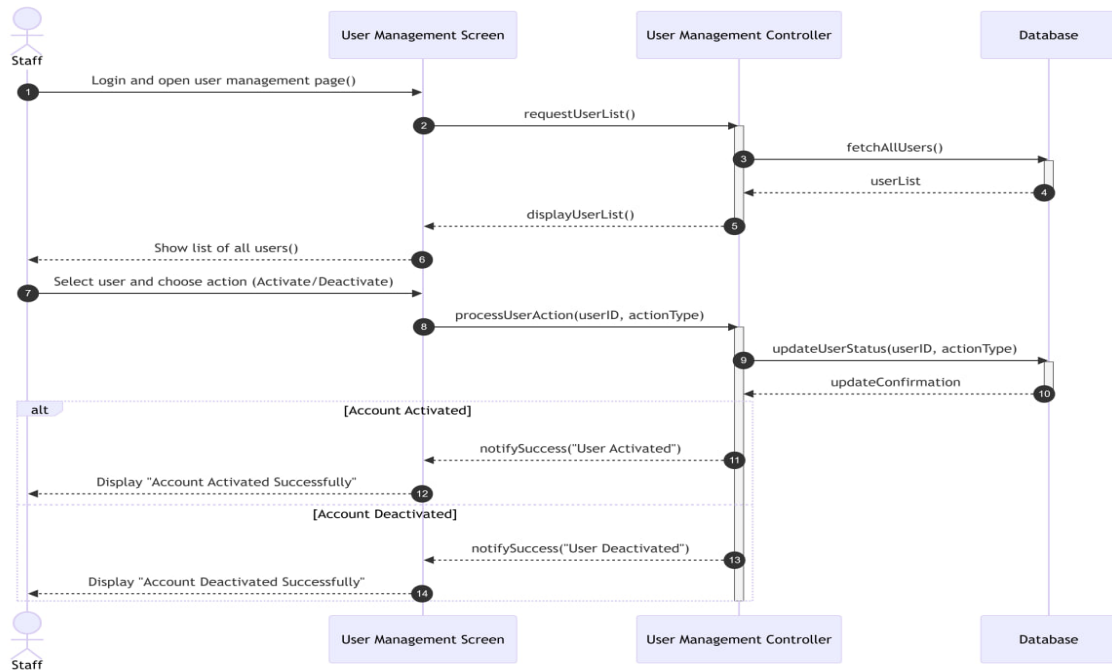


Figure 8: Sequence diagram of Staff Manages User Accounts

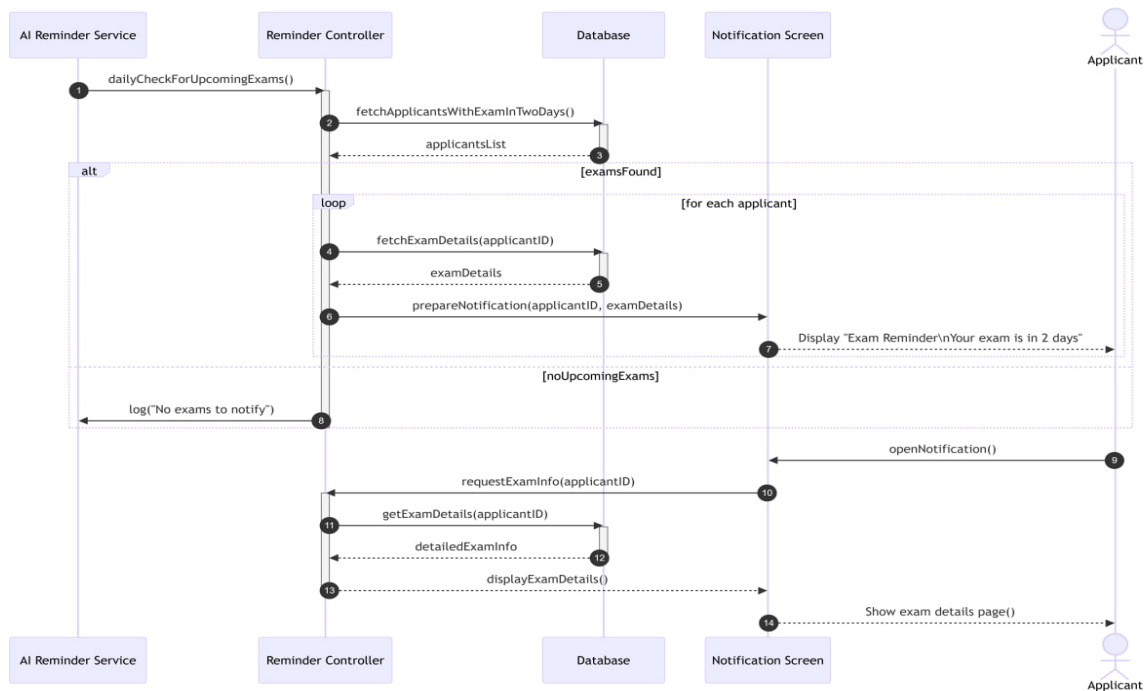


Figure 9: Sequence diagram of System send Exam reminder notification

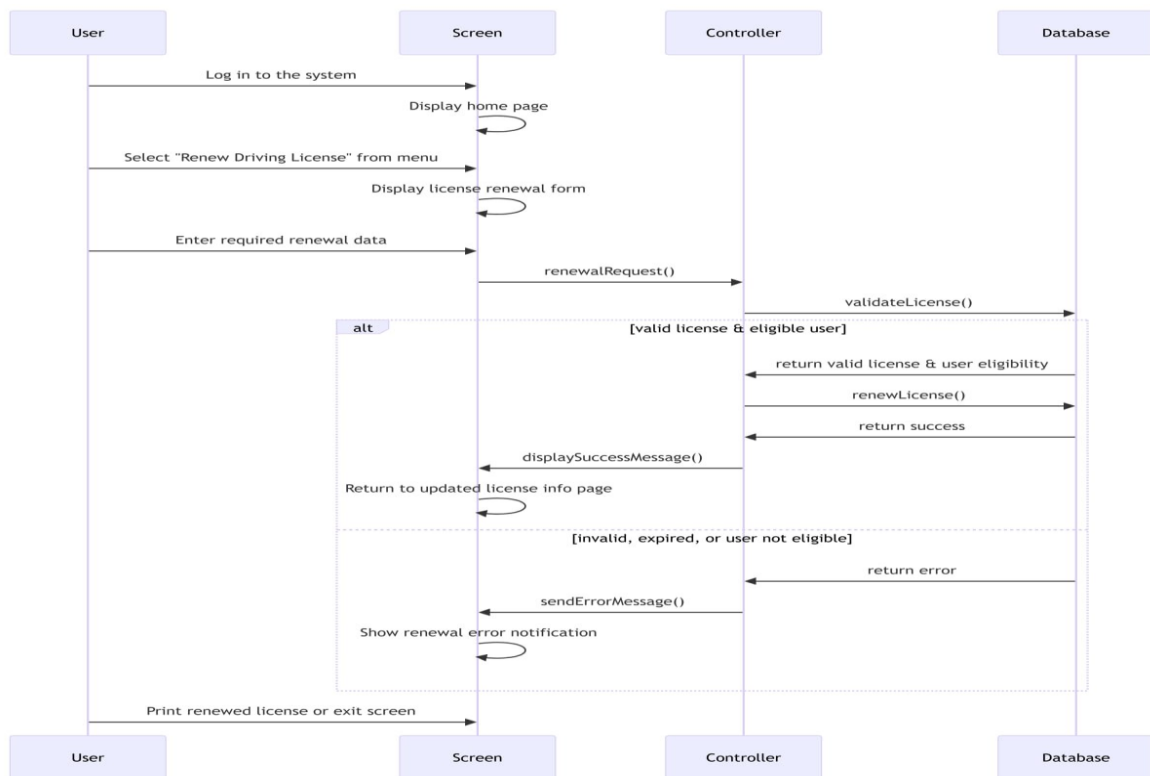


Figure 10: Sequence diagram of License Renew Process

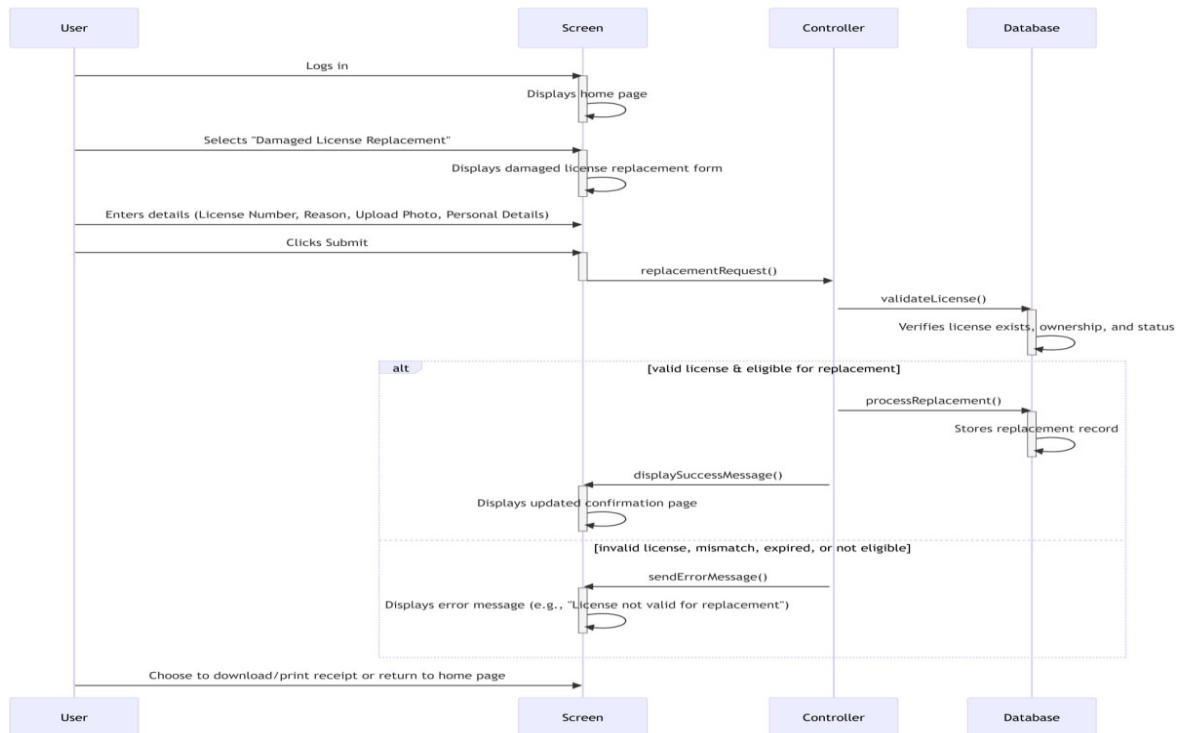


Figure 11: Sequence diagram of Damaged and Lost

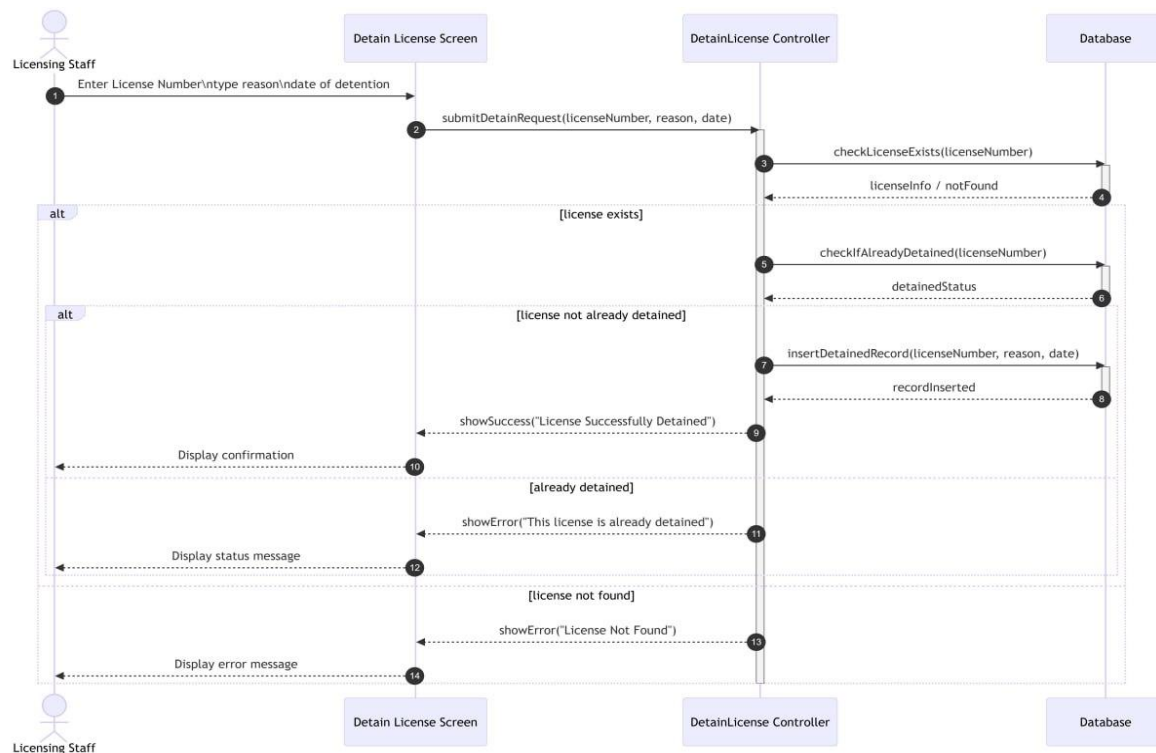


Figure 12: Sequence diagram of Detained License Process

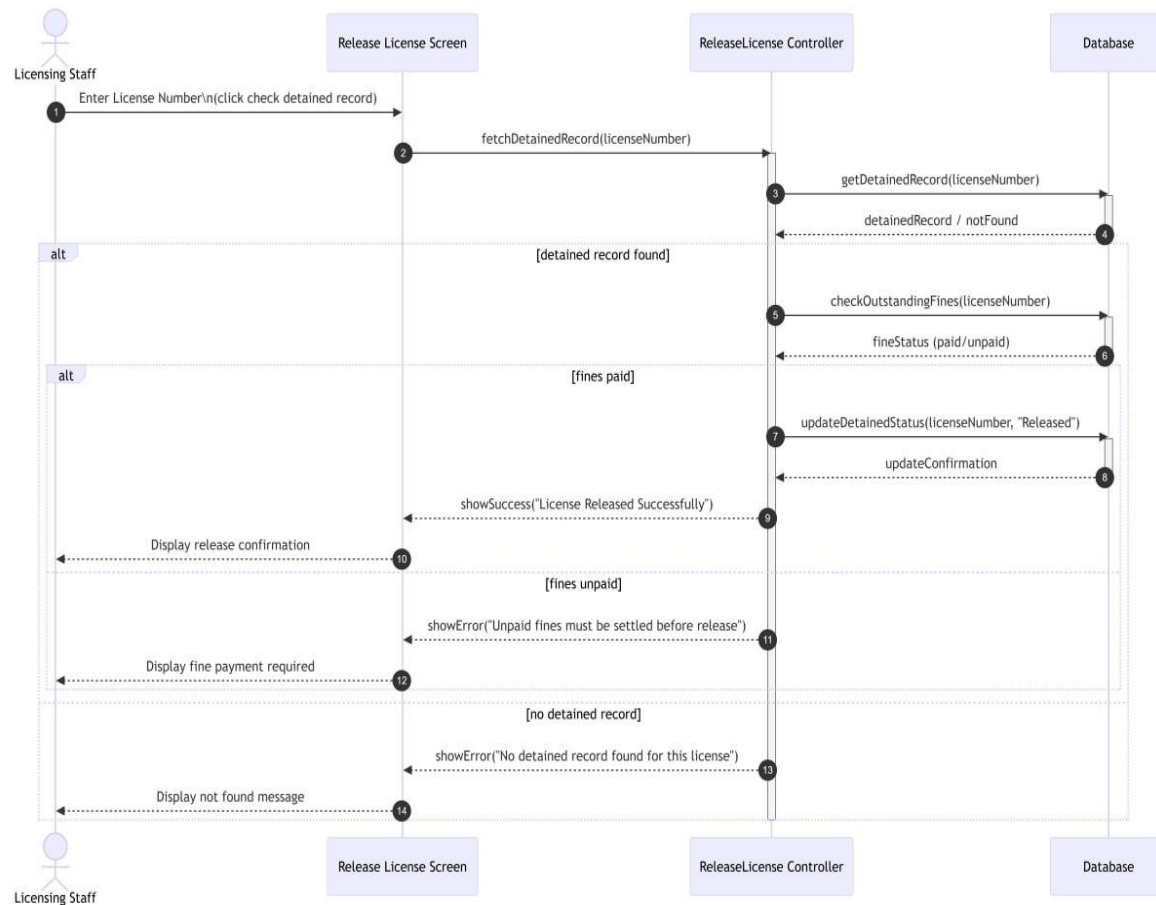


Figure 13: Sequence diagram of Release Detained License

3.6.2. Activity diagram

The Activity Diagram represents the flow of activities and decision points within the DVLD system, offering a structured view of how different tasks are carried out from start to finish. It emphasizes the step-by-step execution of processes, including validations, approvals, test outcomes, and alternative paths such as rejections or retakes, ensuring that all possible scenarios are accounted for. By mapping these actions and decisions, the diagram provides a clear understanding of the overall process logic and operational flow, helping stakeholders visualize dependencies, identify potential inefficiencies, and ensure that the system operates in a consistent and reliable manner.

Activity 1: User Registration & Account Creation

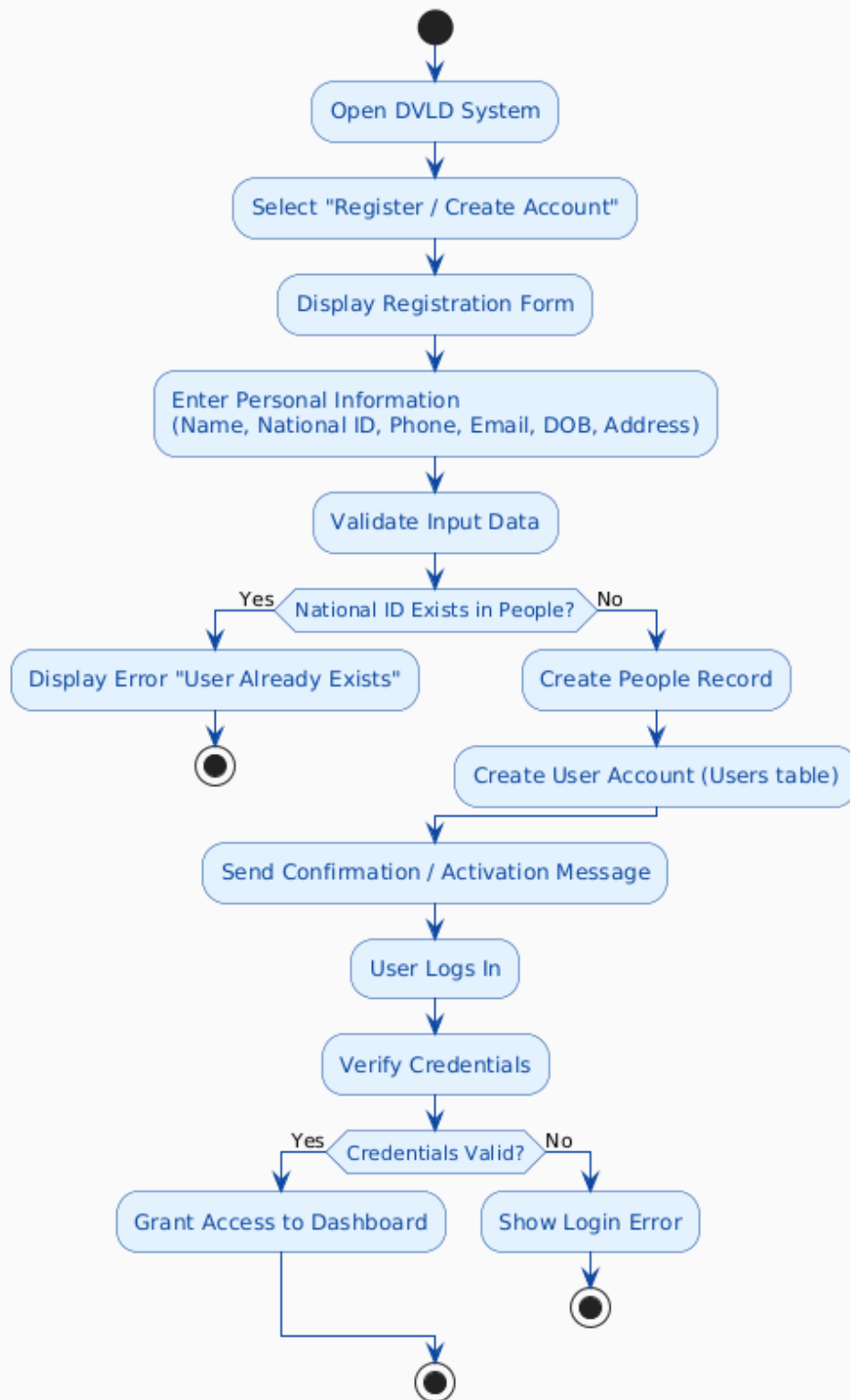


Figure 14: Account Registration and Account Creation

Activity 2: User Profile Completion & Document Upload

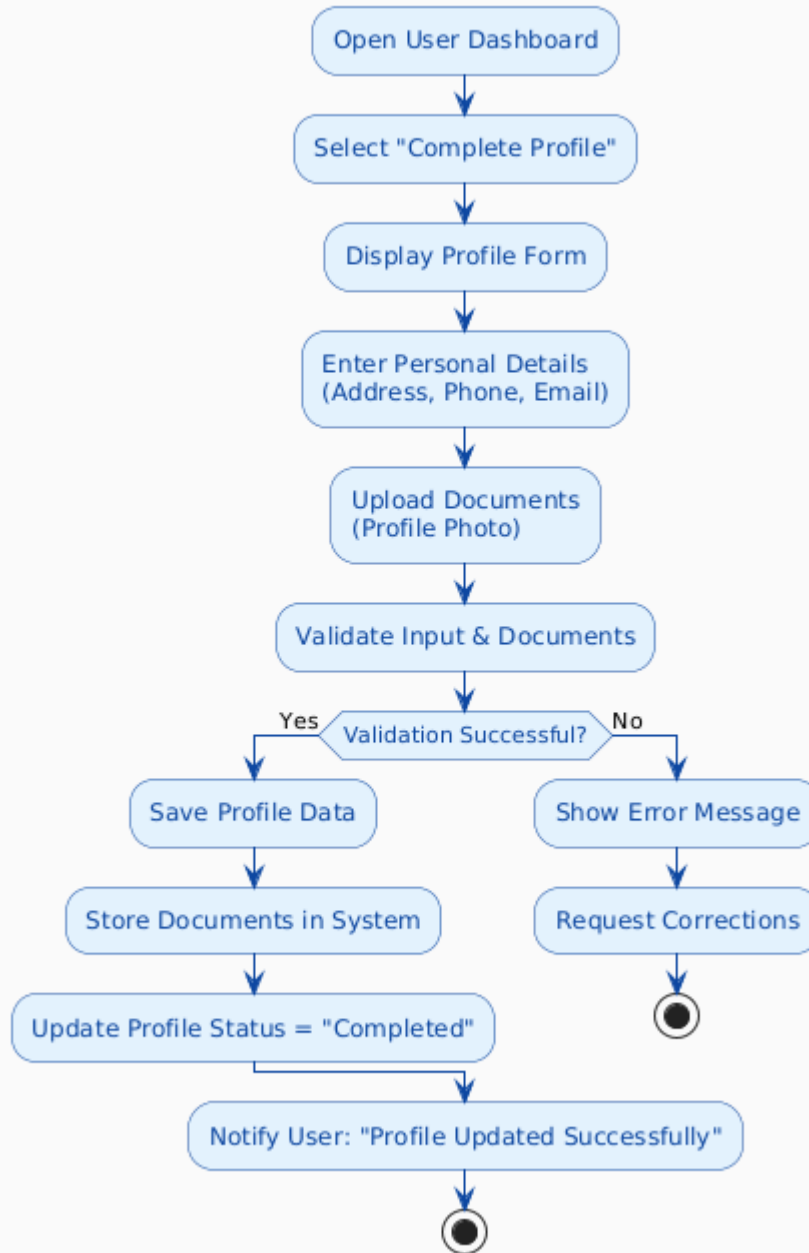


Figure 15: Activity diagram of Profile Compilation

Activity 3: Apply for New Local Driving License

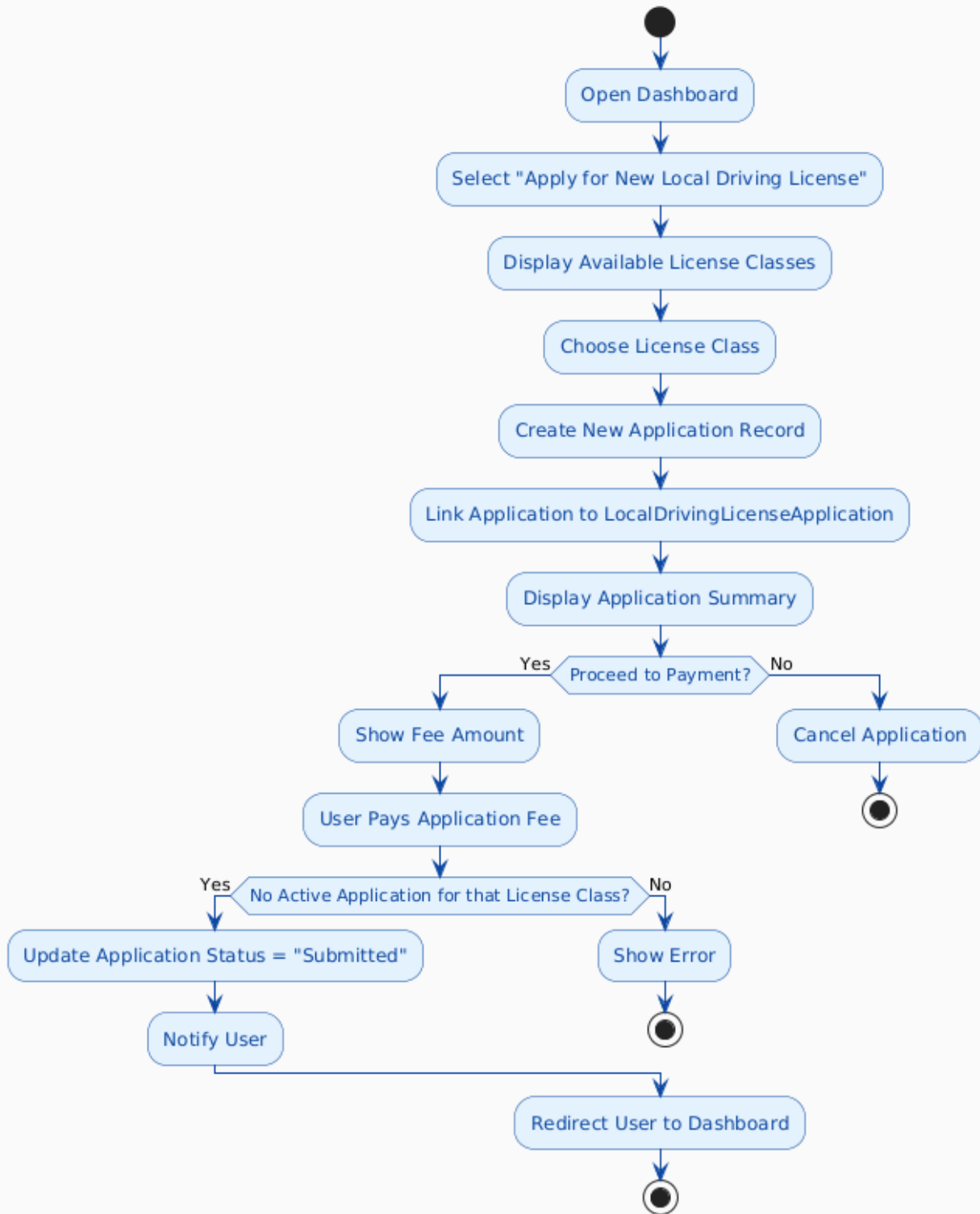


Figure 16: Activity diagram of Apply for New Driving License

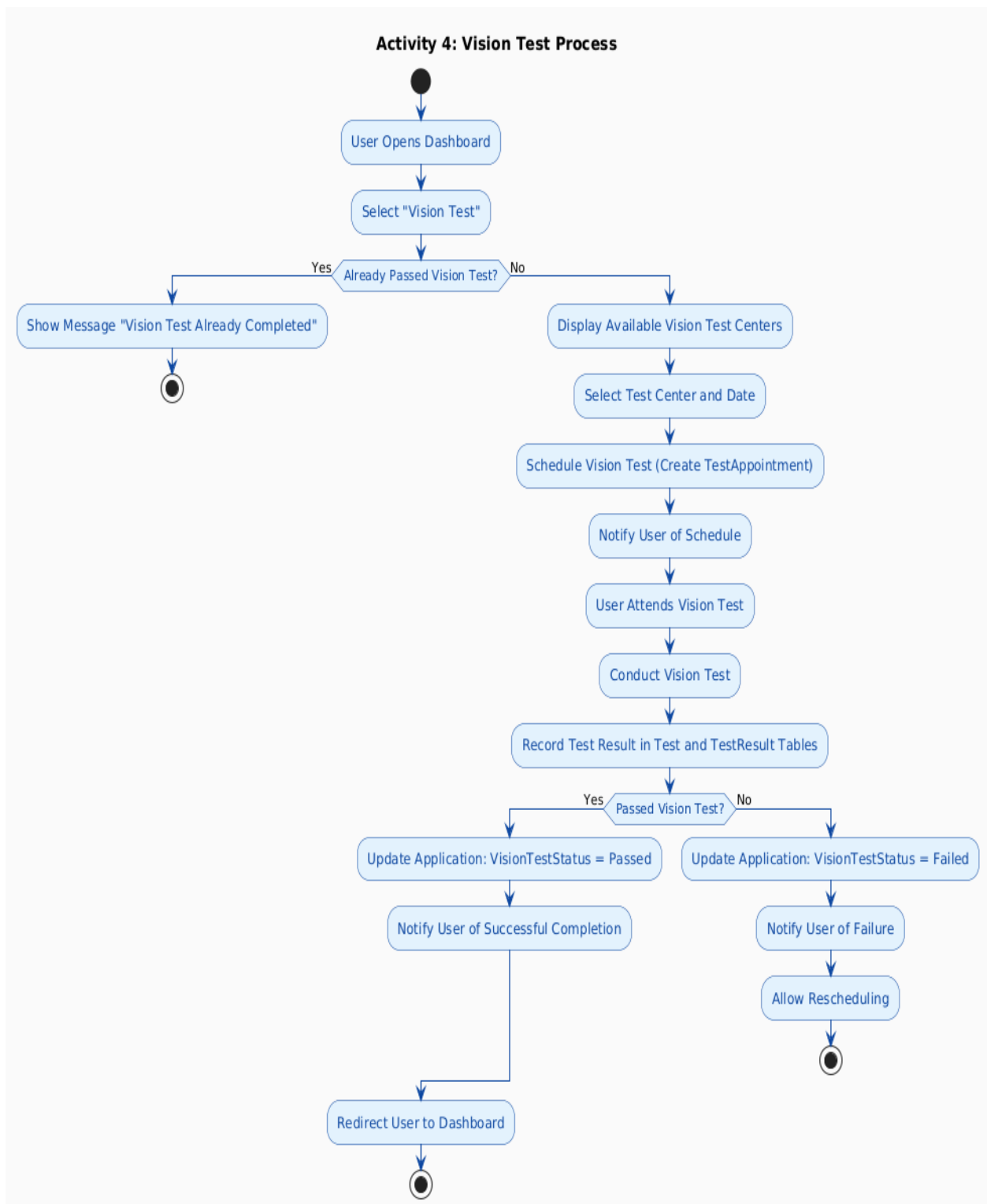


Figure 17: Activity diagram of Vision Test Process

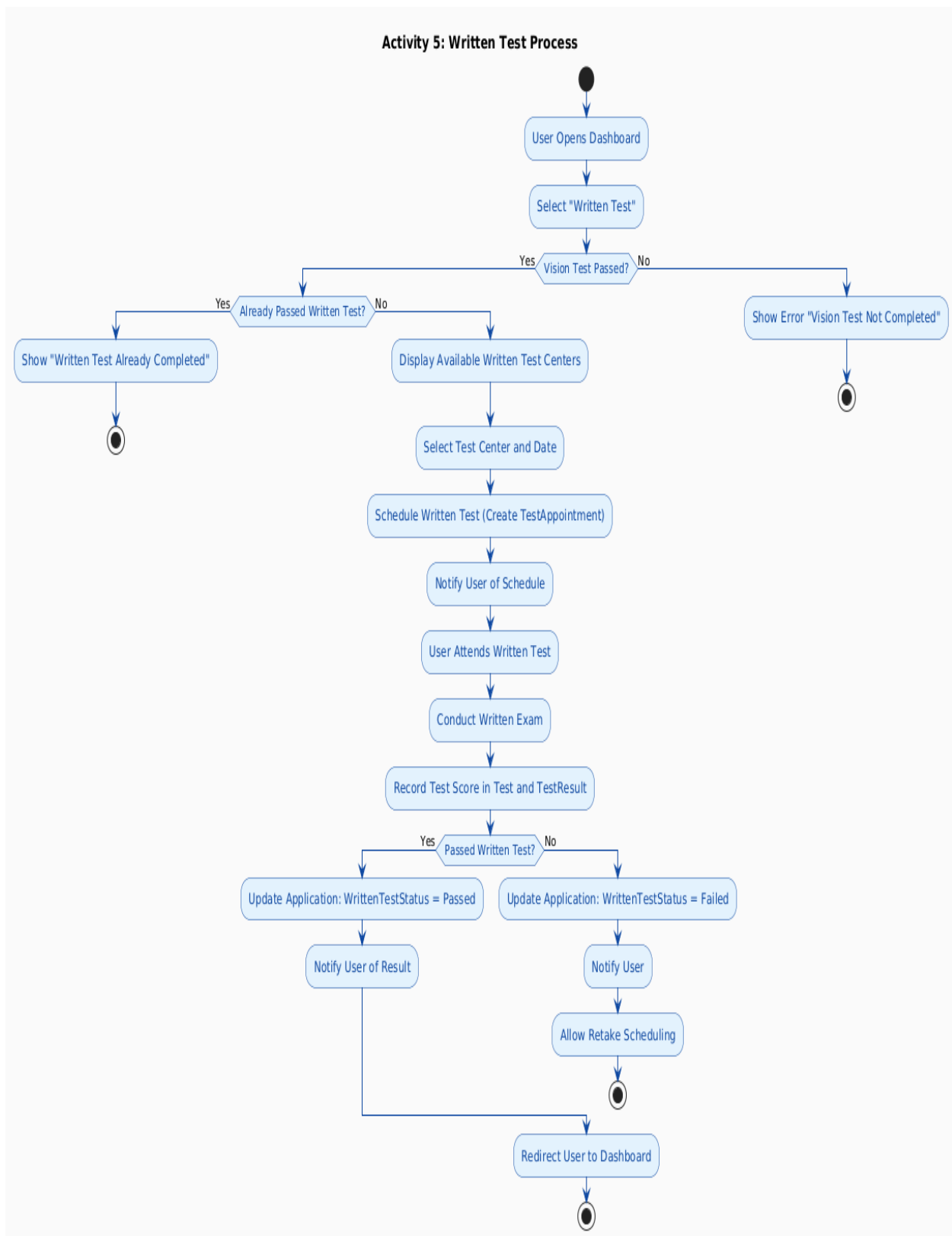


Figure 18: Activity diagram of Written Test Process

Activity 6: Street Test Process

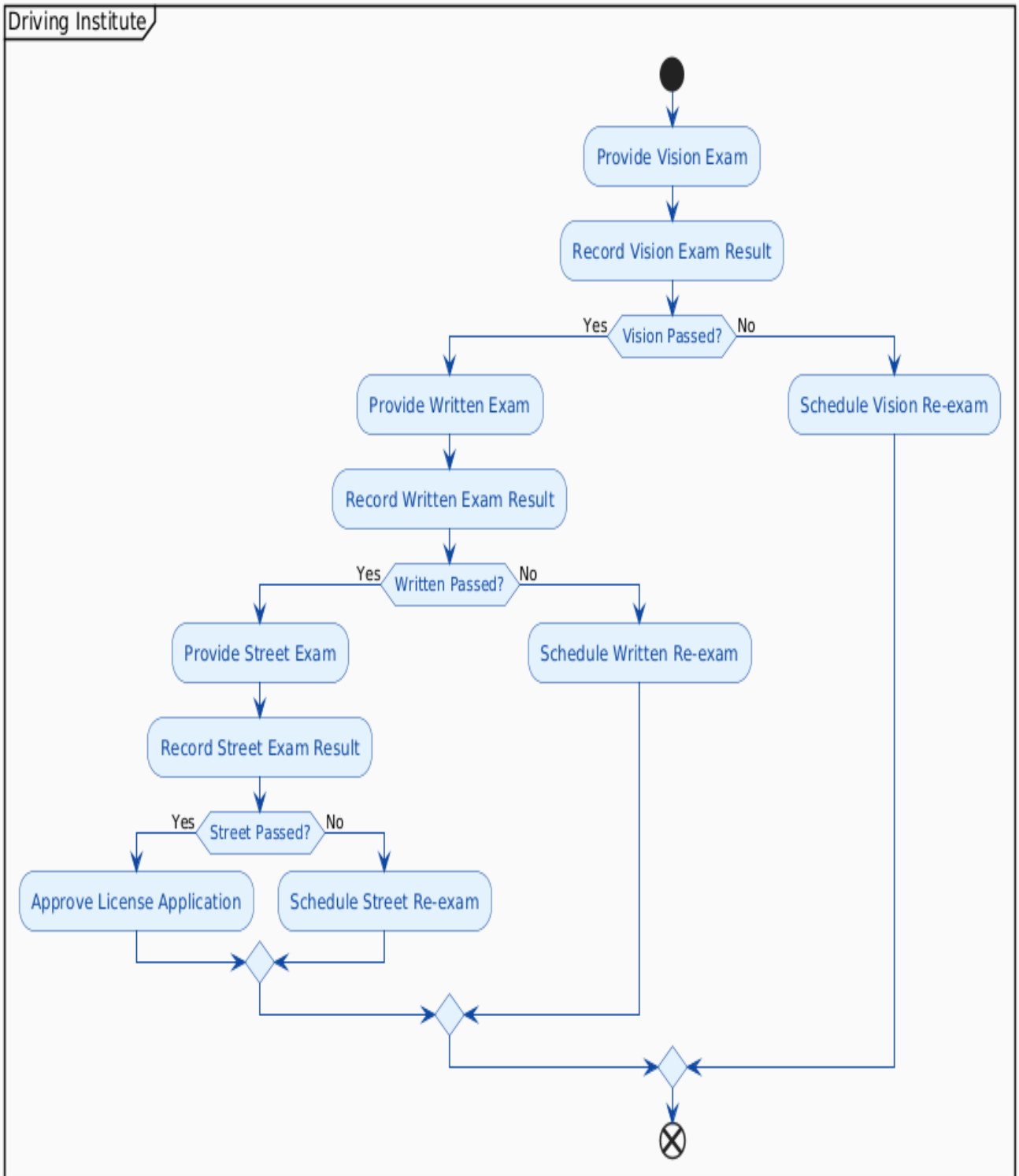


Figure 19: Activity diagram of Street Test Process

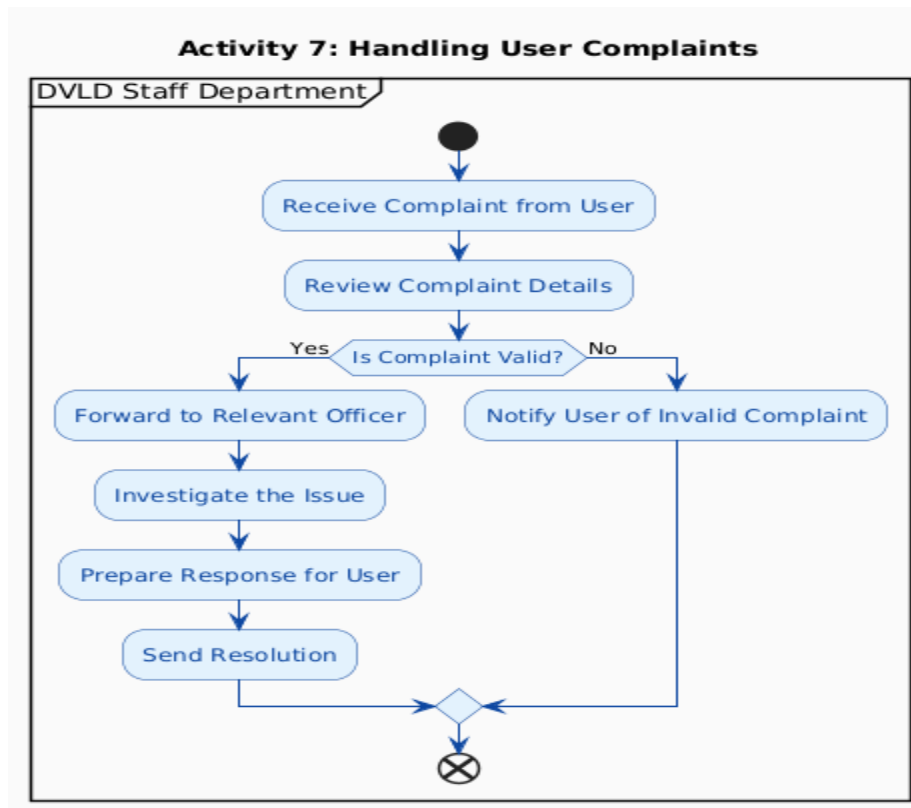


Figure 20: Activity diagram of Handling User Complaints

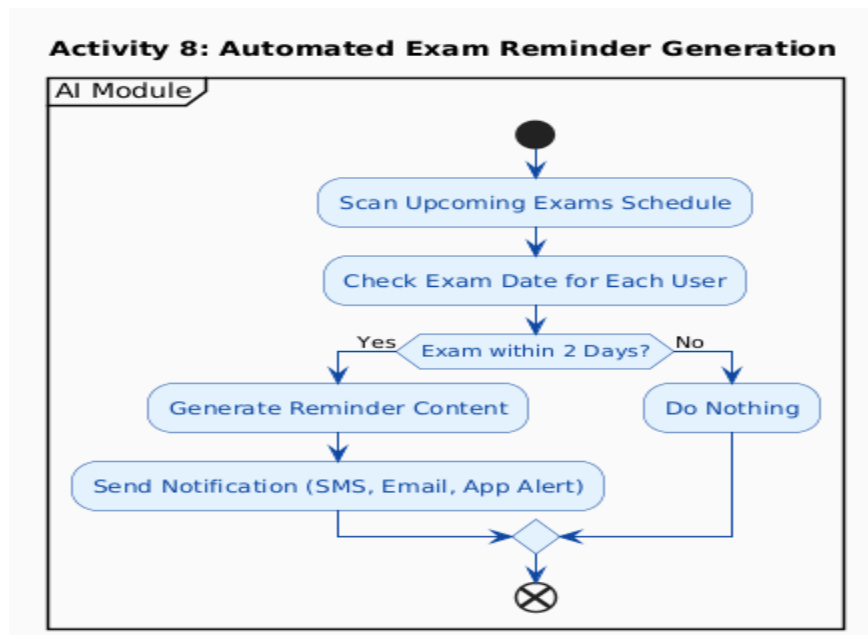


Figure 21: Activity diagram of Automated exam reminder generation

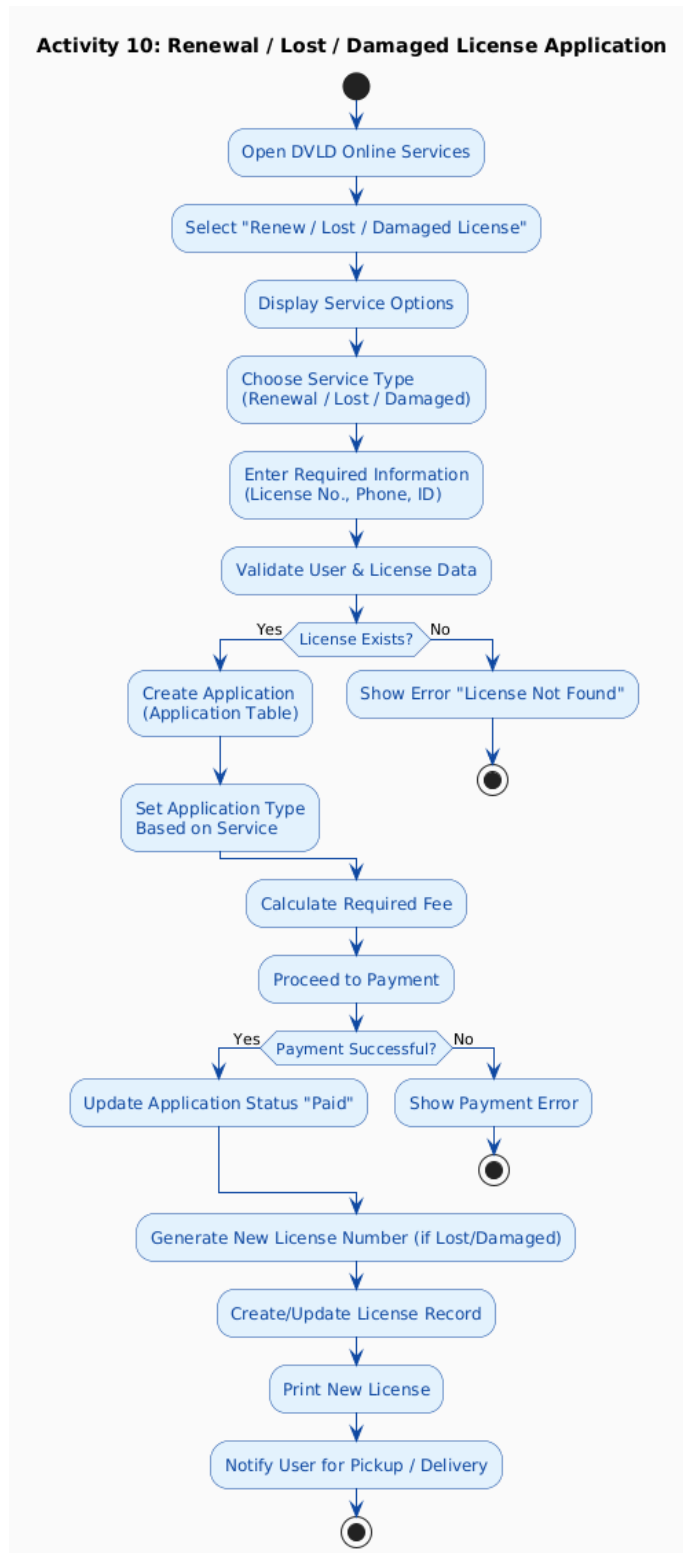


Figure 22: Activity diagram of Renewal/ lost/ damaged licence application

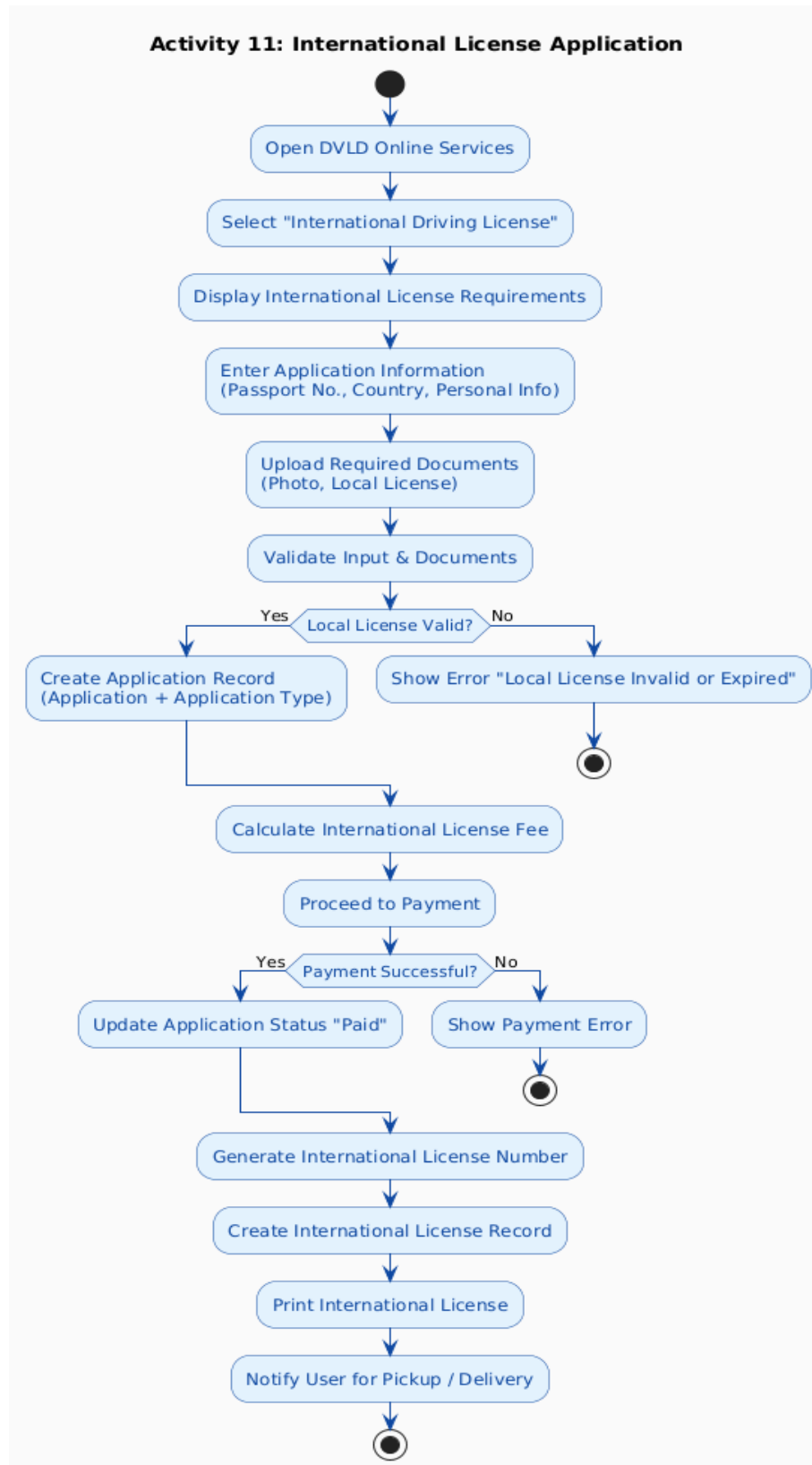


Figure 23: Activity diagram of International license application

Activity 12: License Detainment & Release Process

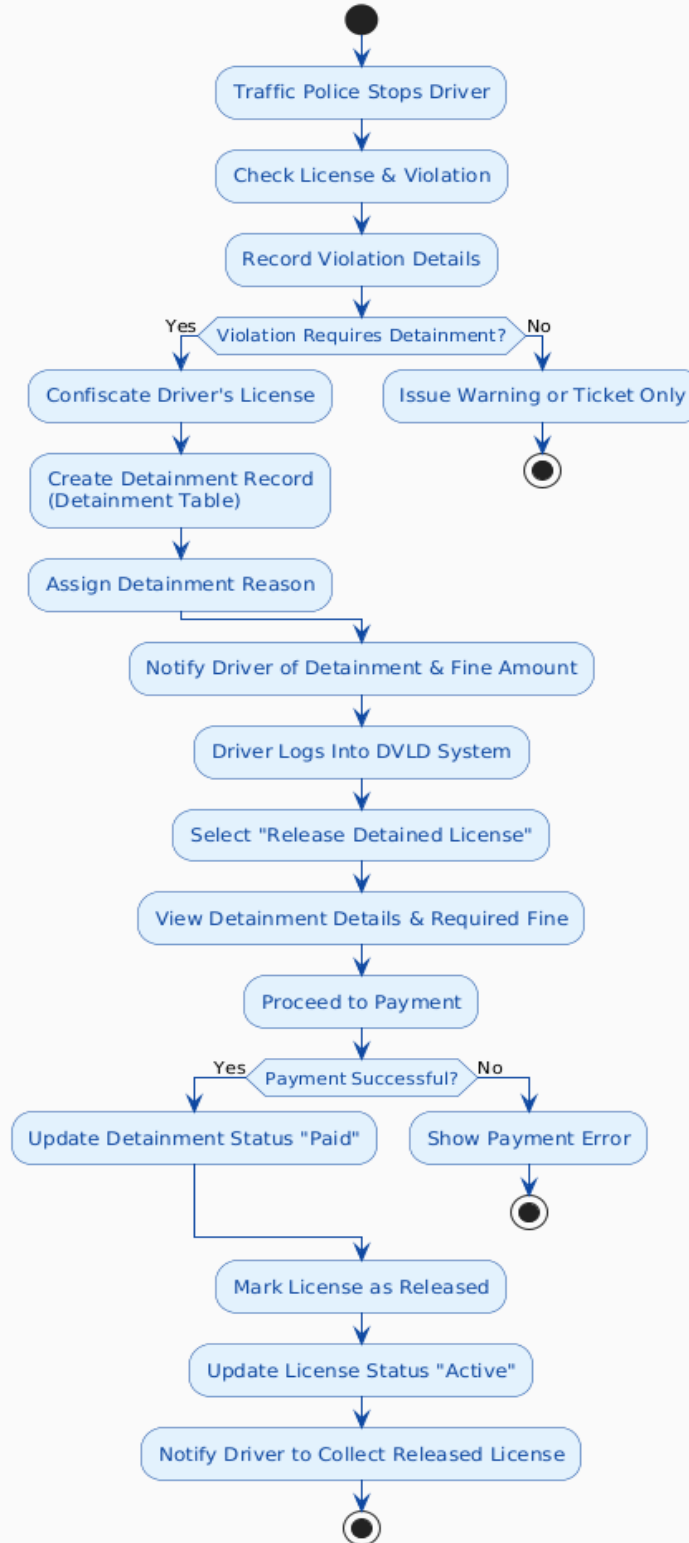


Figure 24: Activity diagram of License Detainment and Release Process

3.6.3. State chart diagram

The State Chart Diagram represents the overall lifecycle of a user's driving-license-related activities within the DVLD System. It illustrates how an applicant moves from one system state to another based on actions such as applying for a license, completing payments, taking exams, or renewing an existing license. The diagram focuses on system behavior rather than user actions and shows how the system responds at every stage. The process begins at the Initial State, where no application exists. When the user initiates a new driving license request, renewal, lost/damaged replacement, or international license application, the system transitions to the Application Created state. From here, the system requires the applicant to upload mandatory documents, moving the process to the Document Submission state. If the documents are incomplete or invalid, the system returns the application to the submission state until all requirements are satisfied.

Once documents are accepted, the system enters the Pending Verification state, where DVLD staff review and approve the documents. If rejected, the system transitions back to Document Submission for corrections. Upon approval, the system advances to the Payment Pending state, where applicants must complete the required fees. A successful payment moves the application to the next phase, while failed payments return the system to the Payment Pending state.

For New License Applications, the system progresses through three exam-related states:

- Vision Test Pending
- Written Test Pending
- Street Test Pending

Each test may lead to a Passed or Failed state. A fail returns the system to scheduling for a retake, whereas passing all three tests transitions the system to All Tests Completed. Renewal, lost/damaged, and international applications bypass the exam stages.

Overall, the State Chart Diagram provides a comprehensive view of the driving license application lifecycle, demonstrating state transitions, approval loops, verification processes, and final license issuance. It ensures clarity in how the DVLD system manages different license services and maintains consistent, rule-based progression through all stages.

State Chart: User Registration Process

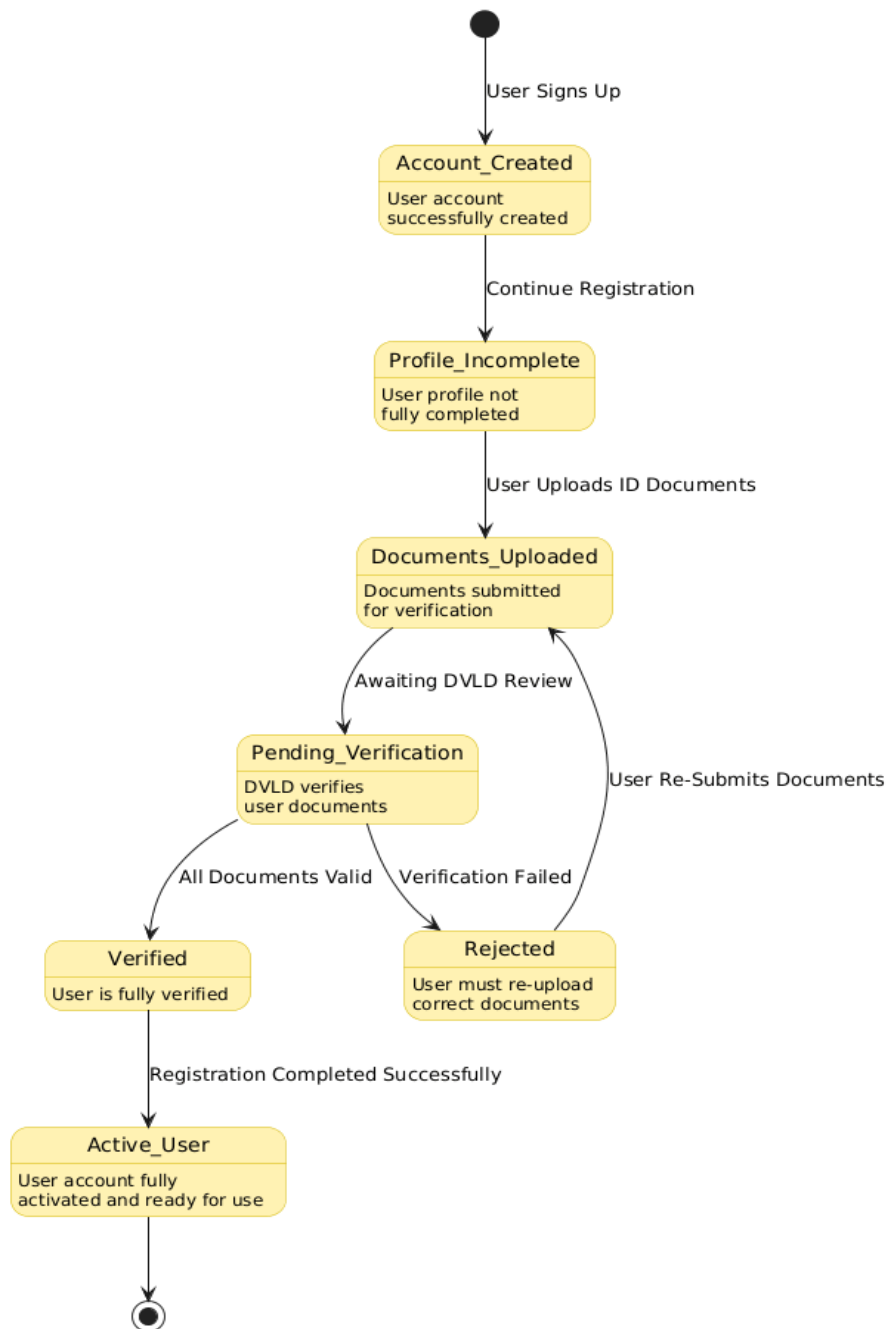


Figure 25: State chart diagram of user registration process

State Chart: Profile Completion & Document Upload

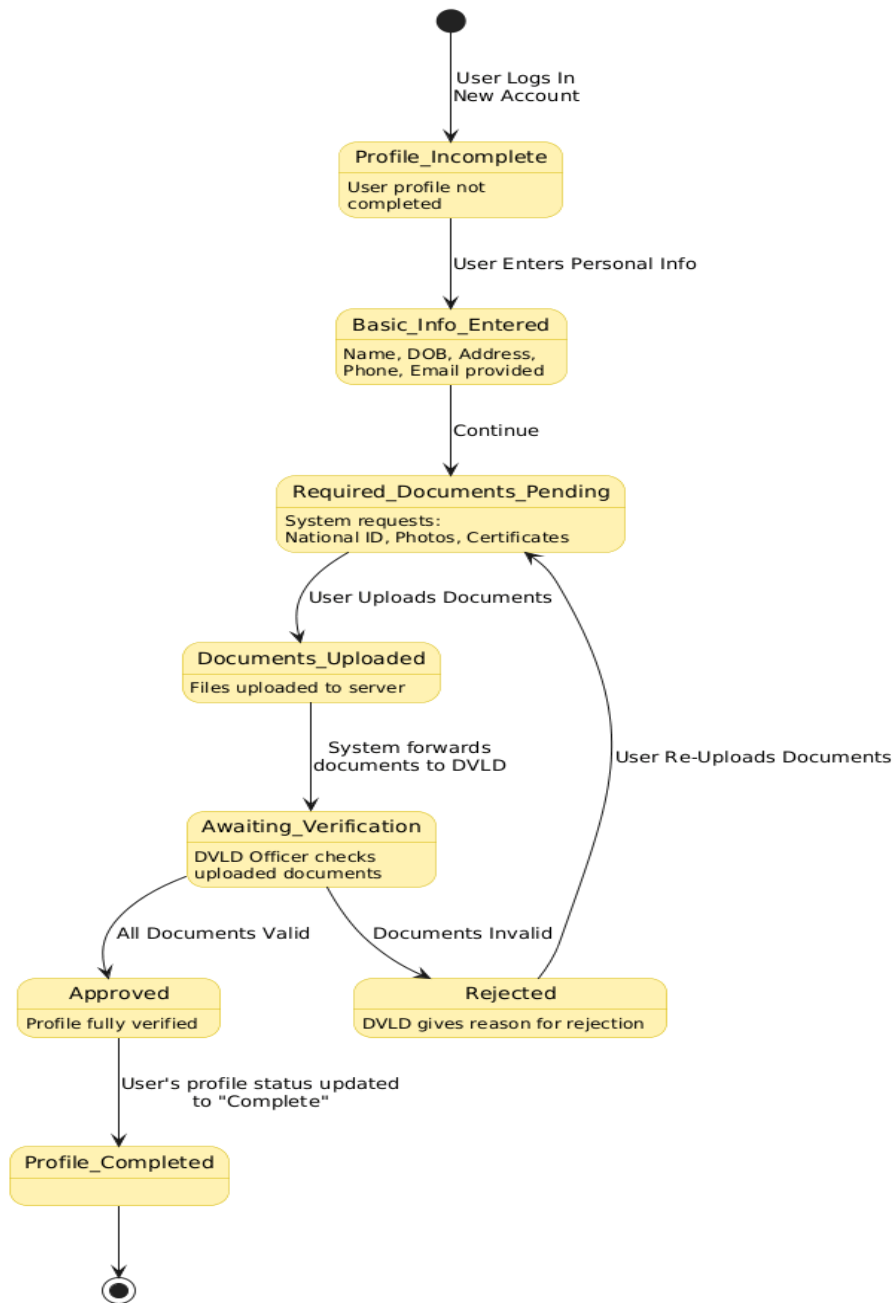


Figure 26: State chart diagram of profile completion and document upload

State Chart: Driving License Application (New Registration)

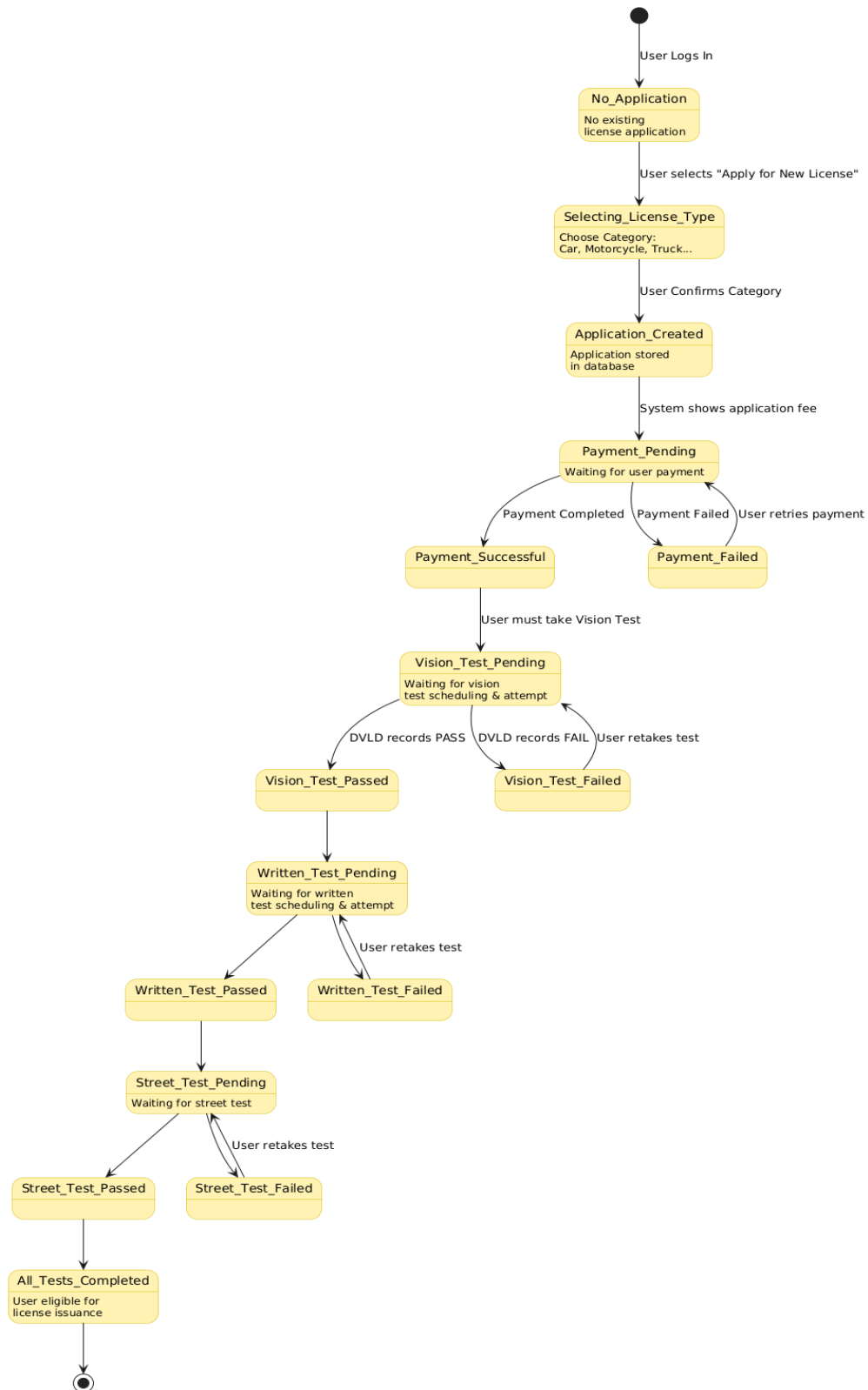


Figure 27: State chart diagram of driving license application

State Chart: Exam Scheduling & Completion

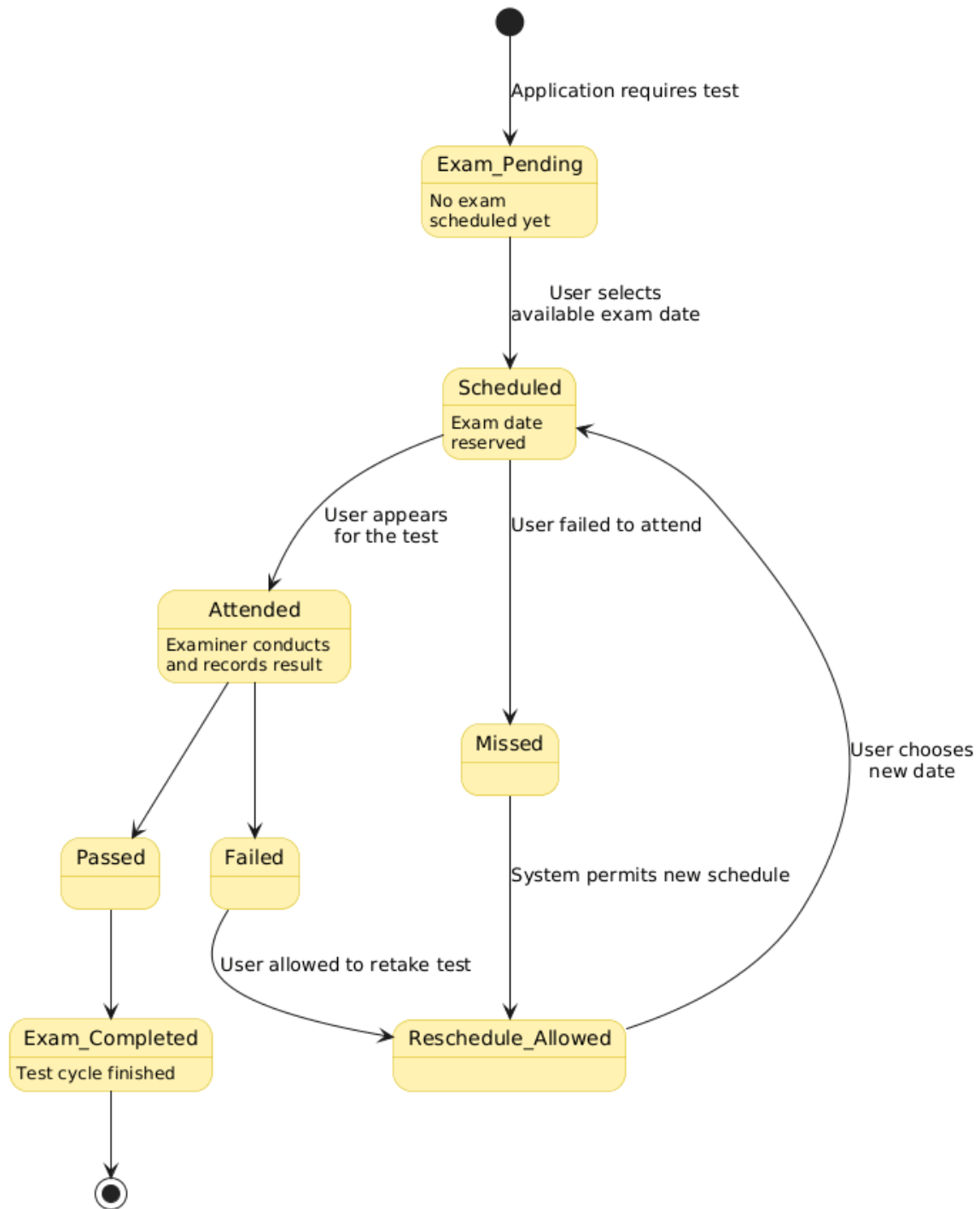


Figure 28: State chart diagram of exam scheduling and completion

State Chart: International License Application

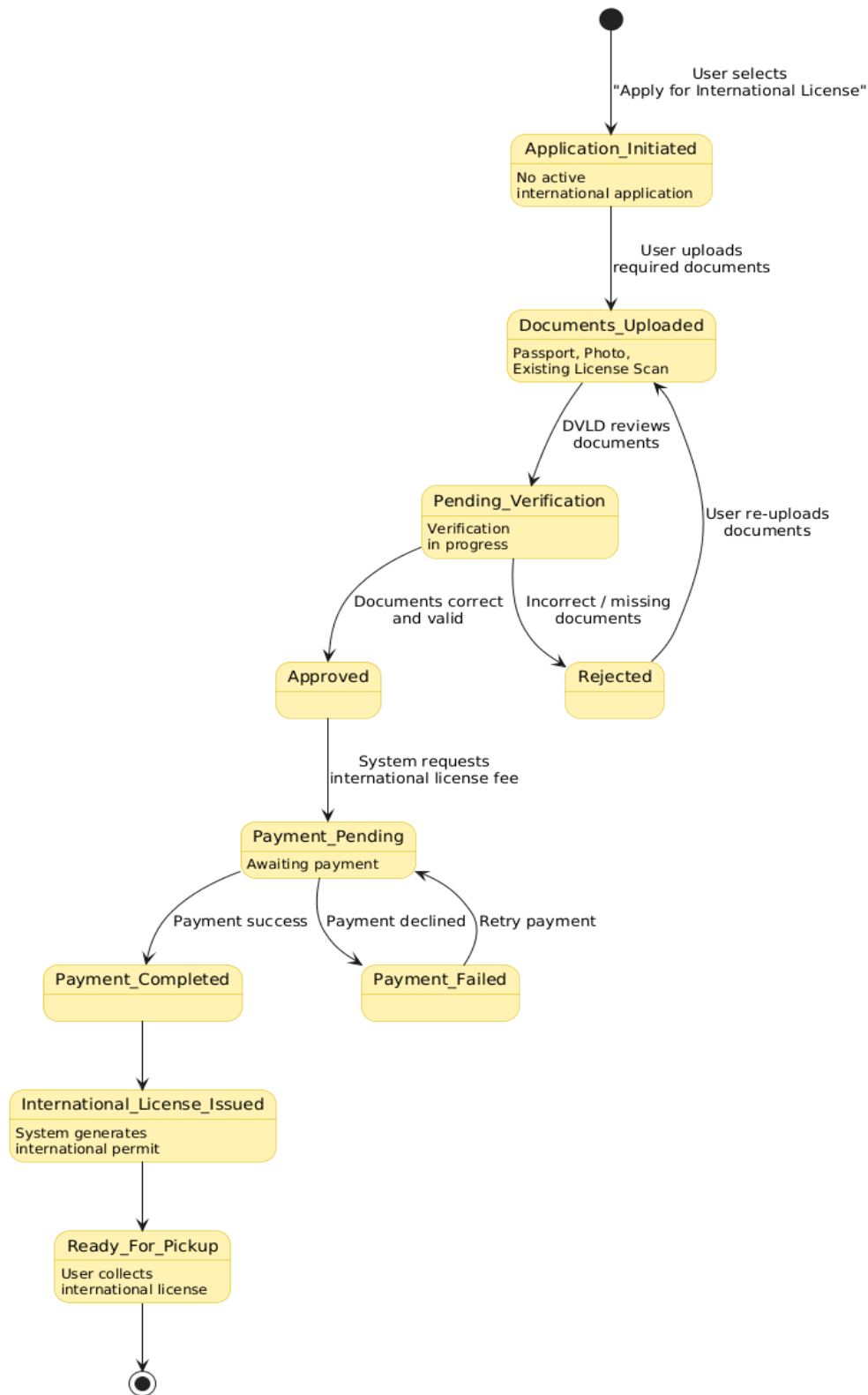


Figure 29: State chart diagram of international license application

State Chart: Renewal / Lost / Damaged License Application

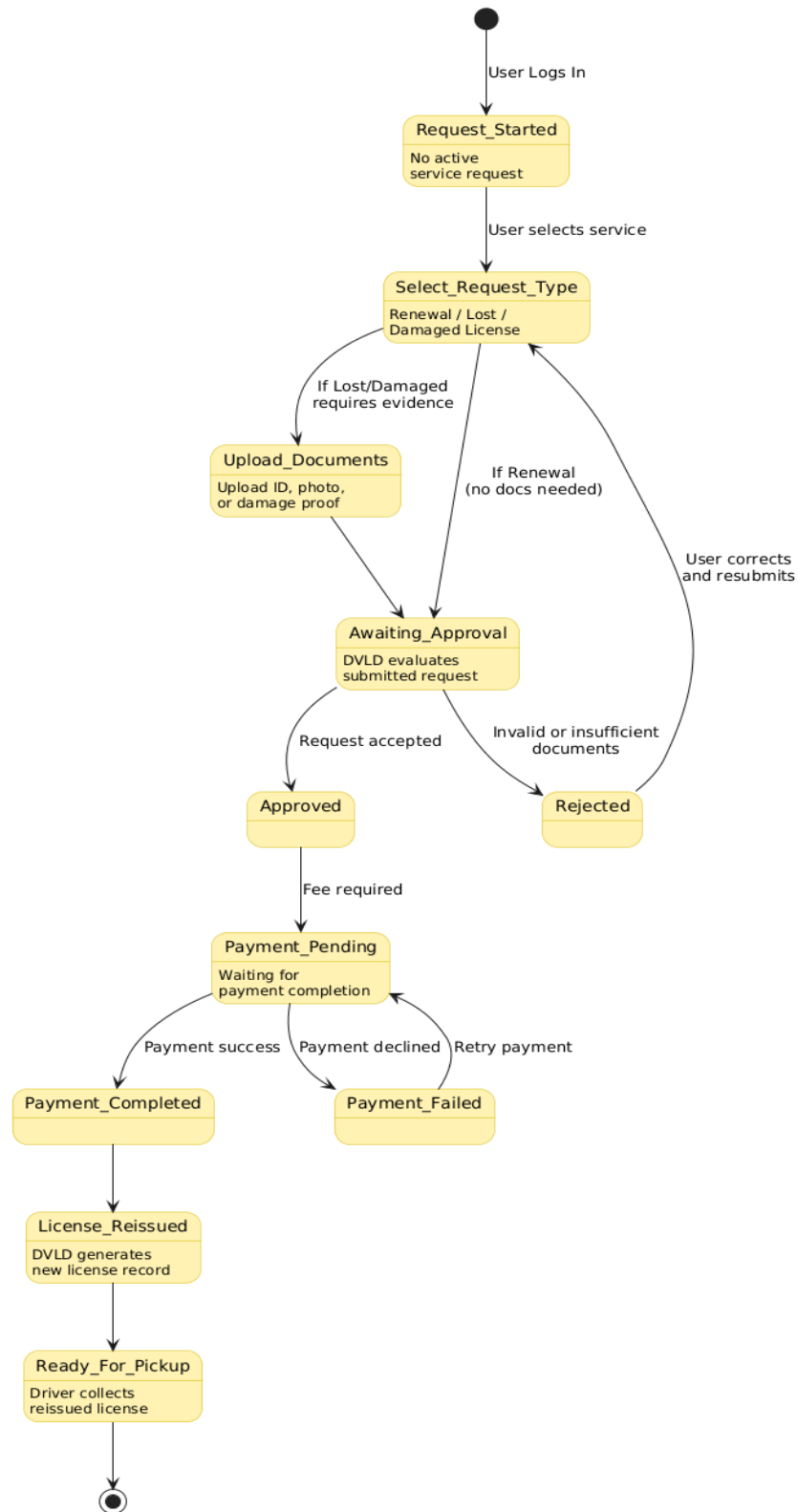


Figure 30: State chart diagram of Renewal/ lost/ damaged license application

State Chart: License Detainment & Release

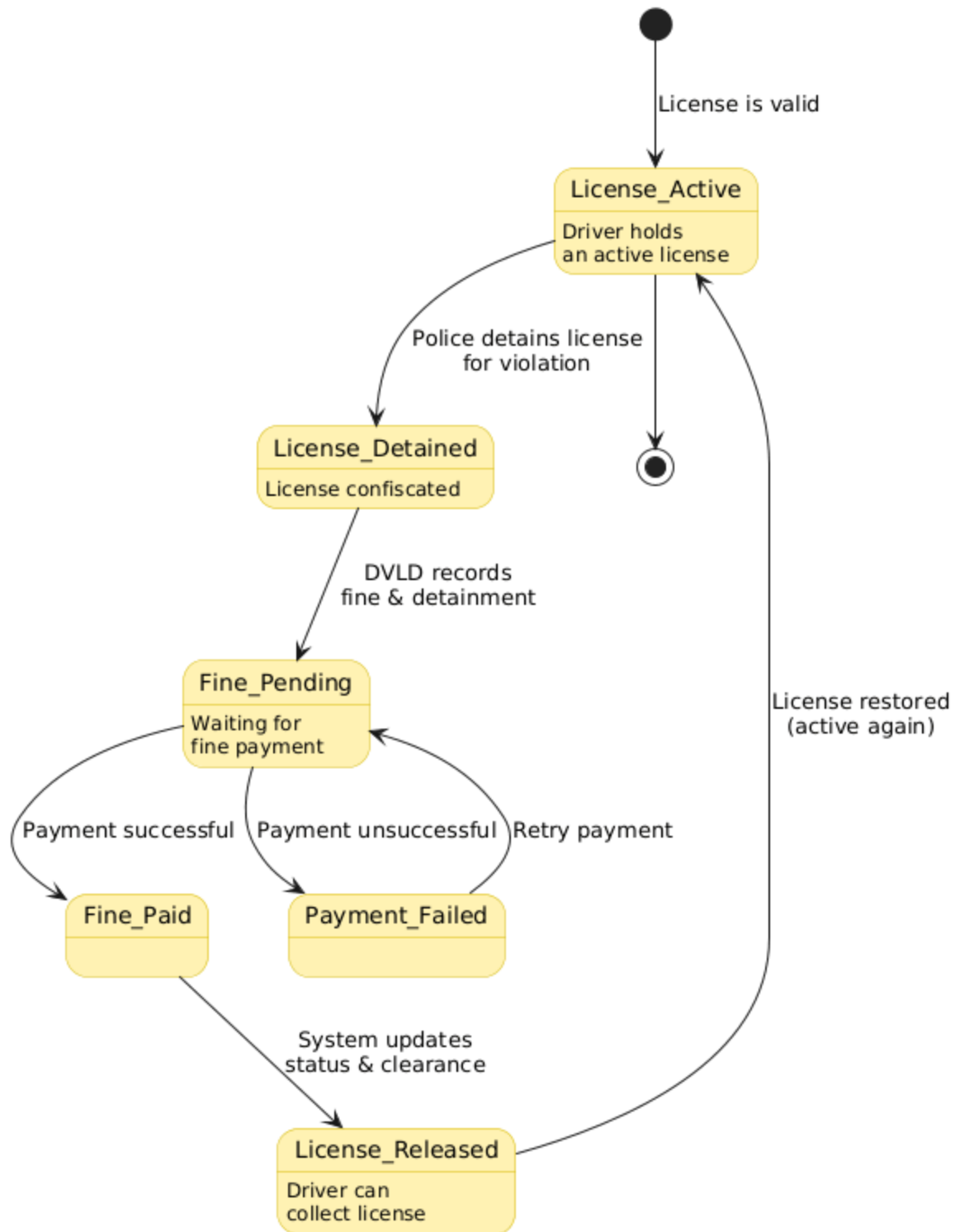


Figure 31: State chart diagram of License detainment and release process

Chapter 4: System design

4.1. Overview

This section provides an overview of the system design for the proposed Driving License Management System (DVLD). The system is designed to support the complete lifecycle of driving license management, including application submission, mandatory training, testing, license issuance, renewal, and administrative control.

The DVLD system adopts a layered and modular design, consisting of a presentation layer, business logic layer, and data layer. The presentation layer includes a single desktop application used by DVLD staff and driving institute personnel, as well as a mobile application used by applicants. Access to system functionalities within the desktop application is controlled using role-based access control (RBAC), ensuring that each user interacts only with the features relevant to their role.

Multiple stakeholders are supported within the same system environment, including system administrators, registration officers, test officers, license issuers, institute managers, and institute trainers. Although these users share the same desktop application, the user interface and available menus are dynamically adjusted based on the authenticated user's role. This approach avoids duplication of applications, simplifies deployment, and improves system maintainability.

The system also integrates driving institutes as part of the license issuance workflow. Training is modeled as a mandatory stage within the new driving license application process, rather than as a separate application type. Applicants select an institute and training batch using the mobile application, while institute staff manage batches, training progress, and attendance through the desktop application. Completion of training is a prerequisite for enabling driving test scheduling by DVLD officers.

External systems, such as payment services and notification mechanisms, may interact with the DVLD system. However, training fees are handled externally by driving institutes and are not processed as part of DVLD financial transactions.

Overall, the system design emphasizes security, scalability, role separation, and real-world alignment, providing a clear and practical blueprint for implementation while supporting efficient coordination between DVLD authorities, driving institutes, and applicants.

4.2. Purpose of the System Design

The purpose of the system design for the Driving License Management System (DVLD) is to translate the identified functional and non-functional requirements into a structured, implementable, and maintainable solution. This design serves as a blueprint that guides system development, integration, testing, and future enhancement.

The system design aims to ensure that all functional requirements such as license application processing, mandatory training management, test scheduling, license issuance, and renewal are supported in a clear and logical workflow. By integrating driving institute training as a mandatory stage within the new driving license application process, the design ensures real-world accuracy and enforces regulatory compliance before applicants proceed to driving tests.

From a non-functional perspective, the design focuses on security, scalability, maintainability, and usability. The use of role-based access control (RBAC) ensures that users can access only the system functions relevant to their responsibilities, thereby protecting sensitive data and reducing the risk of unauthorized actions. A single desktop application with dynamic role-based interfaces minimizes system complexity and simplifies deployment and maintenance.

The system design also supports the system model and object-oriented model by clearly defining system components, subsystems, classes, and their interactions. Business logic is separated from the presentation to promote modularity and reduce coupling, making the system easier to test, debug, and extend. New features, such as training management, can be added without affecting core licensing and testing logic.

Finally, the system design facilitates effective implementation and testing by providing well-defined interfaces, workflows, and data structures. This structured approach ensures that each subsystem can be developed and verified independently while maintaining seamless integration across the entire DVLD system.

4.3. Design Goals

The design of the Driving License Management System (DVLD) is guided by a set of well-defined design goals and object-oriented principles to ensure that the system is robust, scalable, secure, and easy to maintain. These goals influence architectural decisions, subsystem decomposition, and implementation strategies.

Modularity is a primary design goal of the DVLD system. The system is divided into distinct subsystems such as user management, license applications, training management, testing, and license issuance. Each subsystem performs a specific function, allowing independent development, testing, and maintenance without affecting other parts of the system.

High cohesion and low coupling are emphasized throughout the design. Components within a subsystem are closely related and focused on a single responsibility, while dependencies between subsystems are minimized. For example, training management logic is isolated from licensing and testing logic, ensuring that changes in training processes do not impact core DVLD operations.

Abstraction and encapsulation are applied to hide internal implementation details and expose only necessary interfaces. Business rules, validation logic, and data access operations are encapsulated within the business layer, preventing direct access from the presentation layer. This improves system security and reduces complexity.

The system design also follows object-oriented principles such as inheritance and polymorphism where appropriate. Common attributes and behaviors are abstracted into base classes, enabling code reuse and extensibility. Polymorphism allows the system to handle different user roles and processes using unified interfaces while executing role-specific behavior.

Reusability and scalability are key design goals. Avoiding duplicated applications and adopting a single role-based desktop application allows shared components to be reused across multiple user roles. The architecture supports future expansion, such as adding new roles, services, or modules, with minimal changes to existing components.

Finally, security and usability are central to the system design. Role-based access control ensures secure operation, while dynamic user interfaces tailored to each role improve usability and reduce

user errors. Together, these design goals ensure that the DVLD system is efficient, reliable, and aligned with real-world operational requirements.

4.4. Proposed System Architecture

The proposed architecture of the Driving License Management System (DVLD) is based on a layered architectural style, which separates the system into clearly defined layers to improve maintainability, scalability, and security. This architecture ensures that responsibilities are well organized and that changes in one layer have minimal impact on others.

The presentation layer is responsible for user interaction and includes two main interfaces: a desktop application and a mobile application. The desktop application is used by DVLD staff and driving institute personnel and operates as a single executable with role-based access control (RBAC). Based on the authenticated user's role, the system dynamically loads the appropriate menus and screens. The mobile application is designed for applicants and allows them to apply for licenses, select driving institutes and training batches, and track training and application status.

The business logic layer contains the core system functionality and enforces all business rules. It handles processes such as license application management, training coordination, test scheduling, result evaluation, and license issuance. Training-related services are implemented as separate business components and are integrated into the new driving license application workflow without interfering with licensing or testing logic. This layer acts as an intermediary between the presentation and data layers, ensuring consistency and validation of system operations.

The data access layer manages persistent data storage and retrieval. It provides controlled access to the database and ensures data integrity through well-defined data models and relationships. Core entities such as applicants, applications, licenses, tests, training batches, and attendance records are stored in the database, with training data logically linked to license applications.

The DVLD system architecture also supports interaction with external services, such as payment gateways and notification systems. While DVLD-related payments are processed within the system, training fees are managed externally by driving institutes and are not handled as part of DVLD financial transactions.

Overall, this layered architecture promotes separation of concerns, enhances system security, supports role-based operations, and provides a flexible foundation for future enhancements to the DVLD system.

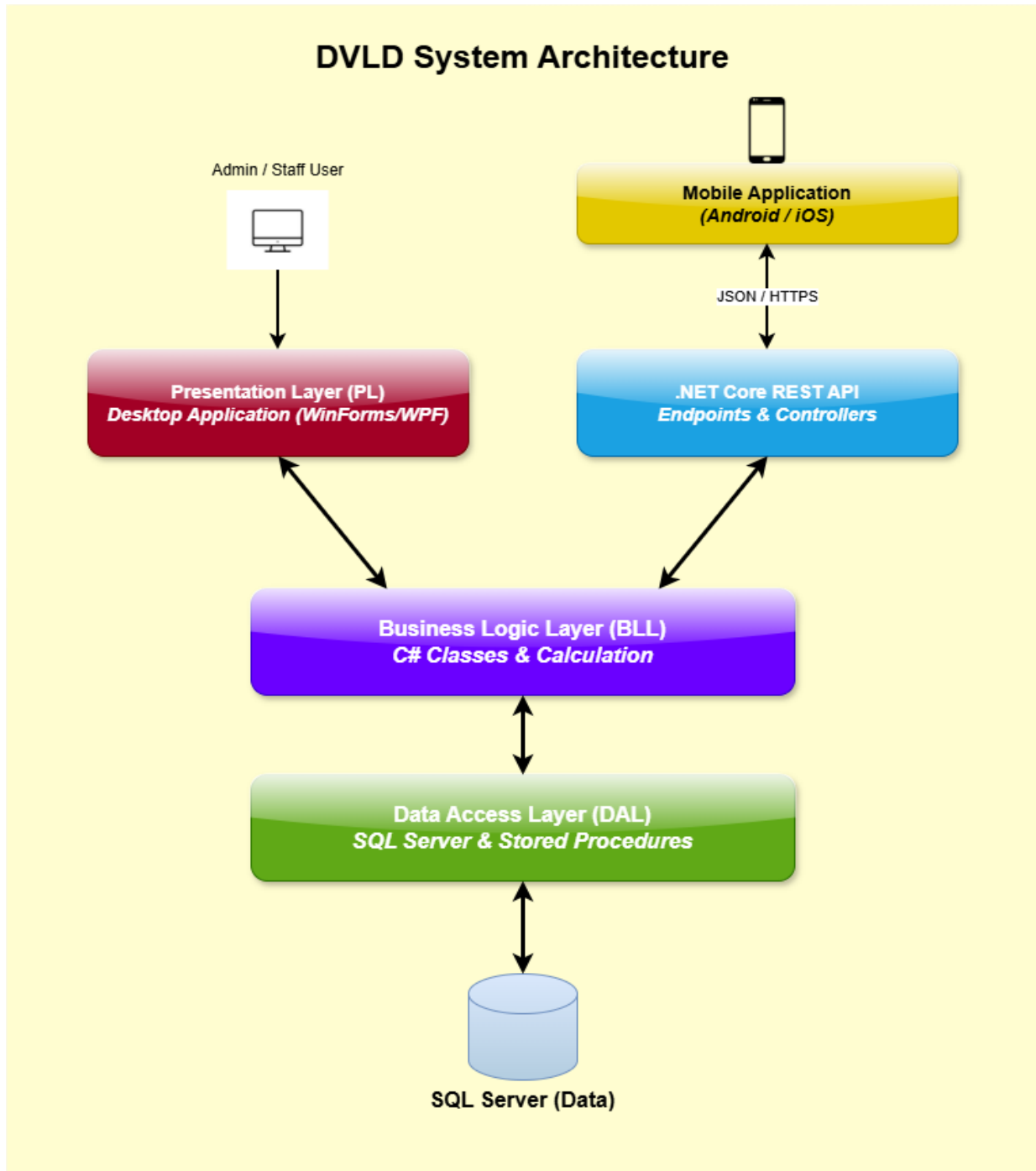


Figure 32: System Process Diagram

4.5. Subsystem Decomposition

This section describes the decomposition of the Driving License Management System (DVLD) into well-defined subsystems based on functional responsibilities. The objective of this decomposition is to group related system components and modules into logical subsystems that simplify system design, development, testing, and maintenance.

The DVLD system is decomposed using a functional and hierarchical approach, where each subsystem represents a major operational domain of the system. Each subsystem is designed to be highly cohesive, performing a specific set of related functions, while maintaining low coupling with other subsystems through clearly defined interfaces.

Based on the proposed system architecture and functional requirements, the DVLD system is decomposed into the following primary subsystems:

1. **User and Access Management Subsystem**

This subsystem groups all components related to user authentication, authorization, and role-based access control. It manages system users, roles, and permissions for DVLD staff and driving institute personnel, ensuring secure and controlled access to system functionalities.

2. **License Application Management Subsystem**

This subsystem is responsible for creating, processing, and tracking driving license applications. It manages application types such as new driving license applications, renewal requests, replacement requests for lost or damaged licenses, and international license applications. It acts as the central workflow coordinator and interacts with training, testing, and license-related subsystems.

3. **Training Management Subsystem**

This subsystem groups components related to driving institute operations, including institute management, training batch creation, applicant enrollment, attendance tracking, and training completion verification. Training is modeled as a mandatory stage for new driving license applications and must be completed before applicants can proceed to testing.

4. Testing Management Subsystem

This subsystem manages all driving test-related activities, including test scheduling, test execution, result recording, and eligibility verification. It also supports test retake management, allowing applicants who fail a test to reschedule and retake the required tests according to DVLD regulations.

5. License Issuance Subsystem

This subsystem includes components responsible for issuing new driving licenses and international driving licenses after applicants successfully complete all required stages. It generates license records and maintains license issuance history.

6. License Renewal and Replacement Subsystem

This subsystem manages license renewal processes and handles replacement of lost or damaged licenses. It verifies eligibility, validates required conditions, and updates license status without repeating unnecessary application stages.

7. Payment and Notification Subsystem

This subsystem groups components responsible for managing DVLD-related payment records associated with system services such as license applications, test fees, renewals, and license replacements. Payments are recorded and maintained internally within the system database, where each payment is directly linked to the corresponding process or entity that requires payment. The subsystem does not integrate with external banking or payment gateway systems in the current implementation.

4.6. Subsystem Description

This section describes the major subsystems of the Driving and Vehicle License Department (DVLD) Management System. Each subsystem is presented with its function, interfaces, interactions, dependencies, and main components. Together, these subsystems ensure efficient management of applicants, driving schools, examinations, and license services.

1. User and Identity Management Subsystem

Function:

Manages user registration, authentication, authorization, and role-based access for applicants, DVLD staff, and driving school personnel.

Interface:

- Mobile application interfaces for applicants
- Desktop application interfaces for DVLD staff
- Desktop interfaces for driving schools

Interaction:

Interacts with all other subsystems to verify user identity and permissions before allowing access to system functions.

Dependencies:

Depends on the centralized SQL Server database for storing user credentials, roles, and account status.

Component	Description
Registration Module	Handles applicant and staff registration
Authentication Module	Manages login and session validation
Role-Based Access Control Module	Enforces permissions based on user roles
User Activity Logging Module	Records login attempts and system activities

2. Applicant and Application Management Subsystem

Function:

Handles applicant data, driving license applications, document submissions, and application status tracking.

Interface:

- Mobile app for applicants
- Desktop application for registration officers

Interaction:

Communicates with the Test Management and License Management subsystems to update application progress.

Dependencies:

Relies on the database subsystem for applicant records and document storage.

Component	Description
Applicant Profile Module	Stores personal and identification details
Application Processing Module	Manages submission, review, and approval
Document Management Module	Handles upload and verification of documents
Application Status Tracker	Tracks progress across all stages

3. Driving School and Training Management Subsystem

Function:

Manages driving schools, training programs, schedules, instructors, and attendance records.

Interface:

- Desktop/web interface for driving school administrators
- Read-only access for DVLD administrators

Interaction:

Shares training completion data with the Test Management subsystem and DVLD administration.

Dependencies:

Depends on user and database subsystems for school accounts and training data.

Component	Description
Driving School Management Module	Registers and manages schools
Training Schedule Module	Creates and manages training plans
Attendance Tracking Module	Handles upload and verification of documents
Training Completion Reporting Module	Submits reports to DVLD

4. Test Management Subsystem

Function:

Manages vision, theory, and practical driving tests, including scheduling, execution, results, and retakes.

Interface:

- Desktop application for test officers
- Mobile notifications for applicants

Interaction:

Receives eligibility data from the Application subsystem and sends results to the License Management subsystem.

Dependencies:

Relies on centralized database and user roles for secure operation.

Component	Description
Vision Test Module	Records vision test results
Theory Test Module	Manages online or scheduled theory exams
Practical Test Module	Handles road test scheduling and evaluation
Retake Management Module	Manages failed test retries

5. License Management Subsystem

Function:

Handles issuance, renewal, replacement, detention, and release of driving licenses.

Interface:

Desktop application used by license issuers and DVLD administrators.

Interaction:

Receives verified test results and application approvals before issuing licenses.

Dependencies:

Depends on Test Management and Applicant subsystems for eligibility verification.

Component	Description
License Issuance Module	Issues new driving licenses
License Renewal Module	Processes renewals
Lost/Damaged License Module	Handles replacements
License Detainment Module	Manages detained licenses
License Detainment Module	Restores released licenses

4.7. Persistent Data Management

This section explains how the DVLD Management System stores, manages, and protects persistent data. Reliable data management ensures accuracy, security, and long-term availability of licensing records.

4.7.1 Data Storage Approach

The DVLD system uses **Microsoft SQL Server** as its centralized relational database management system. SQL Server was chosen for its strong transactional support, reliability, and suitability for government-scale administrative systems.

Data is stored in normalized relational tables to maintain consistency and eliminate redundancy.

4.7.2 Core Data Entities

The system persistently stores the following core entities:

- **User Data:** User accounts, roles, login credentials, permissions
- **Applicant Data:** Personal details, identification documents
- **Application Data:** License applications, types, statuses
- **Driving School Data:** School details, instructors, training schedules
- **Test Data:** Vision, theory, and practical test records
- **License Data:** Issued licenses, renewals, replacements, detainments
- **Notification Logs:** Alerts and communication history
- **Audit Logs:** Administrative actions and system changes

4.7.3 Data Access and Retrieval

Data access is performed through the **Data Access Layer (DAL)** using **ADO.NET**.

The system supports full CRUD operations:

- Create
- Read
- Update
- Delete

All database interactions use parameterized queries and stored procedures to ensure security and performance.

4.7.4 Data Security and Access Control

Persistent data is protected using multiple security mechanisms:

- Secure password hashing
- Role-Based Access Control (RBAC)
- Parameterized SQL queries
- Controlled database connections
- Restricted access through DAL only

Sensitive operations are logged for auditing and accountability.

4.7.5 Data Maintenance and Scalability

SQL Server supports efficient scaling as the number of applicants, licenses, and institutions grows. Routine maintenance includes:

- Regular database backups
- Archiving old records
- Monitoring query performance
- Index optimization

This persistent data management strategy ensures the DVLD system remains **secure, reliable, and scalable** while supporting long-term national deployment.

4.8. Component Diagram

This section presents the component diagram of the proposed Driving License Management System (DVLD). The component diagram illustrates the high-level software components of the system and shows how these components interact through defined interfaces and connections.

The DVLD system is organized into components that correspond to the layered architecture, ensuring a clear separation between user interfaces, business logic, and data access. Each component encapsulates a specific set of responsibilities and communicates with other components through well-defined interfaces.

Main System Components

Presentation Components

The presentation layer consists of two main components:

- **Desktop Application Component**, used by DVLD staff and driving institute personnel. This component provides role-based user interfaces and interacts with business components to perform system operations.
- **Mobile Application Component**, used by applicants to submit applications, select training institutes and batches, and view application status.

These components do not directly access the database and rely entirely on business components for system functionality.

Business Logic Components

The business layer is composed of several core components, including:

- User and Access Management Component
- License Application Management Component
- Training Management Component
- Testing Management Component

- License Issuance Component
- License Renewal and Replacement Component
- Payment Management Component

Each business component implements specific system rules and exposes interfaces that are used by the presentation components. Interactions between business components occur where process coordination is required, such as verifying training completion before test scheduling.

Data Access Components

The data layer includes components responsible for persistent data handling, such as:

- User Data Component
- Application and License Data Component
- Training Data Component
- Test and Payment Data Component

These components manage database communication and provide data services to the business layer through controlled interfaces.

Database Component

The Database Component represents the Microsoft SQL Server database at the core of the DVLD system. Acting as the central repository, it stores and manages all persistent and operational data required by the system, ensuring reliability, integrity, and availability across all modules, including both the desktop and mobile applications.

Designed using a relational model, the database employs well-defined tables, primary keys, foreign keys, and constraints to accurately capture real-world entities and their relationships within the driving license management domain. This structure guarantees consistency, enforces business rules, and supports seamless integration between different system components.

Key Data Domains

- **People and User Accounts**

Stores personal information, login credentials, roles, and permission data for applicants,

DVLD staff, and institute staff. This ensures secure access control and accountability across the system.

- **Applications and Application Types**

Maintains records of all submitted applications, including new licenses, renewals, replacements, international licenses, and detained license releases. Each record tracks application status, associated fees, and processing history, enabling transparent workflows and efficient case management.

- **Licenses and License Classes**

Contains details of issued licenses, license numbers, validity periods, license classes, and current license status (active, expired, detained, or revoked). This supports accurate monitoring of license lifecycles and compliance enforcement.

- **Test Appointments and Results**

Manages schedules and outcomes for vision, theory, and practical tests. Data includes test scores, results, and retake information, all linked to specific applications to provide a complete applicant profile.

- **Training Institutes, Batches, and Attendance**

Records information about registered driving institutes, training batches, applicant enrollment, and attendance tracking during the training phase. This ensures proper oversight of training programs and applicant readiness.

Component Interaction and Dependencies

The component diagram shows that:

- Presentation components depend on business logic components to execute system functions.
- Business logic components depend on data access components to store and retrieve persistent data.
- Direct interaction between presentation components and the database is not allowed.

This structured component interaction improves system maintainability, supports independent component development, and allows future system enhancements with minimal impact on existing components.

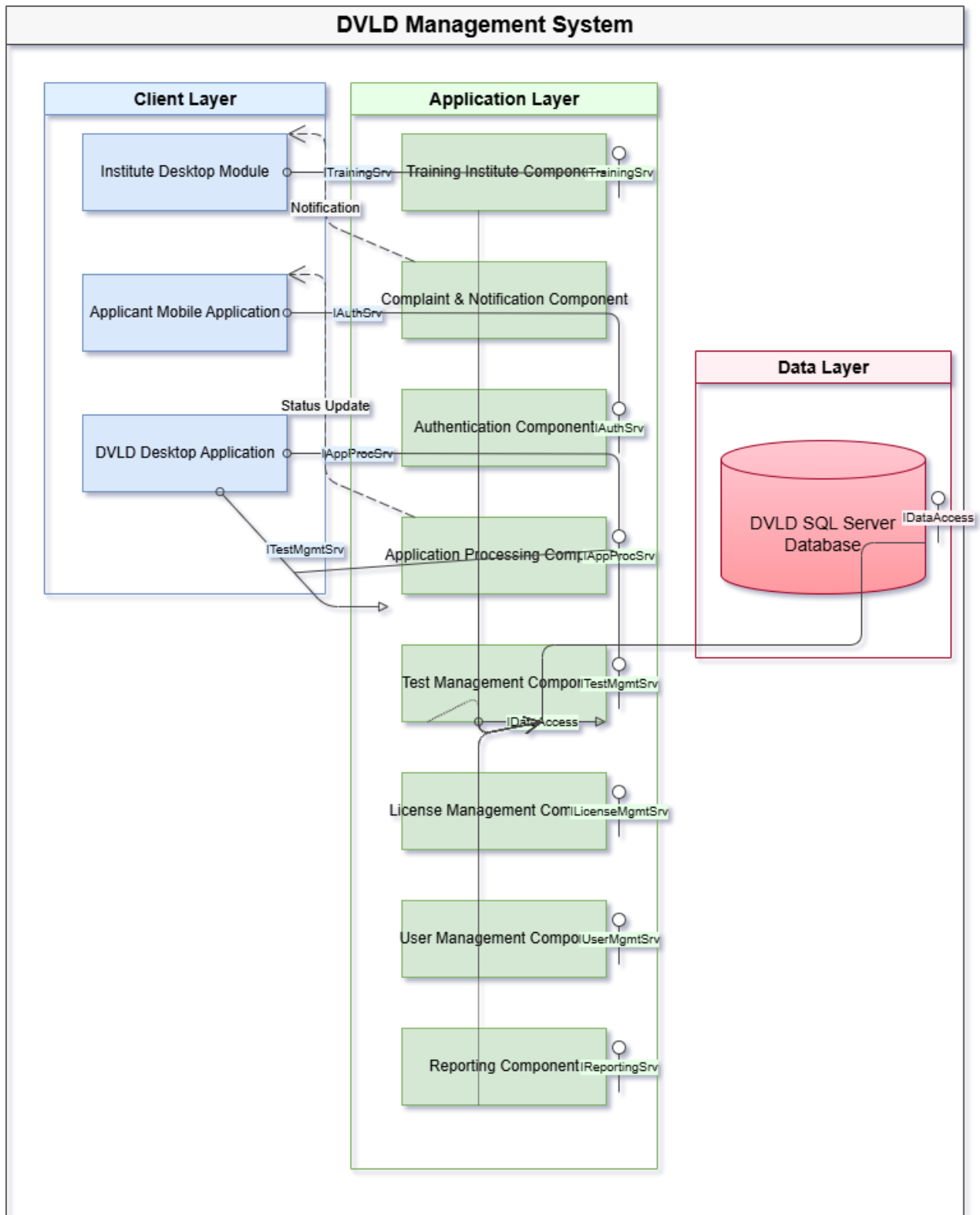


Figure 33: Component Diagram

4.9. Database Diagram

This section presents the database diagram of the proposed Driving License Management System (DVLD). The database diagram illustrates the structure of the system database, including the main tables, their attributes, primary keys, foreign keys, and the relationships between them. It provides a clear view of how system data is organized and interconnected.

The DVLD database is designed using a relational database model to ensure data integrity, consistency, and efficient data retrieval. Each table represents a specific system entity, and relationships are established to support system workflows such as application processing, training management, testing, and license issuance.

4.9.1. Entities and Attributes

Entity	Attributes
User	• user_id (pk) • person_id (fk → People.person_id) • username • password • is_active
People	• person_id (pk) • national_no • first_name • middle_name • last_name
People	• date_of_birth • gender • address • phone • email • nationality_country_id (fk → Countries.country_id) • image_path
Countries	• country_id (pk) • country_name

Drivers	• driver_id (pk) • person_id (fk → People.person_id) • created_by_user_id (fk → Users.user_id) • created_date
LicenseClasses	• license_class_id (pk) • class_name • class_description • minimum_allowed_age • default_validity_length - class_fees
Applications	• application_id(pk) • applicant_person_id (fk → People.person_id) • application_date • application_type_id (fk → ApplicationTypes.application_type_id) • application_status • last_status_date • paid_fees • created_by_user_id (fk → Users.user_id)
ApplicationTypes	• application_type_id (pk) • application_type_title • application_fees
LocalDrivingLicenseApplications	• local_driving_license_application_id (pk) • application_id (fk → Applications.application_id) • license_class_id (fk → LicenseClasses.license_class_id)
Licenses	• license_id (pk) • application_id (fk → Applications.application_id) • driver_id (fk → Drivers.driver_id) • license_class • issue_date • paid_fees • is_active • issue_reason - created_by_user_id (fk → Users.user_id)

DetainedLicenses	• detained_id (pk) • license_id (fk → Licenses.license_id) • detained_date • fine_fees • created_by_user_id (fk → Users.user_id) • is_released • released_by_user_id (fk → Users.user_id) • release_application_id (fk → Applications.application_id)
TestTypes	• test_type_id (pk) • test_type_title • test_type_description • test_type_fees
TestAppointments	• test_appointment_id (pk) • test_type_id (fk → TestTypes.test_type_id) • local_driving_license_application_id (fk → LocalDrivingLicenseApplications.local_driving_license_application_id) • appointment_date • paid_fees • created_by_user_id (fk → Users.user_id) • is_locked • retake_test_application_id (fk → Applications.application_id)
Tests	• test_id (pk) • test_appointment_id (fk → TestAppointments.test_appointment_id) • test_result • notes • created_by_user_id (fk → Users.user_id)
Attendance	• attendance_id (pk) • application_id (fk → Applications.application_id) • course_id (fk → InstituteCourses.course_id) • attendance_date • is_present
InstituteCourses	• course_id (pk) • institute_id (fk → DrivingInstitutes.institute_id) • course_name • duration_days • course_fees

DrivingInstitutes

- institute_id (pk)
- created_by_user_id (fk → Users.user_id)
- institute_name
- address
- phone
- email
- is_active

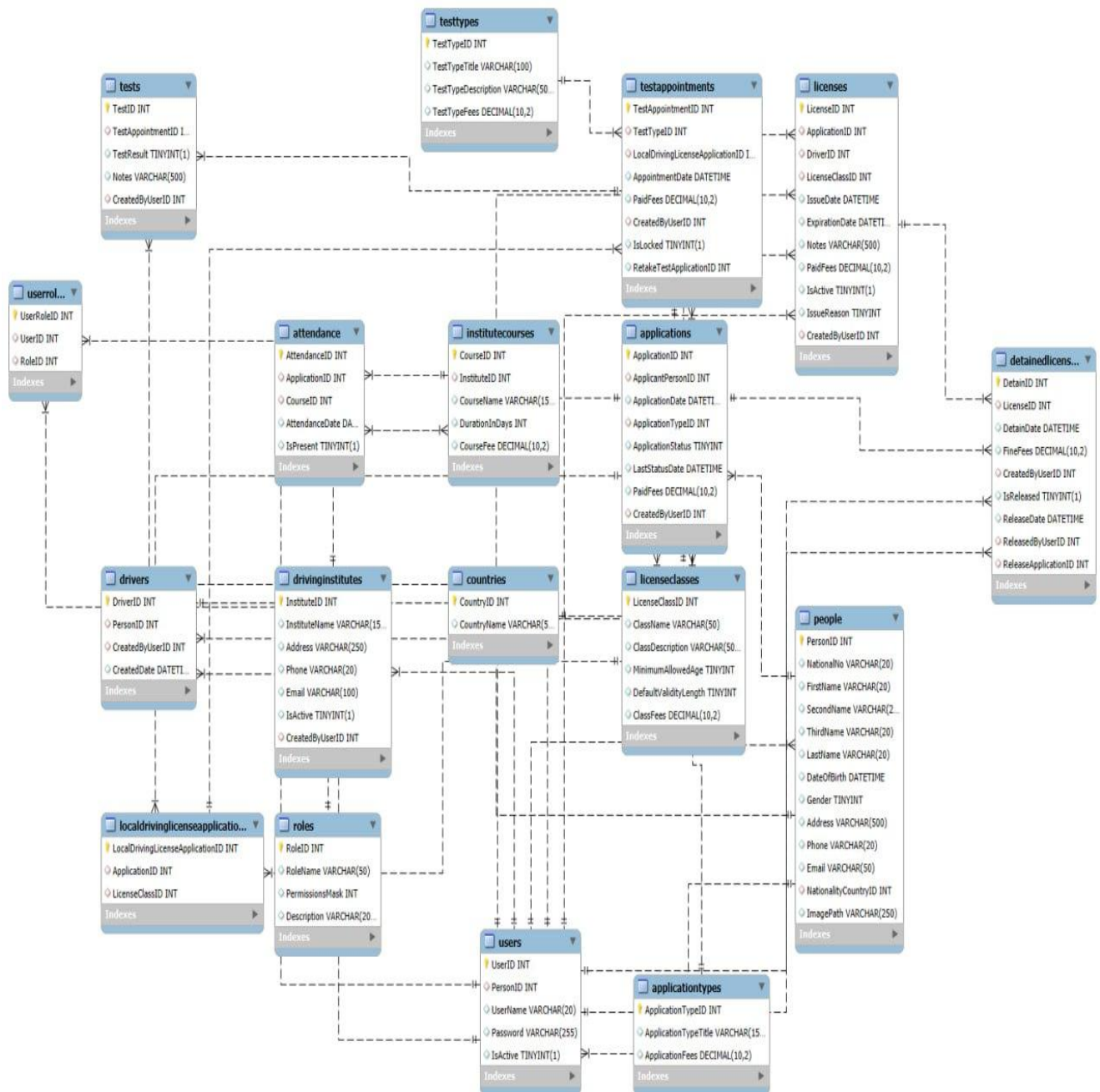


Figure 34: Database Diagram

4.10. Access Control

The **Driving License Management System (DVLD)** incorporates robust access control mechanisms to safeguard system security, protect sensitive data, and ensure proper authorization of users. Access control is a fundamental aspect of the system's architecture, designed to restrict functionalities based on user identity and assigned responsibilities. By enforcing strict boundaries between roles and permissions, the DVLD system minimizes risks of unauthorized actions and ensures that each user interacts only with the features necessary for their duties.

At the core of this design is **Role-Based Access Control (RBAC)**, the primary mechanism used to manage permissions. Under RBAC, every system user is assigned a specific role, and permissions are granted according to that role. This structured approach guarantees that staff, institute personnel, and applicants operate within clearly defined boundaries, reducing complexity while maintaining flexibility.

User Roles and Permissions

The DVLD system supports multiple user roles, each with predefined permissions tailored to their responsibilities:

- **System Administrator** – Oversees the entire system, manages user accounts, configures permissions, and ensures smooth operation of all modules.
- **Registration Officer** – Handles applicant registration, manages submitted applications, and verifies applicant details.
- **Test Officer** – Schedules and supervises vision, theory, and practical tests, records results, and manages retakes.
- **License Issuer** – Responsible for issuing new licenses, renewals, replacements, and handling detained or revoked licenses.
- **Institute Manager** – Manages driving institutes, training batches, and ensures compliance with DVLD standards.
- **Institute Trainer** – Tracks applicant attendance, supervises training sessions, and records progress during the training phase.

- **Applicant** – Accesses limited functions through the mobile application, such as submitting applications, booking test appointments, and viewing license or application status.

This role-based structure ensures that DVLD staff can manage applications, tests, and licenses, while institute staff focus exclusively on training-related activities. Applicants, meanwhile, are restricted to self-service features, preventing them from accessing administrative or sensitive data.

Authentication and Authorization

Access control begins with authentication, which is carried out through a secure login mechanism designed to verify user credentials before granting access to the system. This process ensures that only legitimate users are able to enter, protecting sensitive data and preventing unauthorized activities. By enforcing strong authentication measures, the system maintains integrity, safeguards user information, and establishes a trusted foundation for all subsequent operations.

Once authenticated, **authorization checks** are applied to determine which menus, screens, and operations are available to the user. The system dynamically loads user interfaces based on the authenticated user's role, ensuring that restricted features remain inaccessible. For example, a Test Officer will see test scheduling and result management options, while an Applicant will only see application submission and status tracking features. This dynamic interface design enhances usability while reinforcing security.

Data Access Control

Beyond functional restrictions, the DVLD system enforces strict and role-based data access control to safeguard sensitive information. Critical data such as license details and applicant records are accessible only to authorized DVLD staff whose responsibilities require direct interaction with these records. In contrast, institute trainers are limited to training-related information, ensuring they have the resources necessary for their duties without exposure to confidential applicant data. This separation of access not only protects privacy but also strengthens accountability, reduces the risk of misuse, and maintains the integrity of the overall system.

To further strengthen security, the DVLD system prohibits any direct access to the database from the presentation layer, ensuring that user-facing components cannot bypass established safeguards.

Instead, all data operations are routed exclusively through the business layer, which enforces strict access rules, applies business constraints, and validates requests before they reach the database. This layered architecture not only prevents unauthorized queries but also guarantees consistency across transactions, reduces the risk of data misuse, and provides a clear separation of responsibilities between system components, thereby enhancing both reliability and maintainability.

Security and Future Enhancements

The access control design significantly enhances overall system security by minimizing unauthorized actions and reducing the likelihood of data breaches. By carefully aligning user permissions with specific roles and responsibilities, the DVLD system ensures that each actor has access only to the information and functions necessary for their tasks. This role-based approach not only strengthens accountability and transparency across all operations but also helps maintain data integrity, protect sensitive records, and build trust among stakeholders who rely on the system's reliability and security.

The architecture is also forward-looking, allowing for **future enhancements** without major structural changes. Potential improvements include:

- More granular permission levels for fine-tuned control.
- Comprehensive auditing of user activities to track and analyze system usage.
- Integration with external authentication services such as biometric verification or single sign-on (SSO).

These enhancements would further strengthen the system's resilience by ensuring it can withstand potential disruptions and continue operating reliably under varying conditions. At the same time, they preserve the system's scalability, allowing it to handle growing numbers of users, institutions, and services without performance degradation. By maintaining adaptability, the system remains flexible enough to integrate new features, support evolving requirements, and align with future expansion goals, ultimately providing a secure and sustainable foundation for long-term success.

Chapter 5: Implementation

5.1. Overview

This chapter describes the overall implementation approach of the Driving and Vehicle License Department (DVLD) Management System, explaining how the proposed system design and requirements are transformed into a functional software solution. The implementation follows a modular, multi-tier architecture that clearly separates the Presentation Layer, Business Logic Layer, and Data Access Layer to enhance maintainability, scalability, and reliability.

The system is primarily implemented as a desktop-based application for DVLD staff and driving institutes, while a mobile application is planned for applicants to support registration, application tracking, and communication. The system is designed to support multiple user roles, including Applicants, DVLD Staff, Test Officers, License Issuers, Driving Institutes, and System Administrators, each with clearly defined responsibilities and access privileges enforced through role-based access control.

Microsoft SQL Server is used as the backend database to ensure secure and reliable data storage, while ADO.NET is used for database connectivity and transaction management. The modular design allows the system to be extended in the future to support additional features such as online payments, advanced reporting, and AI-assisted fraud detection.

5.2 Coding Standard

To ensure high code quality, maintainability, and consistency, the DVLD Management System follows well-defined coding standards across all layers of the application. These standards are applied throughout the Presentation Layer, Business Logic Layer, and Data Access Layer.

Consistent formatting rules are enforced, including proper indentation, meaningful variable naming, and structured exception handling. Hard-coded values are avoided where possible, and configuration values such as database connection strings are stored securely in configuration files. These coding practices help reduce errors, improve readability, and simplify future maintenance and enhancement of the system.

5.2.1 Naming Conventions

Clear and descriptive naming conventions are used consistently across the entire system to improve code readability and understanding.

- **Variables and Methods:** camelCase
(e.g., *applicationDate*, *saveApplication()*)
- **Classes and User Controls:** PascalCase
(e.g., *clsApplication*, *frmAddUpdateLicense*, *ctlPersonCard*)
- **Constants and Enumerations:** UPPER_SNAKE_CASE
(e.g., *MAX_LOGIN_ATTEMPTS*, *APPLICATION_STATUS_NEW*)
- **Files and Folders:** PascalCase or meaningful naming based on functionality

These conventions help ensure uniformity across the codebase and make it easier for developers to understand and extend the system.

5.2.2 Code Structure and Organization

The DVLD system follows a **layered and modular architecture**, ensuring a clear separation of concerns:

- **Presentation Layer (PL):**
Contains Windows Forms, user controls, and UI-related logic responsible for interacting with system users.
- **Business Logic Layer (BLL):**
Contains core business rules, validations, workflows, and domain models such as applications, licenses, tests, and users.
- **Data Access Layer (DAL):**
Handles all database interactions using ADO.NET, including executing SQL commands, managing connections, and handling transactions.

The project structure emphasizes:

- Clear separation between UI, logic, and data

- Reusable classes and user controls
- Consistent folder organization
- Maintainable and extensible design

This structure allows each layer to evolve independently without affecting the others.

5.2.3 Documentation Practices

To support maintainability and ease of understanding, proper documentation practices are followed throughout the project:

- Inline comments are used to explain complex logic or critical operations.
- Classes and methods are designed with clear responsibilities and self-explanatory names.
- A project-level documentation file (README) is prepared to describe the system overview, setup instructions, and architectural design.
- Database tables and relationships are documented using ER diagrams to support clarity and future maintenance.

These practices ensure that the system can be easily understood and extended by other developers or future teams.

5.2.4 Security Best Practices

Security is a key consideration in the implementation of the DVLD Management System. Several best practices are applied to protect system data and user access:

- **Secure Authentication:**
User authentication is enforced through a login system, with passwords stored in encrypted or hashed form using secure algorithms.
- **Role-Based Access Control (RBAC):**
System access is restricted based on user roles and permissions, ensuring that users can only access features relevant to their responsibilities.

- **Secure Database Access:**

ADO.NET parameterized queries are used to prevent SQL injection attacks and ensure safe data handling.

- **Input Validation:**

User input is validated and sanitized at both the UI and business logic levels to prevent invalid or malicious data submissions.

- **Configuration Security:**

Sensitive information such as database connection strings is stored securely and not hard-coded into the application.

These measures help protect the system from unauthorized access and common security threats.

5.3 Development Tools

The following tools and technologies are used for the development, testing, and management of the Driving and Vehicle License Department (DVLD) Management System. These tools support efficient development, collaboration, debugging, and long-term maintainability of the system.

Integrated Development Environment (IDE)

Microsoft Visual Studio is used as the primary development environment for building the DVLD desktop application. Visual Studio provides strong support for C#, Windows Forms, ADO.NET, and SQL Server integration, along with advanced debugging, refactoring, and UI design tools. Its built-in designers simplify the development of complex user interfaces and user controls used throughout the system.

Database Management Tools

Microsoft SQL Server is used as the primary relational database management system for storing all system data, including users, applications, licenses, tests, payments, and institutes.

SQL Server Management Studio (SSMS) is used to:

- Design and manage database tables and relationships
- Execute SQL scripts and queries

- Inspect stored data and constraints
- Perform backup and restore operations

Data Access and Connectivity Tools

ADO.NET is used for database connectivity and data manipulation. It enables:

- Secure communication between the application and SQL Server
- Execution of parameterized queries to prevent SQL injection
- Transaction management for critical operations
- Efficient handling of data readers and data tables

Debugging and Testing Tools

Several tools and techniques are used to ensure correct system behavior:

- Visual Studio Debugger for step-by-step execution, breakpoints, and variable inspection
- Logging mechanisms to track application behavior and errors
- Manual testing of business workflows such as application submission, test scheduling, and license issuance
- Database-level validation using SQL queries to verify stored data accuracy

These tools help identify logic errors early and ensure reliable system performance.

API and Mobile Integration Tools

To support mobile access and system extensibility, the DVLD Management System includes a .NET Core-based RESTful Web API that exposes selected backend services to external clients. The API is primarily designed to serve the applicant mobile application and provides secure endpoints for authentication, application tracking, test information, and notifications.

The following tools and technologies are used or planned for API and mobile integration:

- **ASP.NET Core Web API** for implementing RESTful services and handling client-server communication

- **Postman** for testing and validating API endpoints, request/response formats, and authentication mechanisms
- **OpenAPI/Swagger** for automatic API documentation and interface definition
- **Flutter or Xamarin** (planned) for developing the applicant mobile application that consumes the API services

This approach ensures a clear separation between the presentation layer and backend services, enabling easier maintenance, improved scalability, and support for future clients such as web or third-party systems.

Version Control System

Git is used as the version control system, with GitHub serving as the remote hosting platform. This setup enables effective collaboration and code management throughout the development lifecycle.

Key benefits include:

- Tracking changes to source code
- Supporting collaboration among team members
- Branching for feature development and bug fixes
- Rollback to previous stable versions when needed

Productivity and Collaboration Tools

- GitHub Projects for task management and progress tracking
- Google Docs for project documentation and report writing
- Online communication platforms (e.g., Telegram or Slack) for team coordination

5.4 Prototype

The prototype of the Driving and Vehicle License Department (DVLD) Management System is implemented as a **desktop-based application developed using Microsoft Visual Studio**. The desktop prototype is built with **C# and Windows Forms (WinForms)** under the .NET Framework and demonstrates the core system functionalities, including user authentication, application processing, test management, license issuance, and reporting.

DVLD - Secure Login

DVLD Management System

Driving and Vehicle License Department

Username

Enter your username

Password

Enter your password

Login As

Select your role

☐ Remember me on this computer

Login

Cancel

© 2024 DVLD Management System. All rights reserved.

Version 2.0.0 | Support: support@dvld.gov

Figure 35: Login Page Design(planned)

FileEditViewToolsHelp

DVLD Management System v2.0

Admin User

DVLD Management System

Dashboard

Applicants

Applications

Test Management

Licenses

Vehicles

Reports

Complaints

Notifications

Settings

Dashboard

Applicants

Applications

Test Management

Licenses

Vehicles

Reports

Complaints

Notifications

Settings

Dashboard

Welcome back! Here's what's happening today.

TOTAL APPLICANTS

12,458

+12% from last month

PENDING APPLICATIONS

847

-5% from last week

LICENSES ISSUED TODAY

156

+23% from yesterday

TESTS SCHEDULED

234

+8% from last week

Pending Tasks

5 pending

Review Application #A-2024-8847

Licence Application

high

Today

Schedule Theory Test - John Doe

Test Management

medium

Tomorrow

Process Renewal #R-2024-1122

Licence Renewal

high

Today

Verify Documents - App #A-2024-8850

Document Verification

low

Now 7%

Quick Actions

New Applicant

New Application

Issue License

Schedule Test

Renew License

Find Person

Recent Applications

View All

App ID	Applicant	Type	Status	Action
A-2024-8851	Sarah Johnson	New Licence	Pending	
A-2024-8858	Michael Chen	Renewal	Processing	
A-2024-8849	Emma Williams	New Licence	Approved	
A-2024-8848	James Brown	Replacement	Rejected	

System Alerts

4 now

Admin User (Admin)

Connected

Tue, Dec 23, 2025 03:28 PM

Figure 36: Dashboard Page Design(planned)

121

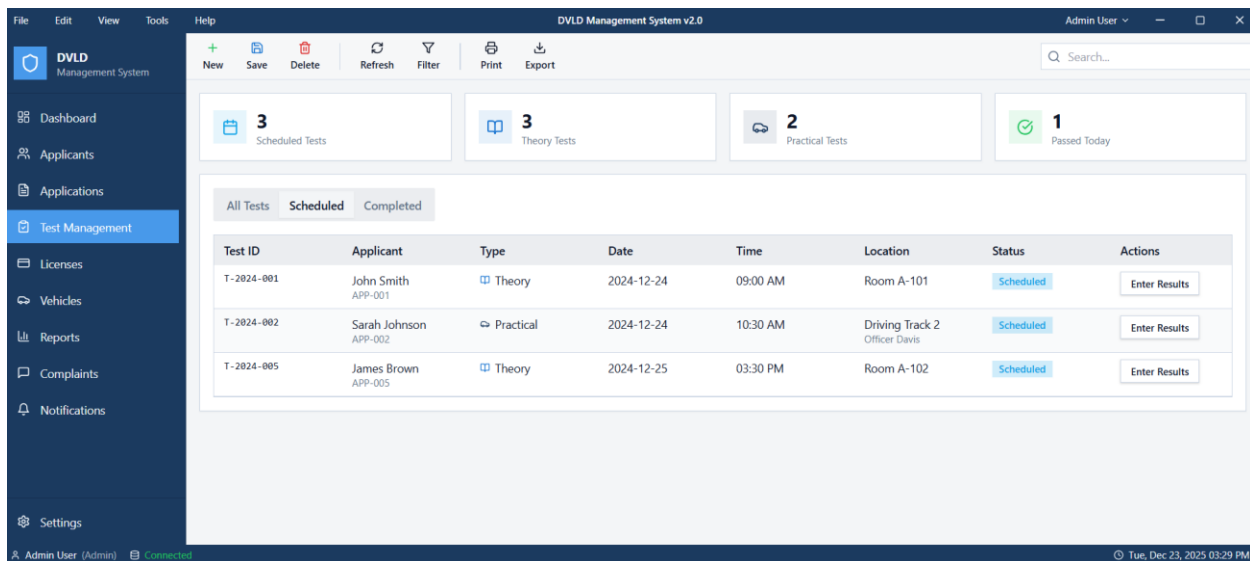


Figure 37: Test Management Page Design(planned)

5.5 Implementation Detail

This section describes how the Driving and Vehicle License Department (DVLD) Management System was practically developed, highlighting the technologies used, architectural structure, and key implementation approaches applied in both the desktop client and server-side components. It explains how system functionalities were translated into working software modules to ensure reliability, security, and maintainability.

5.5.1 Client-Side Implementation

The client-side of the Driving and Vehicle License Department (DVLD) Management System is implemented as a **multi-client environment**, consisting of a **desktop-based application** for internal DVLD staff and driving institutes, and a **mobile application** for applicants. This design ensures accessibility for different user groups while maintaining centralized control and data consistency.

The desktop client is developed using **C# and Windows Forms (WinForms)** within the Microsoft .NET Framework, providing a robust, event-driven environment suitable for administrative and transaction-heavy workflows. The mobile client enables applicants to access selected DVLD services remotely through secure API communication.

Both clients are designed to interact with the same core system logic, ensuring uniform enforcement of business rules and system behavior. The implementation strictly follows **object-oriented programming (OOP) principles**, where each major system function is encapsulated within dedicated classes and modules, promoting code reusability, maintainability, and scalability.

5.5.1.2. Client-Side Technologies Used

The DVLD Management System employs stable and enterprise-proven technologies to ensure reliability, performance, and long-term maintainability.

C# (.NET Framework)

Overview:

C# is a modern, object-oriented programming language developed by Microsoft and widely used for enterprise-grade desktop and backend systems.

Benefits:

- **Strong Object-Oriented Support:** Enables accurate modeling of real-world entities such as users, applications, licenses, tests, and drivers.
- **Type Safety:** Reduces runtime errors through compile-time validation.
- **Rich Standard Library:** Provides built-in support for UI development, security, file handling, and database communication.
- **Maintainability:** Encourages modular, readable, and well-structured code.

Windows Forms (WinForms)

Overview:

Windows Forms is a graphical user interface framework for building Windows desktop applications using the .NET platform.

Benefits:

- **Event-Driven Architecture:** Supports responsive interactions through events such as button clicks, form loading, and data selection.

- **Rapid Application Development:** Visual designers simplify UI creation and iteration.
- **Native Windows Integration:** Seamless interaction with operating system features.
- **Suitability for Administrative Systems:** Well-suited for data-intensive, form-based applications such as DVLD workflows.

Mobile Client Application

Overview:

The mobile application is designed for DVLD applicants to access system services remotely using smartphones.

Capabilities:

- Application submission and tracking
- Viewing test schedules and results
- Receiving license status updates and notifications

The mobile client does not access the database directly and communicates exclusively with the backend through secure RESTful APIs.

ADO.NET

Overview:

ADO.NET is used for database connectivity and communication with Microsoft SQL Server.

Benefits:

- **Efficient Data Access:** Enables secure and high-performance CRUD operations.
- **Parameterized Queries:** Protects against SQL injection attacks.
- **Transaction Management:** Ensures data consistency during multi-step operations such as license issuance or test result submission.
- **Tight SQL Server Integration:** Optimized for enterprise-level relational databases.

5.5.1.3 Client-Side Architecture

The client-side architecture follows a layered and object-oriented design, aligning with the overall system architecture.

Form-Based Modular Design

Each Windows Form corresponds to a specific business function, including:

- User and role management
- Driving license application processing
- Test scheduling and result management
- License issuance, renewal, and detention handling
- Reporting and statistical analysis

This modular structure enables independent development, testing, and maintenance of individual system features.

Business Entity Representation

Core system entities such as:

- Users
- Applications
- Licenses
- Tests
- Drivers
- Payments

are represented using dedicated C# classes. These classes closely reflect the underlying database schema and business rules, ensuring consistency between the presentation layer, business logic, and data storage.

Role-Aware Interface

The user interface dynamically adapts based on the authenticated user's role:

- Administrators have full system access

- Registration officers manage applications
- Test officers conduct and record tests
- License issuers process license creation and renewal

Unauthorized features are hidden or disabled at the user interface level, providing an additional layer of access control alongside server-side validation.

5.5.1.4 User Interface (UI) Design

The DVLD system's UI is designed with a focus on **usability, clarity, and operational efficiency**, particularly for administrative users.

Layout Structure

- Main Navigation Menu: Provides access to core system modules
- Work Area: Displays active forms, lists, and data grids
- Dialog Windows: Used for confirmations, alerts, and detailed data entry

Consistency and Readability

- Uniform color scheme and font usage
- Clear labeling and grouped input fields
- Consistent button placement and naming conventions

Form Validation

- Mandatory fields are clearly indicated
- Input validation is performed before database submission
- User-friendly error messages guide corrective actions

Error Handling

- Informative error dialogs are displayed for invalid operations
- System exceptions are logged to assist debugging and maintenance

5.5.1.5 Interaction with the Server-Side

The desktop client communicates with backend functionality through the **Business Logic Layer (BLL)** and **Data Access Layer (DAL)**.

Communication Method

- Direct method calls between UI forms and business logic classes
- Secure database access via ADO.NET
- Transaction-based execution for critical workflows

Data Format

- Strongly typed C# objects used for internal data exchange
- SQL queries and stored procedures handle database interaction

Mobile Client Interaction

The mobile application interacts with the DVLD system through an **ASP.NET Core Web API**.

- RESTful API endpoints expose selected system functionalities
JSON-based request and response format
- All requests are validated and authorized at the API level
- API invokes shared BLL methods, ensuring consistent business rules
- No direct database access from the mobile client

This approach allows both desktop and mobile clients to operate on the same system logic while using different communication mechanisms.

5.5.1.6 Security and Performance Considerations

Security Considerations

- Authentication: Users must log in before accessing system features.
- Role-Based Access Control: Enforced at both UI and business logic levels.
- Input Validation: Prevents invalid or malicious data entry.

- Secure Database Access: Parameterized queries and controlled connections.

Performance Considerations

- Optimized SQL queries and indexing
- Efficient data loading for large datasets
- Proper event handling to avoid UI blocking
- Structured exception handling to ensure application stability

5.5.2 Server-Side Implementation

The server-side of the Driving and Vehicle License Department (DVLD) Management System is designed using a layered and service-oriented architecture that supports both desktop and mobile clients. The system separates presentation concerns from business logic and data access responsibilities, ensuring high maintainability, security, and scalability.

The DVLD system operates as a multi-client platform, consisting of:

- A Windows Forms desktop application for DVLD staff and driving institutes
- A mobile application for applicants

Both clients interact with a centralized backend through shared business logic and secure data access mechanisms, with the mobile application communicating via ASP.NET Core Web API services.

5.5.2.1 Server Architecture

The DVLD system follows a layered architecture with service exposure, enabling multiple client platforms to operate on the same core system logic.

Main Architectural Components

1. Presentation Layer (PL)

The Presentation Layer consists of two active client applications, each serving a specific user group.

a) Desktop Presentation Layer (Windows Forms)

Function:

Provides a rich desktop interface for DVLD staff and driving institute personnel.

Responsibilities:

- Capture user input through forms and controls
- Display structured data and reports
- Perform UI-level validation
- Restrict access based on user roles
- Forward requests to the Business Logic Layer through direct method calls

b) Mobile Presentation Layer (Applicant Mobile Application)

Function:

Provides applicants with mobile access to DVLD services.

Responsibilities:

- Allow applicants to submit and track applications
- Display test schedules and license status
- Receive system notifications
- Communicate exclusively with the backend via RESTful APIs
- Send and receive data in JSON format

The mobile client does not access the database directly and relies entirely on the Web API for secure communication.

2. Service Layer (ASP.NET Core Web API)

Function:

Acts as a secure communication bridge between the mobile application and the core DVLD system.

Responsibilities:

- Expose DVLD functionalities through RESTful endpoints Authenticate and authorize mobile users
- Validate incoming HTTP requests
- Invoke Business Logic Layer methods Return structured JSON responses
- Enforce security policies and access rules

Key Characteristics:

- Built using ASP.NET Core Web API
- Stateless request handling
- Platform-independent client support
- Reuses existing .NET business logic libraries

3. Business Logic Layer (BLL)

Function:

Represents the core processing engine of the DVLD system and is shared across all client platforms.

Responsibilities:

- Enforce DVLD business rules and workflows
- Validate system operations (eligibility, test requirements, license issuance)
- Coordinate multi-step processes
- Apply role-based permissions
- Serve both desktop and API requests using the same logic

Implementation Detail:

The BLL is implemented as a **.NET Framework C# class library**, allowing it to be:

- Directly referenced by the desktop application
- Consumed by the ASP.NET Core Web API

4. Data Access Layer (DAL)

Function:

Handles all interactions with the Microsoft SQL Server database.

Responsibilities:

- Execute parameterized SQL queries and stored procedures
Perform CRUD operations
- Manage database connections and transactions
- Maintain data consistency and integrity
- Abstract database operations from higher layers

Implementation**Detail:**

The DAL is implemented as a **.NET Framework C# class library**, ensuring consistent and secure data access for both desktop and API-based requests.

Architectural Advantages

- Single Source of Business Logic for all clients
- Secure Mobile Integration via Web APIs
- No Direct Database Access from Clients
- Code Reusability Across Platforms
- Scalable and Maintainable Design

5.5.2.2 Server-Side Technologies and Frameworks

The DVLD system utilizes enterprise-grade technologies suitable for administrative and government systems.

C# (.NET Framework)

- Used for Business Logic Layer and Data Access Layer
- Object-oriented and strongly typed
- Encourages modular and maintainable code
- Shared across desktop and API projects

ASP.NET Core Web API

- Exposes backend services for mobile application
- Supports RESTful communication
- Built-in middleware for authentication and logging
- High performance and scalability

Microsoft SQL Server

- Centralized relational database
- Supports transactions, constraints, and indexing
- Ensures data consistency and durability

ADO.NET

- Used in DAL for database communication
- Parameterized queries to prevent SQL injection
- Transaction handling for critical operations

5.5.2.3 Request Handling Workflow

Desktop Client Request Flow

1. User interacts with Windows Forms UI
2. UI validates input and calls BLL methods
3. BLL applies business rules and permissions
4. DAL executes database operations
5. SQL Server processes the request
6. Results return to UI
7. Errors are handled and logged

Mobile Client Request Flow

1. Mobile app sends HTTP request to Web API
2. API validates authentication and input

3. API calls shared BLL methods
4. BLL interacts with DAL
5. SQL Server processes the data
6. API returns JSON response to mobile client

5.5.2.4 Data Management

Data is centrally managed using Microsoft SQL Server.

Managed Data Includes

- Applicants and DVLD users
- Applications and license requests
- Tests and evaluation results
- Licenses and renewals
- Institutes and training records
- Payments, fines, and complaints

Data Handling Strategy

- Normalized relational schema
- Primary and foreign key enforcement
- Stored procedures for complex workflows
- Parameterized queries for security
- Transaction-based updates

5.5.2.5 Security and Access Control

Authentication

- Secure login for desktop users
- Token-based authentication for mobile users via Web API

Authorization

- Role-based access control enforced in BLL
- UI-level restrictions for desktop users
- API-level permission validation for mobile users

Data Protection

- Encrypted or hashed credentials
- No direct database access from clients
- Centralized validation of all inputs

5.5.2.6 Scalability and Extensibility

- Layered architecture enables easy expansion
- Mobile and desktop clients share core logic
- Web API allows additional clients (web portals, kiosks)
- Database optimized for growing data volume
- New services can be added without restructuring existing layers

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